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Graph-Based Assessment of Carbon Dioxide (CO₂) Migration in Multi-Layered Stratigraphic Systems

Master's thesis in Applied Physics and Mathematics

Supervisor: Jo Eidsvik

Co-supervisor: Andrea Riebler

September 2026



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Nomenclature

The following list provides a quick reference to some of the acronyms and symbols used throughout this thesis. Each term is defined wherever it first appears in the main body, and the thesis may therefore be read independently of the nomenclature.

Acronyms

CCS	Carbon Capture and Storage
cdf	Cumulative Distribution Function
CTP	Capillary Threshold Pressure
DAG	Directed Acyclic Graph
eCTP	Effective Capillary Threshold Pressure
HMC	Hamiltonian Monte Carlo
IP	Invasion Percolation
McMC	Markov chain Monte Carlo
pdf	Probability Density Function
pmf	Probability Mass Function

Chemical Substances

CO ₂	Carbon dioxide
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Non-SI Units

°C	Degrees Celsius ($(x - 273.15) \text{ °C} = x \text{ K } \forall x \geq 0, x \in \mathbb{R}$)
Mtpa	Million tonnes per annum ($1 \text{ Mtpa} \approx 3.171 \cdot 10 \text{ kg/s}$)

Physical Constants

$\Delta\rho$	In situ density difference between CO ₂ and brine
g	Acceleration due to standard gravity ($1 \text{ } g \approx 9.807 \text{ m/s}^2$)

Quantities Related to Layers/Reservoirs

$\mathbf{d}(t) = (d_i(t))$	(Observed) height of the CO ₂ plume within each reservoir/layer at time t , where $d_i(t)$ is the (observed) plume height within the i th reservoir/layer at time t .
$\mathbf{h}(t) = (h_i(t))$	Height of the CO ₂ plume within each reservoir/layer at time t , where $h_i(t)$ is the plume height within the i th reservoir/layer at time t .
$\mathbf{q}^{\text{ent}}(t) = (q_i^{\text{ent}}(t))$	Rate at which CO ₂ enters each reservoir/layer at time t , where $q_i^{\text{ent}}(t)$ is the rate corresponding to the i th reservoir/layer at time t .
$\mathbf{q}^{\text{ext}} = (q_i^{\text{ext}})$	Constant rate at which CO ₂ is injected into each reservoir/layer due to external processes, where q_i^{ext} is the rate corresponding to the i th reservoir/layer.
$\mathbf{q}^{\text{mig}}(t) = (q_i^{\text{mig}}(t))$	Rate at which CO ₂ migrates away from each reservoir/layer at time t , where $q_i^{\text{mig}}(t)$ is the rate corresponding to the i th reservoir/layer at time t .
$\mathbf{q}^{\text{net}}(t) = (q_i^{\text{net}}(t))$	Net rate at which CO ₂ accumulates within each reservoir/layer at time t , where $q_i^{\text{net}}(t)$ is the net rate of accumulation within the i th reservoir/layer at time t .
$\mathbf{T}^{\text{ent}} = (T_i^{\text{ent}})$	Entry time of each reservoir/layer, where T_i^{ent} is the entry time of the i th reservoir/layer.
$\mathbf{T}^{\text{mig}} = (T_i^{\text{mig}})$	Migration time of each reservoir/layer, where T_i^{mig} is the migration time of the i th reservoir/layer.
$\mathbf{V}(t) = (V_i(t))$	Volume of accumulated CO ₂ within each reservoir/layer at time t , where $V_i(t)$ is the volume within the i th reservoir/layer at time t .
k	Number of reservoirs/layers within the multi-layered stratigraphic system.
r_{inj}	Constant rate at which CO ₂ is injected into the bottommost reservoir/layer due to external processes.

Quantities Related to Cap Formations

$\boldsymbol{\alpha} = (\alpha_i)$	(Length of) the lateral semi-axes of $\boldsymbol{\psi}$, where α_i corresponds to the lateral semi-axis of ψ_i .
$\boldsymbol{\beta} = (\beta_i)$	(Length of) the vertical semi-axes of $\boldsymbol{\psi}$, where β_i corresponds to the vertical semi-axis of ψ_i .
$\boldsymbol{\psi} = (\psi_i)$	Cap formation covering each reservoir/layer, where ψ_i is the formation covering the i th reservoir/layer.

$\mathbf{C} = (C_i)$	CTP of ψ , where C_i corresponds to the CTP of ψ_i .
$\mathbf{P}(t) = (P_i(t))$	Pressure head across ψ at time t , where $P_i(t)$ is the pressure head across ψ_i at time t .

Quantities Related to Migration Pathways

$\mathbf{E}(G) = (e_{ij})$	Relative eCTP matrix, where $e_{ij} = e_{i,j}$ if $a_{ij} \in A(G)$, else $e_{ij} = 0$.
$\mathbf{E}^{\text{sat}} = (e_{ij}^{\text{sat}})$	Relative eCTP matrix for a saturated system, is equivalent to $\mathbf{E}(G^{\text{sat}})$.
$\mathbf{e}^{\text{sat}} = (e_{ij}^{\text{sat}})$	$\mathbf{E}^{\text{sat}}$ in row-major form, i.e. $\mathbf{e}^{\text{sat}} = (e_{12}^{\text{sat}}, e_{13}^{\text{sat}}, \dots, e_{1k}^{\text{sat}}, e_{23}^{\text{sat}}, \dots, e_{(k-1)k}^{\text{sat}})$.
$\mathbf{Q}(G) = (q_{ij})$	Relative migration rate matrix, where $q_{ij} = q_{i,j}$ if $a_{ij} \in A(G)$, else $q_{ij} = 0$.
$\mathbf{Q}^{\text{sat}} = (q_{ij}^{\text{sat}})$	Relative migration rate matrix for a saturated system, is equivalent to $\mathbf{Q}(G^{\text{sat}})$.
$\mathbf{q}^{\text{sat}} = (q_{ij}^{\text{sat}})$	$\mathbf{Q}^{\text{sat}}$ in row-major form, i.e. $\mathbf{q}^{\text{sat}} = (q_{12}^{\text{sat}}, q_{13}^{\text{sat}}, \dots, q_{1k}^{\text{sat}}, q_{23}^{\text{sat}}, \dots, q_{(k-1)k}^{\text{sat}})$.
$\mathbf{T}^{\text{path}} = (t_{ij}^{\text{path}})$	Pathway specific migration times, where t_{ij}^{path} is the time when CO ₂ first migrates along the pathway between the i th and j th reservoirs. If no such pathway exists, t_{ij}^{path} is assumed to be infinite.
$E(G) = \{e_{i,j}\}$	Set of all $e_{i,j}$, where $e_{i,j}$ is the relative eCTP of the pathway between the i th and j th reservoirs, affecting when CO ₂ is allowed to migrate along the pathway.
$Q(G) = \{q_{i,j}\}$	Set of all $q_{i,j}$, where $q_{i,j}$ is the relative rate at which CO ₂ migrates along the pathway between the i th and j th reservoirs, once the i th reservoir is compromised.

Quantities Related to the System as a Whole

$\mathbf{A}(G) = (a_{ij})$	Adjacency matrix of G , where a_{ij} is the number of arcs in G having $v_i \in V(G)$ as its head and $v_j \in V(G)$ as its tail.
$A(G) = \{a_{i,j}\}$	Set of arcs in G , where $a_{i,j}$ represents an inter-reservoir migration pathway from the i th to the j th reservoir.
$G = (V(G), A(G), \vartheta_G)$	Strict connected DAG representing a multi-layered stratigraphic system.

$V(G) = \{v_i\}$	Set of vertices in G , where v_i represents the i th reservoir.
$\epsilon(G)$	Cardinality of $A(G)$.
$\nu(G)$	Cardinality of $V(G)$.
G^{sat}	Similar to G , but where the stratigraphic system has been saturated, in the sense that for all $i = 1, 2, \dots, k-1$ and $j = 2, 3, \dots, k$, such that $i < j$, we have that $a_{i,j} \in A(G^{\text{sat}})$.
$V_{\text{sink}}(G) \subset V(G)$	Set of sinks in the network induced by G and c .
$V_{\text{source}}(G) \subset V(G)$	Set of sources in the network induced by G and c .

Quantities Related to the Simulator

$\mathbf{x}(t) = (x_i(t))$	Scaled height of the CO ₂ plume within each reservoir/layer at time t , where $h_i(t)$ is the scaled plume height within the i th reservoir/layer at time t .
$\mathbf{y}(t) = (y_i(t))$	Scaled volume of accumulated CO ₂ within each reservoir/layer at time t , where $y_i(t)$ is the scaled volume within the i th reservoir/layer at time t .
$\mathbf{Y}^{\text{crit}} = (y_{ij}^{\text{crit}})$	Critical scaled volume for the pathways of the stratigraphic system, where y_{ij}^{crit} is the critical scaled volume for the pathway between the i th and j th reservoirs. If $y_i(t) > y_{ij}^{\text{crit}}$, then CO ₂ may migrate along the corresponding pathway.
$\mathbf{z}(t) = (z_i(t))$	Scaled net rate at which CO ₂ accumulates within each reservoir/layer at time t , where $z_i(t)$ is the scaled net rate of accumulation within the i th reservoir/layer at time t .
$\mathcal{S} : t \mapsto \mathbf{h}(t)$	Differentiable simulator that to each time t associates a plume height $\mathbf{h}(t)$ to each reservoir.
τ^{act}	Parameter controlling the smoothness of pathway activations.
τ^{min}	Parameter controlling the smoothness of simulator minimisations, used to compute the time until pathway activations.

Quantities Related to Bayesian Inference

$\theta = (\alpha, \beta, C, \mathbf{q}^{\text{sat}}, \mathbf{e}^{\text{sat}})$	Collection of continuous parameters, on which we make inference.
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$\mathbf{h}^{\text{thres}} = (h_i^{\text{thres}})$	Detection threshold for the likelihood model, where h_i^{thres} is the detection threshold of the i th reservoir/layer.
$\mathcal{D} = \{\mathbf{d}^{(\ell)}(t_\ell)\}$	Set of all observations, where $\mathbf{d}^{(\ell)}(t_\ell)$ is the ℓ th observation, made at time t_ℓ , of the plume heights in the reservoirs of the stratigraphic system. Note that $\ell \leq D$.
α_{accept}	Metropolis-Hastings acceptance probability, used when sampling from the full conditional distributions.
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.
Ω_X	General symbol used to denote the sample space of some random variable X .
D	Number of observations in the observation set $\mathcal{D}$.
$p(\boldsymbol{\theta} \mid \mathcal{D}, G)$	Full conditional distribution of the continuous parameters $\boldsymbol{\theta}$.
$p(\boldsymbol{\theta})$	Prior distribution on the continuous parameters $\boldsymbol{\theta}$.
$p(\boldsymbol{\theta}, G \mid \mathcal{D})$	Posterior distribution on the model parameters $\boldsymbol{\theta}$ and G .
$p(\mathcal{D} \mid \boldsymbol{\theta}, G)$	Joint likelihood of the model parameters $\boldsymbol{\theta}$ and G , given every observation in the observation set $\mathcal{D}$.
$p(G \mid \mathcal{D}, \boldsymbol{\theta})$	Full conditional distribution of the DAG G .
$p(G)$	Prior distribution on the DAG G , representing the stratigraphic system of interest.

Other Quantities not Listed Elsewhere

$\vartheta_G : A(G) \rightarrow V(G) \times V(G)$	Incidence function of G that to each arc in $A(G)$ assigns an ordered pair of vertices in $V(G)$.
$c : A(G) \rightarrow \mathbb{N} \cup \{0\}$	Capacity (weight) function that defines the weights in a network.
ϕ	Angle of contact between a fluid and a surface.
i, j, ℓ	Indices used in connection with various quantities.
t	Time.

Abstract

Technologies related to the sequestration of CO_2 have gained renewed interest over the past decades, as a way of mitigating the negative effects of global warming. To ensure both the safety and viability of current, as well as future, CO_2 sequestration projects, it is of vital importance that we have a thorough understanding of how CO_2 behaves once sequestered. Only then can we be reasonably certain that the CO_2 stays sequestered long enough to have a positive effect on the climate. To that extent, probabilistic surrogate models have been proposed to model the inherent uncertainty of inter-layer CO_2 migration in multi-layered stratigraphic systems. Current formulations place a Markovian assumption on the migration of CO_2 , hence only accounting for the migration of CO_2 that occurs between subsequent layers. Multiple studies do however indicate that the migration of CO_2 between distant reservoirs can play a substantial role for reliably capturing the uncertainty of such systems.

In this thesis, we propose a novel framework to model the uncertainty associated with inter-layer CO_2 migration in multi-layered stratigraphic systems. Our approach differs from previous attempts at constructing similar surrogate models, by leveraging concepts from graph theory to account for possible CO_2 migration between distant reservoirs. We develop a differentiable simulator for the migration process, to allow for fast computation of multi-layered CO_2 migration, whilst also providing flexibility and control over the connections between reservoirs in the storage system. Bayesian inversion is then used to examine how seismic measurements affect the uncertainty of inter-layer CO_2 migration, where simulation from the posterior model is made computationally efficient through the use of Hamiltonian Monte Carlo. Our experiments consider a situation with 5 storage reservoirs and single-well injection in the deepest layer for a time period of 100 years. We discuss effects of gathering monitoring data at year 2.5 and 25. Our findings indicate that the migration of CO_2 between distant reservoirs has a pronounced effect on the posterior uncertainty of inter-layer migration in multi-layered stratigraphic systems. Additionally, our model indicates that rapid CO_2 accumulation in the topmost reservoir is exclusive to multi-layered stratigraphic systems that allow migration of CO_2 between distant reservoirs.

Sammendrag

Teknologi knyttet til karbonlagring har vært av stor interesse de siste tiårene som følger av det pågående arbeidet med å begrense de negative effektene av global oppvarming. For å sikre levedyktigheten av nåværende så vel som fremtidige lagringsprosjekter, kreves det en grundig forståelse av hvordan CO₂ oppfører seg etter at klimagassen er injisert i bakken. Kun da kan vi med rimelig sikkerhet vite at den injiserte gassen forblir lagret lenge nok til at karbonlagring har en positiv effekt på klimaet. Forskere har nylig utviklet sannsynlighetsbaserte surrogatmodeller for å beskrive den iboende usikkerheten knyttet til vertikal CO₂ migrasjon i geologiske systemer bestående av flere lag. Disse modellene benytter seg av Markovprosesser for å beskrive migrasjonen av CO₂, noe som betyr at de kun kan ta høyde for migrasjon mellom direkte etterfølgende lag. Forskning viser imidlertid at migrasjon mellom fjerne reservoarer kan være av stor betydning for realistisk modellering.

I denne masteroppgaven utvikler vi et nytt rammeverk for å modellere usikkerheten knyttet til vertikal CO₂ migrasjon i geologiske flerlagssystemer. Metoden vår skiller seg fra tidligere forsøk på å konstruere liknende surrogatmodeller ved å benytte seg av konsepter fra grafteori for å beskrive den mulige forflytningen av CO₂ mellom fjerne geologiske lag. Ved å utvikle en deriverbar simulator for migrasjonsprosessen, tillater vi hurtig beregning av flerlagsmigrasjonen samtidig som vi beholder en høy grad av fleksibilitet, samt kontroll, over forbindelsene mellom reservoarene i lagringssystemet. Bayesisk inversjon blir deretter benyttet for å undersøke hvordan seismiske målinger påvirker usikkerheten i flerlagssystemet, hvor simulering fra posteriorfordelingen gjøres på en beregningseffektiv måte gjennom bruk av "Hamiltonian Monte Carlo". Eksperimentene våre tar utgangspunkt i et system bestående av 5 reservoarer der CO₂ blir injisert i det dypeste reservoaret over en periode på 100 år. Vi diskuterer deretter effektene av målinger foretatt etter 2.5 og 25 år. Funnene våre indikerer at vertikal migrasjon mellom fjerne geologiske lag har en uttalt effekt på usikkerheten knyttet til flerlagssystemer a posteriori. I tillegg indikerer modellen vår at en hurtig oppsamling av CO₂ i det øverste reservoaret kun er forenlig med geologiske systemer hvor CO₂ kan migrere mellom fjerne reservoarer.

1. Introduction

As part of the global effort to achieve net zero emissions within the year 2050, the United Nations' (UN) Intergovernmental Panel on Climate Change (IPCC) reiterated, in their special report of 2018, the need to quickly adopt industrial scale technologies for the collection and sequestration of CO₂, if we are to limit global warming to below 1.5 °C above pre-industrial levels (Ma et al. 2022). The process of collecting and sequestering CO₂ in subsurface geological formations is often referred to by its acronym, carbon capture and storage (CCS), and serves as a way to reduce the increase in global temperature due to climate change, by limiting the amount of CO₂ that enters the atmosphere. With the adoption of the Kyoto protocol in 1997, industrial nations, including Norway and members of the European Community (EC), now known as the European Union (EU), pledged their commitment to "implement and/or further elaborate (...) research on, promotion, development and increased use of (...) carbon dioxide sequestration technologies" (United Nations 1997, Article 2(1)(a)). Statements from both the International Energy Agency (IEA) and the Carbon Sequestration Leadership Forum (CSLF) do however indicate that the current state of CCS is vastly insufficient, if the goals of limiting global warming to below 1.5 °C are to be met (Ma et al. 2022).

Although the European Parliament and Council of the EU (2009) have amended the regulatory and legal frameworks across Europe to allow implementation of industrial scale CCS projects, the regulations require comprehensive analyses on the suitability of potential storage sites before they can be used to sequester CO₂. Moreover, the regulations require any operator of such a project to continuously monitor the storage site for potential leakage and other environmentally ill effects that may follow as a result of the sequestration. When Statoil, now known as Equinor, and its partners began sequestering CO₂ from the gas-condensate production at the Sleipner field (block 15/9 in the Norwegian North Sea) in 1996, the Norwegian regulations concerning CCS was limited (Furre et al. 2017). In a bid to avoid the Norwegian emissions tax, Statoil decided to inject CO₂ from the production of natural gas at the Sleipner field into the saline aquifers of the Utsira formation, thus commencing the worlds 1st large scale (offshore) CCS project (Furre et al. 2017; Ma et al. 2022; Zhang et al. 2022). To this day, Equinor continues to sequester CO₂ in the Utsira formation at an approximate rate of 0.9 million tonnes per annum (Mtpa), which has resulted in the sequestration of over 20 million tonnes to date. As the project was the 1st of its kind, it has provided valuable insight into the behaviour of CO₂ once sequestered, and forms the basis for the monitoring guidelines used by operators of today's CCS projects (Chadwick et al. 2006; Furre et al. 2024; Furre et al. 2017).

One of the main findings of the CCS project at the Sleipner field, centres at the use of saline aquifers to sequester CO₂. Historically, the sequestration of CO₂ have only been utilised for the purposes of enhanced oil recovery (EOR), in a technique known as water alternating gas (WAG), where water and CO₂ is alternately injected into an oil reservoir to reduce the amount of residual oil trapped within the reservoir (Feder 2021; Ma et al. 2022; Merchant 2017). To model the dynamics of such reservoirs, finite element methods (FEM) based on Darcy flow are often used, as seen in for instance Ali et al. (2018) and Li et al. (2015), which require well posed boundary conditions to allow for the

simulation of multi-phase flow within the reservoirs. As for saline aquifers, microscopic fractures are common in the rock elements covering the reservoirs (Wang et al. 2021), and so it becomes difficult to develop boundary conditions for FEM that respects the physical properties of the reservoirs. In the case of the Sleipner field, these fractures originally stunned researchers, as seismic monitoring indicated that CO₂ had migrated to shallower reservoirs within the Utsira formation much faster than what was predicted, had the thin layers of shale within the Utsira formation acted as seals for the migrating CO₂ (Cavanagh and Haszeldine 2014).

To properly model the uncertainty in multi-layered stratigraphic systems, where saline aquifers are considered for CO₂ sequestration, a surrogate model might therefore be necessary. Recent attempts at implementing such models that explain the uncertainty associated with inter-layer CO₂ migration, including for instance Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear), use a Markovian process to model the migrating CO₂. Consequently, the migration gets restricted to occur between subsequent layers only, which in the case of the Sleipner field, is non-realistic (Furre et al. 2019; Vinje et al. 2025; Williams and Chadwick 2021). In this thesis, we instead propose a probabilistic framework based on graph theory (Bondy and Murty 1982), to try and explain the uncertainty of inter-layer CO₂ migration in a general multi-layered stratigraphic system. Using a simulation based approach, we then ask the question of whether it possible to reduce this uncertainty through the acquisition of seismic measurements and subsequent Bayesian inversion.

In Chapter 2, we present the necessary prerequisites to understand how graph theory might be used to model the inter-layer migration of CO₂ in multi-layered stratigraphic systems, after which, we develop a differentiable simulator to characterise the migration of CO₂ within these systems. Observations from the Sleipner field are here used as a motivator for the choices we make throughout this chapter, although the framework is in general intended to work for any multi-layered stratigraphic system. In Chapter 3, we introduce prior uncertainty in our model, and describe how seismic measurements can be obtained through the same process Statoil used to monitor the sequestered CO₂ at the Sleipner field, a process that is easily replicated for other offshore CCS projects. We then characterise the posterior uncertainty of our model by applying Bayes' theorem, and obtain samples from the posterior distribution efficiently using Hamiltonian Monte Carlo (HMC). In Chapter 4, we summarise our findings and point to the remaining work ahead. We note that the thesis is a continuation of the report by Grødeland (2026), who used a similar graph based model to characterise the uncertainty in simpler multi-layered stratigraphic systems. As part of this thesis, we treat the underlying theory of the framework in much greater detail, referring to the report only when strictly needed to highlight some of the limitations of the simpler model proposed by Grødeland (2026). Note in particular, that the thesis may be read independently of the report in question.

2. Graph-Based Framework for CO₂ Migration Modelling

2.1. Dynamics of CO₂ Migration

2.1.1. Invasion Percolation Theory

The framework we use in this thesis to model the migration of CO₂ in multi-layered stratigraphic systems, relies heavily on the theory of invasion percolation (IP), as it enables the specification of criteria under which the migration of CO₂ can take place. The theory can be traced back to a symposium on Monte Carlo methods in 1954, where Broadbent and Hammersley (1957) began working on a novel scheme to describe the process in which a fluid moves through a medium. Their methodology, now known as (regular) percolation theory, is distinct from previous analyses on diffusion processes, in the sense that the medium is endowed with properties that govern the transportation of a fluid (as opposed to the fluid itself). Broadbent and Hammersley (1957) showed that by modelling the medium as an interconnected structure, both intrinsic and random characteristics could be accounted for. Consequently, the migration of fluid throughout the medium could be fully determined by the properties of the medium alone.

Drawing inspiration from the hydraulic radius theories of Scheidegger (1974, ch. 6.3), Chandler et al. (1982) notably elaborated on this result, demonstrating the feasibility of considering certain displacement processes involving immiscible fluids in porous media as percolation processes. In particular, Chandler et al. (1982) considered the process in which water displaces oil in porous rock, preferentially wetted by either water or oil, and modelled the rock as an interconnected structure of pores and throats. In simple terms, preferentially wetting refers to the phenomena where certain fluids tend to stick more closely to a surface, compared to other fluids. When in contact with a surface, a fluid will have an angle of contact ϕ that is either acute ($\phi < \pi/2$) or obtuse ($\pi/2 < \phi < \pi$). If ϕ is acute, as seen in Figure 2.1a, the fluid is non-wetting and if ϕ is obtuse, as seen in Figure 2.1b, it is a wetting fluid (Zhang et al. 2024). When multiple fluids are in contact with a surface at the same time, the surface is said to be preferentially wetted by the fluid whose ϕ is largest. As for Figure 2.1, this implies that the surface is preferentially wetted by the wetting fluid (Display b), as opposed to the non-wetting fluid (Display a).

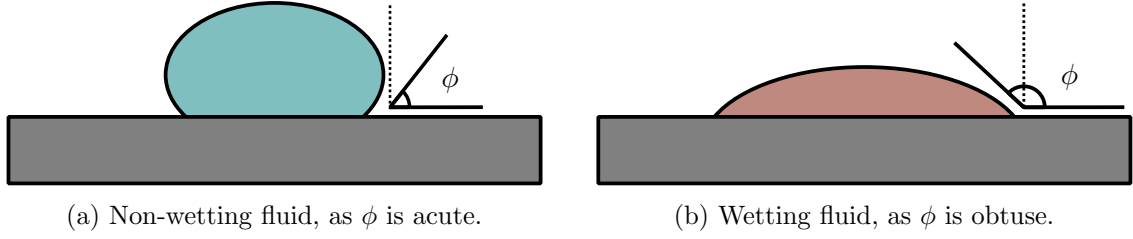


Figure 2.1.: Wetting and non-wetting fluid droplets (red/blue) in contact with a solid surface (grey). The angle of contact between the droplets and the surface is denoted by ϕ .

For the purposes of this thesis, only the latter case will be of importance, i.e. the case where a fluid displaces another in a porous rock preferentially wetted by the latter, a process known as drainage (Al-Wahaibi et al. 2006). Chandler et al. (1982) demonstrated that by applying an external pressure head across the interconnected structure, the displacing fluid would be forced through the throats of the medium, only stopping when the applied pressure reaches an equilibrium with the capillary threshold pressure (CTP) resisting the movement of the displacing fluid. Chandler et al. (1982) argued that on a microscopic scale, surface tension forces tend to dominate viscous forces, resulting in a quasi-static regime with intermittent movement only. As the CTP is inversely proportional to the radius of the throats, Chandler et al. (1982) found that when the displacing fluid expands into the medium, the direction of flow is always along the throat of largest radius.

Wilkinson and Willemsen (1983) later coined the term IP theory, to describe a discrete process in which a fluid displaces another in a porous media, under constant flow rate due to the effects of capillary action. As the theory builds upon (regular) percolation theory, many of the same results first reported by Broadbent and Hammersley (1957) and Frisch and Hammersley (1963) can be shown to hold for IP theory as well. In particular, the so called "fundamental theorem of percolation", as reported on by both Chandler et al. (1982) and Wilkinson and Willemsen (1983). The theorem states that if the connections within the structure making up the medium is blocked or removed with some probability, there exists a critical threshold, such that if the probability is higher than this threshold, the probability of a fluid traversing the entire medium from end to end is 0. In terms of IP theory, the analogous result is that there exist a CTP, such that if the external pressure head across the interconnected structure is lower than this CTP, the displacing fluid will (almost surely) not traverse the entire medium.

For the purposes of this thesis, the notion of residual trapping is also worth mentioning, as it affects the way CO_2 migrates within the strata (Chandler et al. 1982; De Silva and Ranjith 2012; Wilkinson and Willemsen 1983). In Figure 2.2, an external pressure head has been placed over a brine-saturated mudstone (Display a), forcing a drainage process in which brine gets displaced by CO_2 . As the displacing fluid advances within the media, following the throats with the largest radii (Display b), the fluid is likely to branch off in multiple directions, creating separate channels along which the CO_2 migrates (Display c). If these channels later combine, the advancing CO_2 may completely surround and encapsulate parts of the media occupied by the preferentially wetted fluid. As the fluids are assumed incompressible, the encapsulated fluid will be unable to retreat any further, thus trapping it within the medium (Display d). The advancing CO_2 will then be able to react with positive ions in the trapped brine, forming solid carbonates that drop out of the solution, chemically trapping parts of the CO_2 within the strata (Callioli Santi et al. 2025). To keep our model manageable, we assume this effect to be negligible.

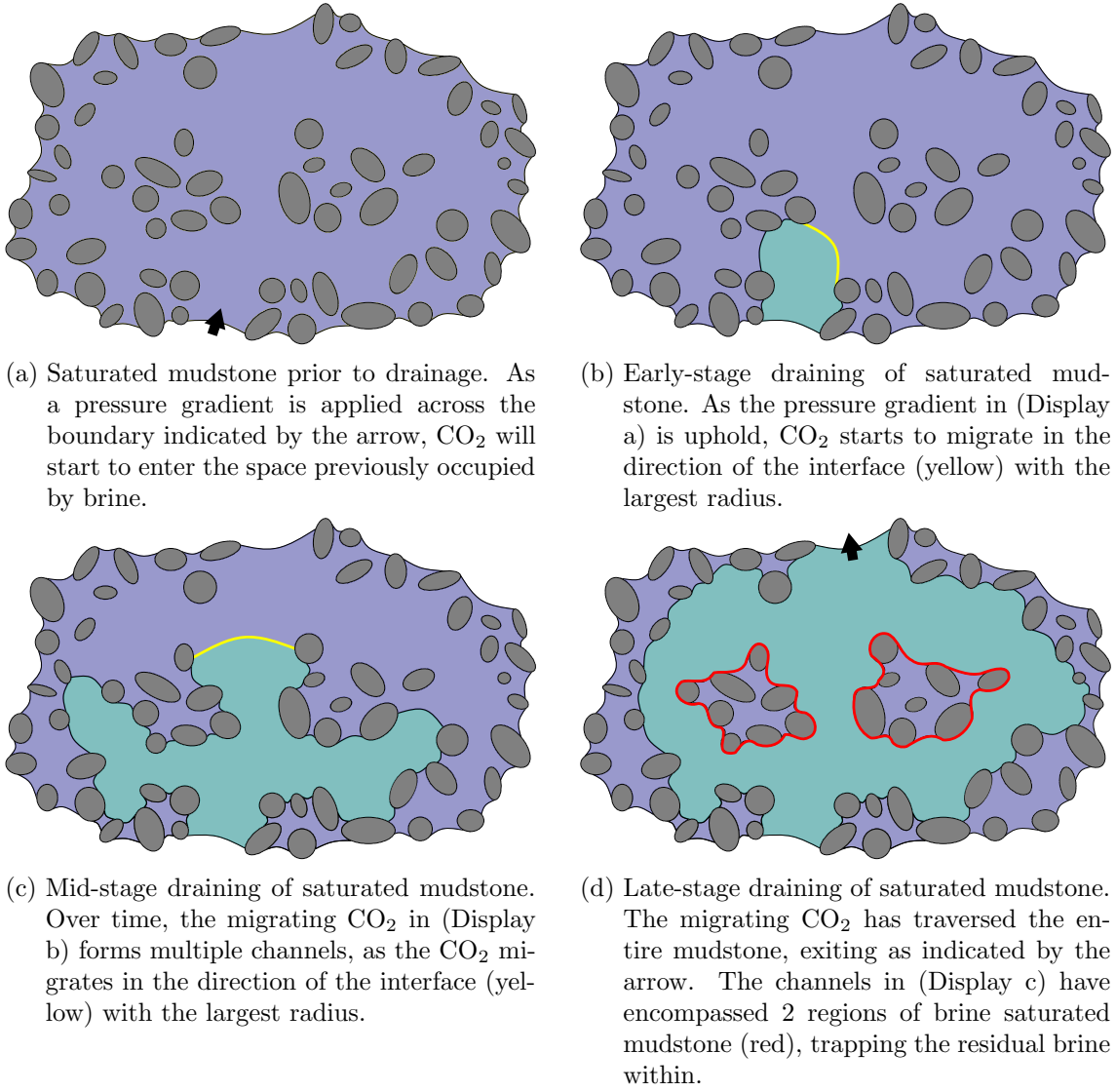


Figure 2.2.: Drainage process in which brine (purple) gets replaced by CO₂ (blue).

2.1.2. Conditions for Inter-Layer CO₂ Migration

Similar to Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear), we assume that our multi-layered stratigraphic system consists of $k \in \mathbb{N}$ layers of permeable brine-saturated rock, separated by regions of low permeability. Specifically, we assume that each layer is covered by a low permeability cap formation, which we denote by ψ_i , for $i = 1, 2, \dots, k$. Following IP theory, as presented in Section 2.1.1, we know that CO₂ will migrate through each ψ_i , thus entering a region of high permeability, when the pressure head $P_i(t)$ across ψ_i at time t surpasses the non-negative CTP C_i of ψ_i . Following the aforementioned research, we assume that $P_i(t)$ is mainly an artifact of the buoyancy forces acting on the CO₂, which tends to be the case in regions far from the injection site (Callioli Santi et al. 2025). From Archimedes' principle, it then follows that $P_i(t)$ is multilinear in the density difference between the two phases, which we denote by $\Delta\rho$, and the height $h_i(t)$ of accumulated CO₂ within the i th layer at time t . Specifically, we have that

$$P_i(t) = \Delta\rho g h_i(t), \quad i = 1, 2, \dots, k, \quad (2.1)$$

where $g \approx 9.807 \text{ m/s}^2$ denotes standard gravity.

Following Dutta et al. (2026), we define the notion of the migration time T_i^{mig} of the i th layer, as the time t when CO_2 first migrates through ψ_i , resulting in the i th layer being compromised. IP theory then provides an expression for T_i^{mig} in terms of a minimum,

$$T_i^{\text{mig}} = \min\{t > 0 : P_i(t) > C_i\} = \operatorname{argmin}_{t \in (0, \infty)} \left\{ h_i(t) > \frac{C_i}{\Delta \rho g} \right\}, \quad i = 1, 2, \dots, k, \quad (2.2)$$

where the 2nd equality follows from a simple substitution with Eq. (2.1).

We note that Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear) all considered inter-layer CO_2 migration as a Markovian process, in which CO_2 had to enter the i th layer before it could enter layer $i + 1$. As part of our thesis, we loosen this assumption and allow our framework to handle non-Markovian systems as well. Since we are not assuming that CO_2 migrates directly into layer $i + 1$ once the i th layer gets compromised, we need a way of describing when CO_2 first enters a particular layer. For that reason, we define the entry time T_i^{ent} of the i th layer, as the time t when CO_2 first migrates into it. As the presence of CO_2 within a layer increases its pressure head following Archimedes' principle, the following relation between T_i^{ent} and $P_i(t)$ exists,

$$T_i^{\text{ent}} = \operatorname{argmin}_{t \in (0, \infty)} \{P_i(t) > 0\} = \operatorname{argmin}_{t \in (0, \infty)} \{h_i(t) > 0\}, \quad i = 1, 2, \dots, k, \quad (2.3)$$

where we once again substitute Eq. (2.1) to obtain the 2nd equality.

Although the Markovian assumption held by Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear) provides a useful simplification, it restricts inter-layer CO_2 migration to consecutive layers only, and as such, does not allow the (direct) migration of CO_2 between distant layers. Seismic surveys from the Sleipner field, does however indicate that such migration patterns are of vital importance, if our model is to be an accurate representation of reality. When Cavanagh and Haszeldine (2014) modelled the inter-layer migration of CO_2 at the Sleipner field, they found that the observed accumulation of CO_2 could only be explained by multiple pre-existing fractures in ψ_i , for all $i = 1, 2, \dots, k$. Subsequent studies have found that these fractures have likely caused the formation of chimney-like structures within the system, channelling the migrating CO_2 past intermediate layers (Furre et al. 2019; Vinje et al. 2025; Williams and Chadwick 2021). Several reasons as to why these fractures exist have been put forth, ranging from overpressure caused by Pleistocene ice sheets to prior migration of natural gas (Cavanagh and Haszeldine 2014; Williams and Chadwick 2021). As such fractures are not unique to the Sleipner field, but commonly occur in most formations considered for CO_2 sequestration (Wang et al. 2021), a Markovian assumption on the inter-layer migration of CO_2 is likely to strict of an assumption to adequately model most multi-layered stratigraphic systems.

Figure 2.3 is a sketch of a (non-Markovian) multi-layered stratigraphic system, comprising $k = 5$ layers in total. By injecting CO_2 into the bottommost layer at constant rate r_{inj} , we have at the time t when the figure is depicted, caused an inter-layer migration process to commence, governed by the rules of IP theory. In particular, the migration from the 2nd layer was made possible because $P_2(t) > C_2$. We notice that at time t , CO_2 has migrated (directly) from the 3rd layer to the 5th layer, implying that $T_3^{\text{mig}} = T_5^{\text{ent}} < t < T_4^{\text{ent}} \leq T_4^{\text{mig}}$, where the last inequality follows from $C_4 \geq 0$. Had we used a Markovian assumption on the migration process, this would not have been possible, as the Markovian assumption implies $T_i^{\text{ent}} = T_{i-1}^{\text{mig}}$, for all $i = 2, 3, \dots, k$. Moreover, we notice that at time t , CO_2 migrates from the 1st layer directly to both the 2nd and the 3rd layers, which in itself, would not have been possible under a Markovian paradigm.

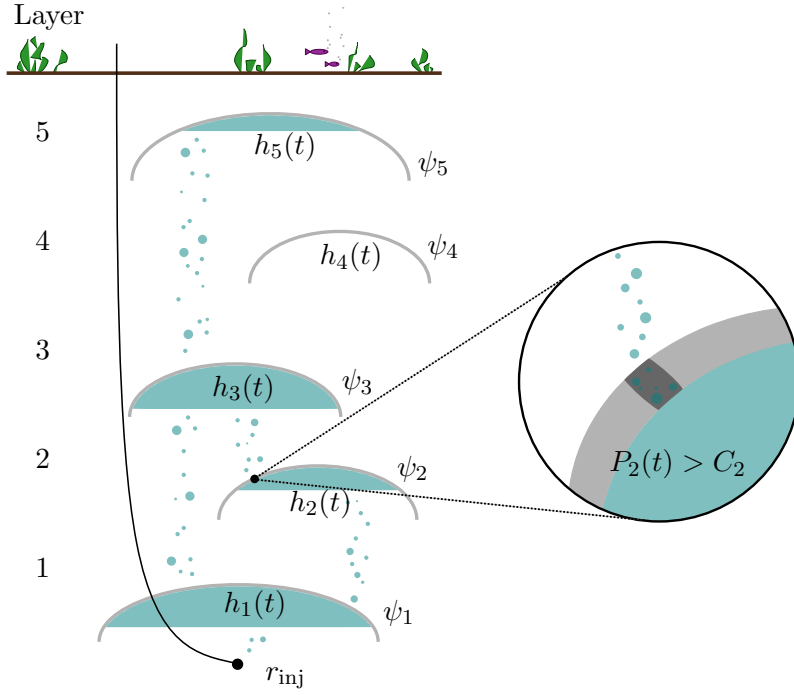


Figure 2.3.: Sketch of a non-Markovian $k = 5$ layered stratigraphic system. As CO_2 continues to be injected into the bottommost layer (layer 1) at a rate equal to r_{inj} , the presence of low permeability cap formations ψ_i hinder vertical migration of CO_2 , resulting in the formation of CO_2 plumes with height $h_i(t)$ at time t . Buoyancy forces acting on the CO_2 then cause a build-up in pressure $P_i(t)$, which allows the CO_2 to migrate through the ψ_i 's. This happens whenever the pressure $P_i(t)$ due to buoyancy surpasses the CTP C_i of the ψ_i 's, as shown in the magnified section of the 2nd layer.

We denote the volume of CO_2 within the i th layer at time t using $V_i(t)$. Note that by knowing the rates at which CO_2 enters- and migrates away from the i th layer at time t , denoted $q_i^{\text{ent}}(t)$ and $q_i^{\text{mig}}(t)$ respectively, it is possible to compute $V_i(t)$ as an integral over time,

$$V_i(t) = \int_0^t q_i^{\text{ent}}(s) - q_i^{\text{mig}}(s) ds = \int_0^t q_i^{\text{net}}(s) ds, \quad i = 1, 2, \dots, k.$$

where we define $q_i^{\text{net}}(t)$ as the net rate of CO_2 accumulation within the i th layer at time t . In particular, if $q_i^{\text{net}}(t)$ remains constant over some intervals leading up to t , e.g. $(t^{(0)}, t^{(1)}]$, $(t^{(1)}, t^{(2)}]$, $\dots$, $(t^{(m-1)}, t^{(m)}]$, where $0 = t^{(0)} < t^{(1)} < \dots < t^{(m-1)} < t^{(m)} = t$, the above integral reduces to a weighted sum over each interval where $q_i^{\text{net}}(t)$ is constant,

$$V_i(t) = \sum_{\ell=1}^n q_i^{\text{net}}(t^{(\ell-1)}) \cdot [t^{(\ell)} - t^{(\ell-1)}], \quad i = 1, 2, \dots, k. \quad (2.4)$$

Although constant injection rates might decrease the storage capacity of CO_2 in stratigraphic systems, notable examples, including the one at the Sleipner field, have operated with (approximately) constant injection rates since the beginning of the injection process (Cavanagh and Haszeldine 2014; Edlmann et al. 2019; Zhang et al. 2022). We therefore assume the injection rate within the i th reservoir due to external processes, denoted q_i^{ext} , to be constant. In cases where the injection process only injects CO_2 in the bottommost reservoir, i.e. $q_1^{\text{ext}} > 0$ and $q_i^{\text{ext}} = 0$ for all $i = 2, 3, \dots, k$, we refer to q_1^{ext} using r_{inj} . This is

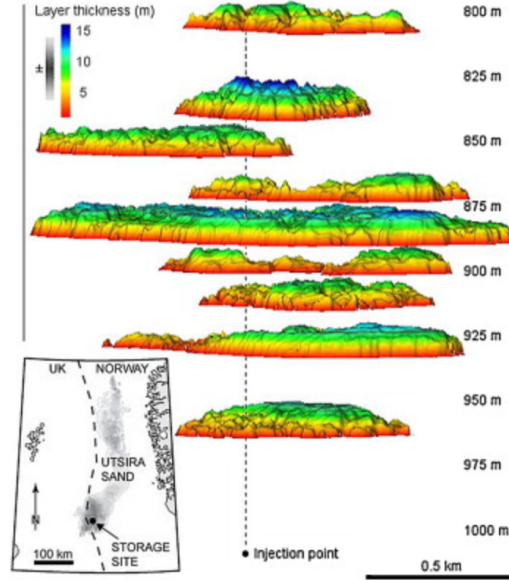


Figure 2.4.: Southeastern profile of the $k = 9$ layered system at the Sleipner field, illustrating the distribution of accumulated CO_2 after the injection of approximately 5 Mt CO_2 . The figure includes a map of the Utsira formation in the lower left corner, showing the location of the Sleipner storage site on the Norwegian continental shelf, as well as annotations on the right hand side indicating the reservoirs' depth below sea level. The figure is taken from Cavanagh and Haszeldine (2014).

done to conform with the notation used by previous researchers (Dutta et al. 2026; Eidsvik et al. to appear), and aligns well with the notation used in Figure 2.3. As q_i^{ext} is constant, $q_i^{\text{net}}(t)$ only changes when the migration pattern changes, i.e. when CO_2 starts to migrate in some novel way in response to a layer being compromised.

2.2. Modelling Elements of a Multi-Layered Stratigraphic System

2.2.1. Representing Reservoirs as Ellipsoidal Caps

We have till now used the term "layer" when referring to a region of high permeability within the stratigraphic system. Researchers typically refer to such regions as reservoirs, a term we adopt for the remainder of this thesis. As mentioned in Section 2.1.2, advancing CO_2 will be trapped within the i th reservoir until $P_i(t) > C_i$. The extent of each reservoir is therefore in part determined by the topology of ψ_i , as CO_2 is forced to accumulate underneath it prior to T_i^{mig} . The topology of ψ_i can in many cases be rather complicated, as shown in Figure 2.4, where the plume profile of the $k = 9$ layered system at the Sleipner field is illustrated. Notice that some of the reservoirs have variable thickness, letting $h_i(t)$ peak at multiple locations.

Recent studies by Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear) explore different alternatives to the problem of modelling the topology of ψ_i , for $i = 1, 2, \dots, k$. The use of ellipsoidal caps as surrogates for the ψ_i 's provided promising results, and we therefore assume a similar model for our thesis. Specifically, we assume

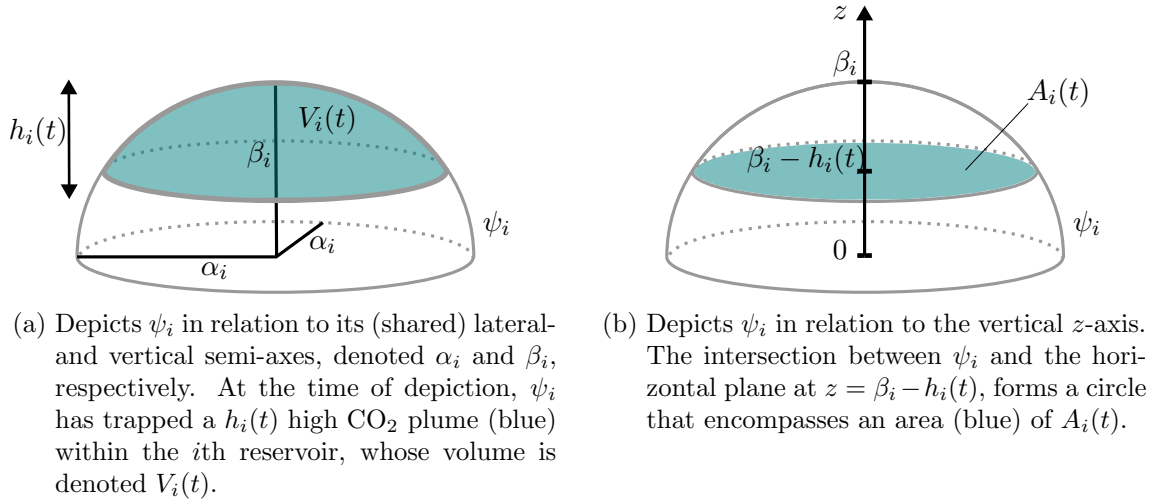


Figure 2.5.: Sketch of the (ellipsoidal) cap formation ψ_i covering the i th layer, at time t .

that ψ_i takes the form of an ellipsoidal cap, parametrised by a (shared) lateral- and a vertical semi-axis, which we denote by α_i and β_i , respectively, as shown in Figure 2.5a. Using set-builder notation, we find that

$$\psi_i = \left\{ (x, y, z) \in \mathbb{R}^3 : \left(\frac{x}{\alpha_i} \right)^2 + \left(\frac{y}{\alpha_i} \right)^2 + \left(\frac{z}{\beta_i} \right)^2 = 1 \wedge z \geq 0 \right\}, \quad i = 1, 2, \dots, k, \quad (2.5)$$

where $(x, y, z) \in \mathbb{R}^3$ denotes a point in Cartesian coordinates. In particular, $x \in R$ denotes the point's easting, $y \in R$ its northing and $z \in R$ its elevation relative to the origin. As we assume ψ_i to be parametrised by a shared lateral semi-axis, as opposed to individual semi-axes in the east- and north facing directions, the shape of ψ_i is more precisely referred to as a spheroidal cap. Note that a spheroid is a special type of ellipsoid, where 2 of its 3 semi-axes are equal (Osserman 2025), and so the use of the term "ellipsoidal cap" is also warranted. The reason for this additional assumption on the semi-axes of ψ_i should become apparent in the sections that follow.

Dutta et al. (2026) stated that by assuming an ellipsoidal cap model similar to that of Eq. (2.5), the following non-linear relation between $V_i(t)$ and $h_i(t)$ could be established,

$$V_i(t) = \frac{\pi \alpha_i^2}{3 \beta_i^2} h_i^2(t) [3 \beta_i - h_i(t)], \quad i = 1, 2, \dots, k. \quad (2.6)$$

Although Dutta et al. (2026) did not include a proof of this statement, it is not particularly difficult to show that the statement is indeed true. If we fix the value of $z \in [0, \beta_i]$ and examine the intersection $\psi_i \cap z$ between ψ_i and the horizontal plane at z , we notice that $\psi_i \cap z$ takes the form of circle. This is illustrated in Figure 2.5b, where we (for a fixed value of t) fixed the value of z by equating it to $\beta_i - h_i(t)$, resulting in a circular intersection that encompasses a total area of $A_i(t)$. By integrating $A_i(t)$ over z from $\beta_i - h_i(t)$ to β_i , we obtain an expression that reduces to Eq. (2.6). This is shown in greater detail in Appendix A, where a full derivation is included. Note that by substituting Eq. (2.4) in the above expression, we can compute $h_i(t)$ using only $q_i^{\text{net}}(t)$ and the semi-axes of ψ_i . This is an important insight that will be covered in greater detail in the sections that follow.

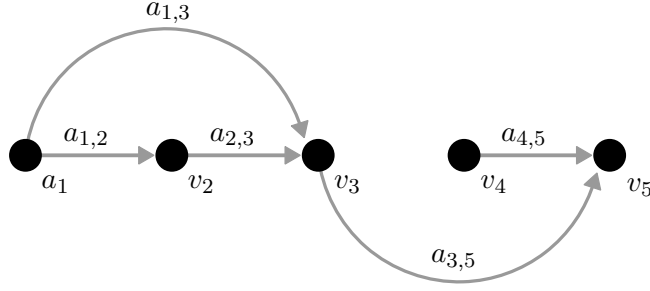


Figure 2.6.: Diagram of a strict digraph $G = (V(G), A(G), \vartheta_G)$ with (black) vertices $V(G) = \{v_1, v_2, v_3, v_4, v_5\}$ and (grey) arcs $A(G) = \{a_{1,2}, a_{1,3}, a_{2,3}, a_{3,5}, a_{4,5}\}$. The incidence function ϑ_G is implicitly defined through the choice of naming convention, i.e. for all $v_i, v_j \in V(G)$ and $a_{i,j} \in A(G)$, we have $\vartheta_G(a_{i,j}) = (v_i, v_j)$.

2.2.2. Representing Migration Pathways Using a Directed Graph

As we loosened the Markovian assumption held by Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear), we need a way of representing the pathways along which CO₂ migrates, once the conditions for inter-reservoir migration are met. For this purpose, we consider the stratigraphic system as a directed graph G , also known as a digraph. Using the definition of Bondy and Murty (1982, ch. 10.1), G is an ordered triplet $(V(G), A(G), \vartheta_G)$, consisting of a non-empty set of vertices $V(G)$, a set of arcs $A(G)$ that is disjoint from the set of vertices, i.e. $A(G) \cap V(G) = \emptyset$, and an incidence function $\vartheta_G : A(G) \rightarrow V(G) \times V(G)$ that to each arc in $A(G)$ associates an ordered pair of vertices in $V(G)$. Specifically, if $a \in A(G)$ is an arc and $v_1, v_2 \in V(G)$ are vertices, a is said to join v_1 and v_2 if $\vartheta_G(a) = (v_1, v_2)$ or $\vartheta_G(a) = (v_2, v_1)$, in which case a is incident to both v_1 and v_2 , whom themselves are said to be adjacent. It is important to note that ϑ_G maps to ordered pairs, so (v_1, v_2) and (v_2, v_1) are not equal in this respect. If $\vartheta_G(a) = (v_1, v_2)$, we say that v_1 and v_2 are the tail- and head of a , respectively. Following the convention set by Bondy and Murty (1982, ch. 10.1), we use $\nu(G)$ and $\varepsilon(G)$ to denote the cardinality of $V(G)$ and $A(G)$, respectively.

As we are interested in modelling the migration pathways between reservoirs, we need only consider digraphs whose arcs join separate vertices and where there is at most 1 arc joining the same 2 vertices (in the same order). Bondy and Murty (1982, ch. 10.1) refer to such digraphs as strict. Specifically, G is said to be strict, if for all vertices $v \in V(G)$ and any 2 arcs $a_1, a_2 \in A(G)$,

$$\vartheta_G(a_1) \neq (v, v) \quad \text{and} \quad a_1 \neq a_2 \implies \vartheta_G(a_1) \neq \vartheta_G(a_2), \quad (2.7)$$

thus making ϑ_G injective (Houston 2009). As ϑ_G is injective for strict digraphs, we can define ϑ_G implicitly using a suitable naming convention on the vertices and arcs of G (Bondy and Murty 1982, ch. 1.2). Specifically, we index both vertices $v_i, v_j \in V(G)$ and arcs $a_{i,j} \in A(G)$ using natural numbers $i, j \in \mathbb{N}$, such that ϑ_G is implicitly given by the relation $\vartheta_G(a_{i,j}) = (v_i, v_j)$. When representing G in terms of a diagram, as shown in Figure 2.6, arrows are used to denote the orientation of each arc, in addition to the above naming convention.

As CO₂ has positive buoyancy within the stratigraphic system, it will always migrate towards shallower reservoirs once the conditions in Section 2.1.2 are met. Representing the stratigraphic system in terms of G , where vertices take on the role of reservoirs, therefore

results in arcs whose tail will always have a lower index compared to its head, i.e. $i < j$ for all $a_{i,j} \in A(G)$. This is known as a topological sorting of G , which is indicative of G being an acyclic digraph, as its not possible to sort cyclic graphs in topological order (Cormen et al. 2009, ch. 22.4).

To understand the concept of cycles within G , we first have to be acquainted with the notion of a directed walk within the context of graph theory. A directed (v_0, v_n) -walk is defined as a finite non-empty sequence $\{v_0, a_1, v_1, \dots, a_n, v_n\}$ from v_0 to v_n of alternating vertices and arcs in G , such that for all $i = 1, 2, \dots, n$, the arc a_i has tail v_{i-1} and head v_i (Bondy and Murty 1982, ch. 10.1). For a (v_0, v_n) -walk, we refer to v_0 and v_n as the walk's origin and terminus, respectively, whilst the other vertices appearing in the sequence are known as internal vertices (Bondy and Murty 1982, ch. 1.6). Moreover, if the arcs appearing in the walk are distinct, i.e. they appear only once, the walk is known as a trail, and if in addition the vertices are distinct, the walk is called a path. A walk that has positive length, i.e. contains more than a single vertex, and whose origin and terminus are equal, is said to be closed (Bondy and Murty 1982, ch. 1.7). A cycle is then used to describe a closed trail whose origin and internal vertices are distinct. In simple terms, this means that we can start at a particular vertex, and by following arcs to other vertices, visiting each vertex only once, return to the vertex we started at. Figure 2.6 has no such cycles, as no matter which vertex we choose as our origin, we are unable to traverse G and somehow terminate our trail at the origin. A digraph containing no cycles is known as a directed acyclic graph (DAG) (Bondy and Murty 1982, ch. 2.1).

Conservation of mass further dictates that CO_2 must have a place to go once a reservoir has been compromised. In practice, this means that there must exist a migration pathway from every reservoir, ultimately leading to the topmost reservoir. An equivalent statement on G is that there exists at least a single path to the topmost vertex $v_k \in V(G)$ from each of the digraph's other vertices $v_1, v_2, \dots, v_{k-1} \in V(G)$. In terms of graph theory, if $u, v \in V(G)$ are vertices and there exists a path having u as its origin and v as its terminus, u and v is said to be connected (Bondy and Murty 1982, ch. 1.6). Moreover, in cases where it is not possible to partition $V(G)$ into non-empty subsets $V_1(G), V_2(G), \dots, V_n(G)$, such that the vertices in any one partition are not connected to the vertices in any of the other partitions, G is said to be connected (Bondy and Murty 1982, ch. 1.6). As for our case, the condition in which $v_1, v_2, \dots, v_{k-1}$ must all be connected with v_k , implies that G is connected.

In practise, when encoding a stratigraphic system it can be difficult to determine where the migration pathways emerging from the i th reservoir are supposed to lie prior to T_i^{mig} . The reason for this is lack of information regarding both composition and number of faults within the strata. When Furre et al. (2019) and Williams and Chadwick (2021) described the existence of chimney-like structures at the Sleipner field, they did so by comparing recent seismic surveys with a baseline survey conducted prior to the start of the the injection process. Without the additional information of recent surveys, the researchers would not have been able to determine whether any chimneys were present in the stratigraphic system, as they would not have been able to determine the destination(s) of the migrating CO_2 . If for some system at time t , CO_2 has not yet migrated from a particular reservoir $i < k$, it can in theory migrate to any of the shallower reservoirs $i + 1, i + 2, \dots, k$ within the stratigraphic system, once $t \geq T_i^{\text{mig}}$. Our assumptions on positive buoyancy and conservation of mass only dictate that at least 1 of these pathways must be available for the migrating CO_2 .

In Figures 2.7b and 2.8b, showing sections of G for 2 stratigraphic systems exhibiting

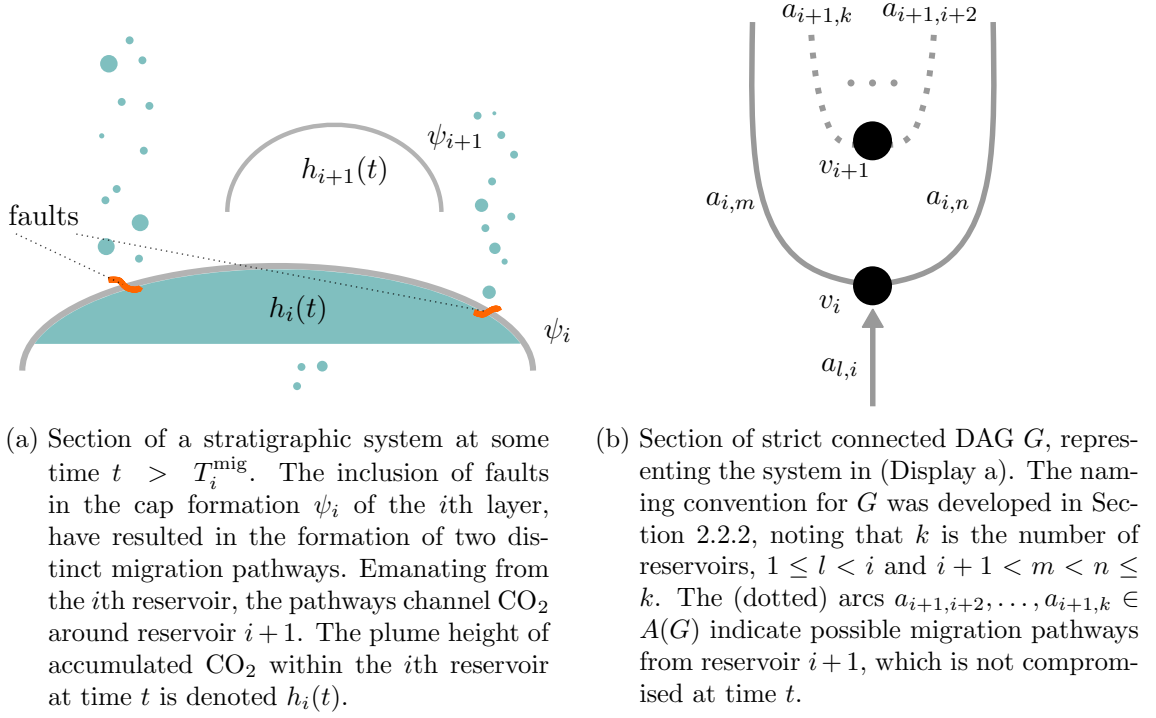


Figure 2.7.: Sketch showing how multiple migration pathways emanating from the same reservoir (Display a) can be represented in terms of a strict connected DAG (Display b).

non-Markovian features, we have chosen to represent this uncertainty using dotted arcs. Specifically, as reservoir $i + 1$ is compromised in neither Figures 2.7a nor 2.8a at the time t of depiction, i.e. $t < T_{i+1}^{\text{mig}}$, each figure depicts a total of $k - (i + 1)$ dotted arcs representing the possible pathways emanating from reservoir $i + 1$. The reservoir's actual migration pathways would then be a subset of the pathways represented by the dotted arcs. Figure 2.7 illustrates how fault-lines and fractures within the ψ_i 's of a stratigraphic system can be represented in terms of G , while Figure 2.8 shows how chimney's can be represented using the same underlying theory. By combining features from both Figures 2.7 and 2.8, we can express chimney-like structures that are a result of fractures and fault lines within the ψ_i 's of our stratigraphic system. This is an important result, as researchers have indicated that such features are likely present within most realistic stratigraphic systems, as mentioned in Section 2.1.2.

In particular, we can use Figures 2.7 and 2.8 in combination with the theory of this section, to represent the stratigraphic system in Figure 2.3 in terms of G , as shown in Figure 2.9. Although the 4th reservoir has not yet been compromised, the fact that the reservoir is 2nd from the top means that there can only be a single pathway for migrating CO₂ to follow, once T_4^{mig} is reached and the reservoir becomes compromised. We note, for the sake of clarity, that $A(G)$ does not represent pathways along which CO₂ migrates at the current time t , but rather a static set of migration pathways along which CO₂ migrates once the conditions in Section 2.1.2 are met. It follows that $a_{4,5} \in A(G)$, even though $t < T_4^{\text{mig}}$. As a side-note, we also mention that $V(G)$, $A(G)$ and ϑ_G are identical across both Figures 2.6 and 2.9b, implying that the illustrated digraphs are equal to one another. This illustrates an important concept within graph theory, in that graphs can appear different from each other when drawn as a diagram, e.g. horizontally vs. vertically, even though they are mathematically equivalent (Bondy and Murty 1982, ch. 1.2).

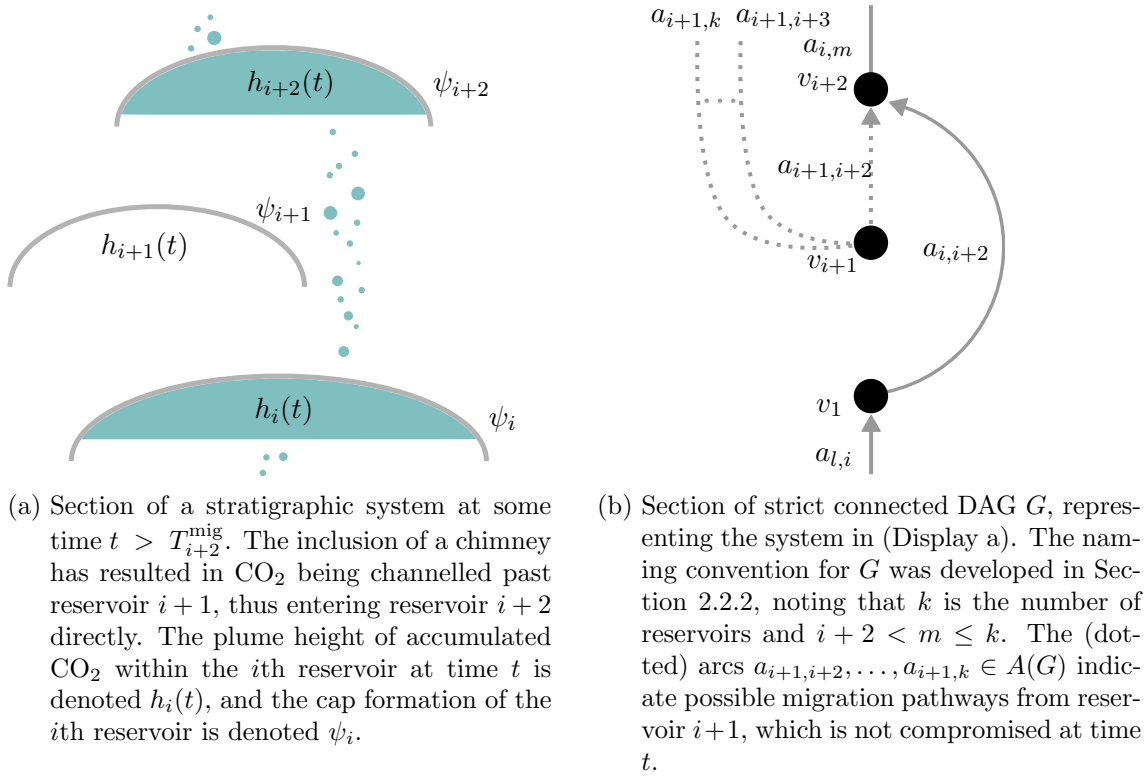


Figure 2.8.: Sketch showing how a chimney-like structure within the strata (Display a) can be represented in terms of a strict connected DAG (Display b).

When representing G on a computer, we use its adjacency matrix $\mathbf{A}(G)$ instead of the usual ordered triplet $(V(G), A(G), \vartheta_G)$. This is done in part to save on memory but also to increase the speed and efficiency of any computation involving G . Bondy and Murty (1982, ch. 1.3) defines the adjacency matrix $\mathbf{A}(G) = (a_{ij})$ of a digraph using its elements. Specifically, they define a_{ij} as the number of arcs in G having $v_i \in V(G)$ as its tail and $v_j \in V(G)$ as its head. Using G in Figure 2.9 as an example, we find that

$$\mathbf{A}(G) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Note in particular that $\mathbf{A}(G)$ is $k \times k$ strictly upper-triangular with binary entries, and that the first $k - 1$ rows all contain at least a single 1 element. This is true in general, as our assumptions on conservation of mass and the positive buoyancy of CO_2 imply that G is both strict, connected and acyclic. We have included a full derivation of this in Appendix A. It follows that no information is stored in the lower triangular part of $\mathbf{A}(G)$, and so an efficient computer implementation can save additional memory by only storing the strictly upper-triangular part of the matrix.

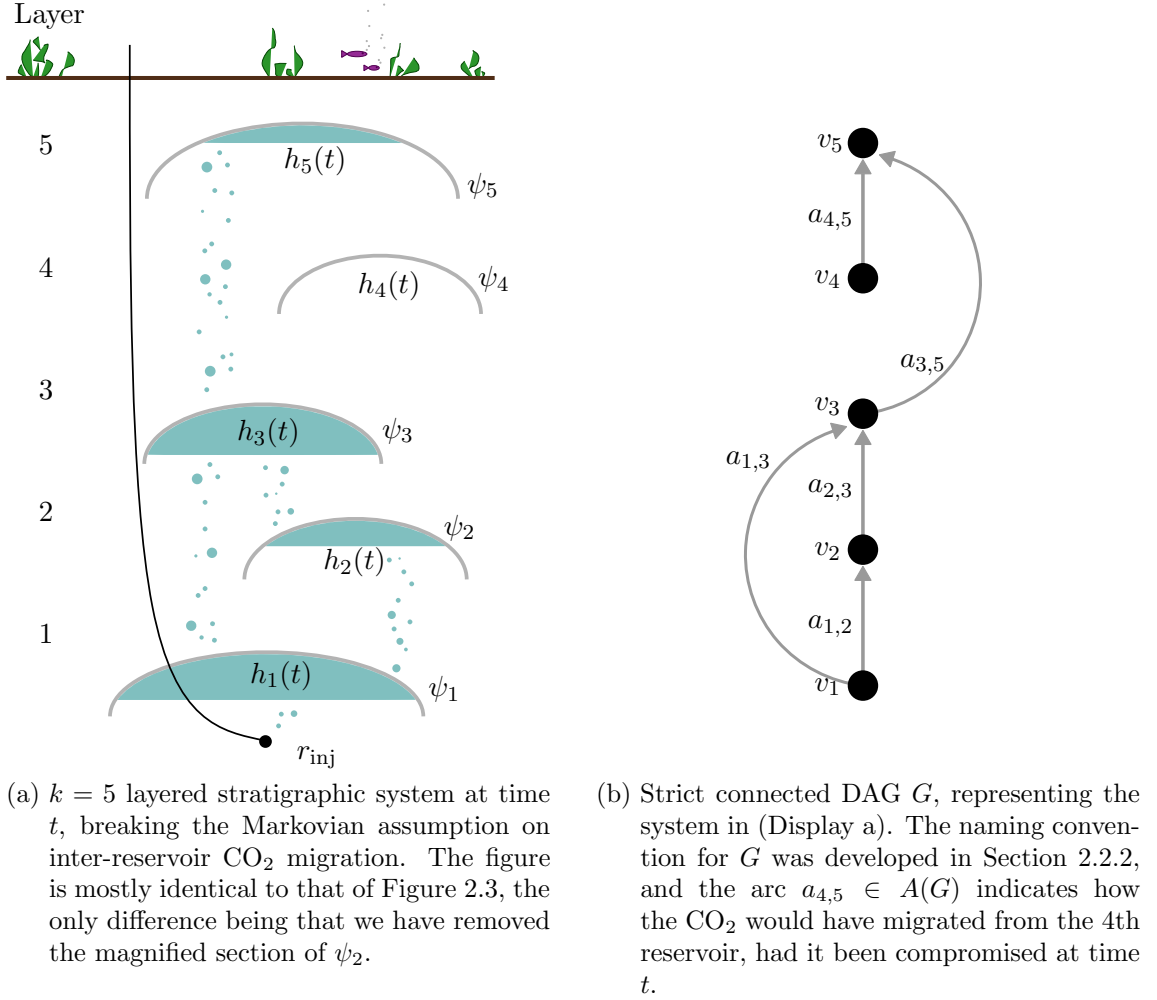


Figure 2.9.: Sketch showing how a non-Markovian stratigraphic system (Display a) can be represented in terms of a strict connected DAG (Display b).

2.3. Accounting for Heterogenous CO_2 Migration

2.3.1. Modelling the Rate of Migration Along Individual Pathways

When we introduced the non-Markovian system illustrated in Figure 2.9a, we were deliberately somewhat vague in our description of how our framework handles multiple migration pathways emanating from the same reservoir. For the model to be physically consistent, it obviously needs to enforce conservation of mass. In Section 2.2.2, this was found to imply that all but the topmost reservoir must have a migration pathway that ultimately leads to the topmost reservoir. For a reservoir $i < k$ having a single pathway only, this is sufficient to ensure conservation of mass. Injected CO_2 accumulates within the i th reservoir until T_i^{mig} , at which time, (additional) CO_2 migrates away from the compromised reservoir, following the (single) pathway indicated in G . Reservoirs with additional migration pathways, e.g. the 1st reservoir in Figure 2.9a, are still problematic however, as the CO_2 now

migrates along multiple pathways once the reservoirs get compromised. If the mass of CO₂ is to stay conserved, we must ensure that the total amount of CO₂ migrating away from any such reservoir equals the amount that is entering reservoirs of higher topological order.

Even Distribution

We could e.g. assume that the migrating CO₂ distributes itself evenly amongst the available migration pathways of each reservoir. As for the 1st reservoir in Figure 2.9a, this implies that $q_2^{\text{ent}}(t) = q_3^{\text{ent}}(t) = q_1^{\text{mig}}(t)/2$, for $T_1^{\text{mig}} \leq t < T_2^{\text{mig}}$, i.e. when the 1st reservoir is compromised but before the 2nd reservoir has been compromised. As we in Section 2.1.1 assumed carbonate precipitation due to residual trapping to be negligible, any (additional) CO₂ entering the i th reservoir once compromised, must contribute to the amount of CO₂ migrating away from the i th reservoir, resulting in $q_i^{\text{net}}(t) = 0$, for $t \geq T_i^{\text{mig}}$. For the system depicted in Figure 2.9a, it follows that $q_2^{\text{ent}}(t) = q_3^{\text{ent}}(t) = r_{\text{inj}}/2$, for $T_1^{\text{mig}} \leq t < T_2^{\text{mig}}$. Note in particular the importance of an upper limit on t in this example. At time $t = T_2^{\text{mig}}$ the 2nd reservoir gets compromised, and so (additional) CO₂ entering the 2nd reservoir must migrate to the 3rd reservoir, following the pathway indicated in Figure 2.9b. It follows that $q_3^{\text{ent}}(t) = r_{\text{inj}} \neq r_{\text{inj}}/2$, for $t \geq T_2^{\text{mig}}$.

Although this assumption may at first seem reasonable, it fails to account for the fact that some pathways may be caused by fault lines with a larger than average transverse extent, and as such, pathways that naturally has a bigger capacity for migrating CO₂. In particular, multiple studies on the migration of CO₂ at the Sleipner field, including for instance Callioli Santi et al. (2025), Furre et al. (2019) and Williams and Chadwick (2021), have concluded with there being a large chimney-like structure between the 1st and 9th reservoirs, along which (most of) the injected CO₂ migrates. If we were to impose the simple assumption of CO₂ being evenly distributed amongst the migration pathways emanating from each reservoir, we would not be able to account for such features within our framework.

Uneven Distribution

To account for the aforementioned features in our model, we instead assign a weight $q_{i,j} \in [0, 1] \subset \mathbb{R}$ to each arc $a_{i,j} \in A(G)$, representing the relative rate at which CO₂ migrates along the pathway between the i th and j th reservoirs, and let $Q(G) = \{q_{i,j}\}$ denote the set of all $q_{i,j}$. To ensure conservation of mass, we make the additional assumption of $\sum_{j:a_{i,j} \in A(G)} q_{i,j} = 1$, for $i = 1, 2, \dots, k-1$, i.e. the sum of $q_{i,j} \in Q(G)$ over the migration pathways of a single reservoir $i < k$ equals 1. Note that we do not need to make an additional assumption on the topmost reservoir, as it does not have any migration pathways emanating from it. Using the 1st reservoir in Figure 2.9a as an example yet again, we get that $q_2^{\text{ent}}(t) = q_{1,2} \cdot r_{\text{inj}}$ and $q_3^{\text{ent}}(t) = q_{1,3} \cdot r_{\text{inj}}$, for $T_1^{\text{mig}} \leq t < T_2^{\text{mig}}$, as $q_1^{\text{net}}(t) = 0$ implies $q_1^{\text{mig}}(t) = r_{\text{inj}}$. Moreover, the additional assumption of $\sum_{j:a_{i,j} \in A(G)} q_{i,j} = 1$ implies that $q_2^{\text{ent}}(t) + q_3^{\text{ent}}(t) = r_{\text{inj}}$, for $T_1^{\text{mig}} \leq t < T_2^{\text{mig}}$, demonstrating that the mass of CO₂ is conserved in the time between the 1st and 2nd reservoirs being compromised. If we then wanted the stratigraphic system in Figure 2.9a to resemble that of the Sleipner field, we could e.g. let $q_{1,3} = 0.9$ and $q_{1,2} = 1 - q_{1,3} = 0.1$, as this would result in a dominating pathway between 2 non-subsequent reservoirs, that being the 1st and 3rd reservoirs in this case.

When representing $Q(G)$ on a computer, we use its associated relative migration rate matrix $\mathbf{Q}(G) = (q_{ij})$, defined element-wise as follows,

$$q_{ij} = \begin{cases} q_{i,j}, & \text{if } q_{i,j} \in Q(G). \\ 0, & \text{else.} \end{cases}$$

Or equivalently, using $\mathbf{A}(G)$,

$$q_{ij} = \begin{cases} q_{i,j}, & \text{if } a_{ij} = 1. \\ 0, & \text{else.} \end{cases}$$

It follows from the structure of $\mathbf{A}(G)$, that $\mathbf{Q}(G)$ is $k \times k$ strictly upper-triangular, and so we only need to store the strictly upper-triangular part of $\mathbf{Q}(G)$ on the computer. For reasons that will become evident later, we sometimes store a saturated relative migration rate matrix $\mathbf{Q}^{\text{sat}} = (q_{ij}^{\text{sat}})$ instead of $\mathbf{Q}(G)$. We define $\mathbf{Q}^{\text{sat}}$ similarly to $\mathbf{Q}(G)$, but on a saturated system where each reservoir has a migration pathway to every other reservoir of higher topological order, i.e. for all $i = 1, 2, \dots, k-1$ and $j = 2, 3, \dots, k$, such that $i < j$, we have that $a_{i,j} \in A(G^{\text{sat}})$, where G^{sat} denotes the strict connected DAG representing the saturated system. When representing $Q(G)$ in this way, we always have that $q_{ij} = q_{ij}^{\text{sat}}$, for all $i, j : q_{i,j} \in Q(G)$. This enables the computation of $\mathbf{Q}(G)$ through a simple Hadamard (element-wise) product,

$$\mathbf{Q}(G) = \mathbf{Q}^{\text{sat}} \odot \mathbf{A}(G),$$

which is easily calculated by a computer. Note that by storing some superfluous weights in $\mathbf{Q}^{\text{sat}}$ not relevant to the stratigraphic system of interest, we have essentially decoupled the weights in $\mathbf{Q}(G)$ from G . This allows us to treat the weights separately from any particular stratigraphic system, an important result with implications for the sections that follow.

2.3.2. Connection to Network Theory

By adding (scalar) weights to each of the arcs in G , we have technically constructed a network in terms of graph theory. In particular, using the definition of Bondy and Murty (1982, ch. 11.1), a network is a digraph G equipped with a non-negative integer valued function $c : A(G) \rightarrow \mathbb{N} \cup \{0\}$ that assigns a capacity $c(a)$ to each arc $a \in A(G)$ and where $V(G)$ contains 2 disjoint non-empty subsets $V_{\text{source}}(G), V_{\text{sink}}(G) \subset V(G)$ of sources and sinks. These represent the production centres and markets of the network, and a vertex $v \in V(G)$ need not belong to either $V_{\text{source}}(G)$ or $V_{\text{sink}}(G)$, in which case it is known as intermediate vertex (Bondy and Murty 1982, ch. 11.1). As $q_{i,j} \in Q(G)$ is assumed to be real valued and in the range between 0 and 1, our weights are not necessarily integer-valued. Experts typically relax this condition anyway, as seen in Henning and van Vuuren (2022, ch. 4.2) who required c to be a positive real-valued function and Estrada (2011, ch. 1.3) who removed the condition of positivity all together, only requiring c to be real-valued. Similar to ϑ_G , it is possible to define c implicitly using a suitable naming convention on each capacity (weight).

Being especially pedantic, it would probably be more correct to say that $q_{i,j} \in Q(G)$ induces a family of networks on G , following a relaxed version of the definition used by Bondy and Murty (1982, ch. 11.1), where we allow the weights to be real as opposed to integer-valued. This follows from the fact that both $V_{\text{source}}(G)$ and $V_{\text{sink}}(G)$ are static sets, whose elements do not change with time. In our case, we have a stratigraphic system that evolves over time. The migration pathways of the system as indicated by G stay

the same, but the role of the reservoirs, hence also that of the vertices, change as the system involves. At time $t < T_i^{\text{mig}}$, the i th reservoir has not yet been compromised and so $v_i \in V_{\text{sink}}(G)$, as CO_2 accumulates within the i th reservoir. Once the reservoir gets compromised, (additional) CO_2 migrates away from it, so $v_i \notin V_{\text{sink}}(G)$ for $t \geq T_i^{\text{mig}}$. Over time, the number of elements in $V_{\text{sink}}(G)$ decreases, thus inducing new networks on the same underlying G .

Note that the connection between our framework and networks is illustrated mainly for the sake of being complete. Network theory is an extensive component of graph theory, studying how various quantities flow between sinks and sources (Bondy and Murty 1982, ch. 11; Estrada 2011; Henning and van Vuuren 2022, ch. 7 – 8). What we neglected to mention in the preceding paragraphs is that it becomes difficult to consider stratigraphic systems, where $q_i^{\text{ext}} > 0$, for some $i \in \{2, 3, \dots, k\}$, as networks, without also making changes to $V(G)$. When we introduced digraphs as a means of modelling migration pathways in Section 2.2.2, we let each vertex in $V(G)$ correspond to a single reservoir. Note that by injecting CO_2 (directly) in a reservoir $i > 1$ of higher topological order, the corresponding vertex would be a source within the network. As $v_i \in V_{\text{source}}(G)$, no arcs can have v_i as their head, and so we can no longer account for CO_2 migrating into the i th reservoir. We could mediate the problem by e.g. assigning a vertex to all of the injection sites, in addition to the reservoirs of the system, letting the vertex of each injection site be a source joined to the vertex of the reservoir in which it resides. The vertices representing reservoirs could then be modelled exclusively as sinks and intermediate vertices, depending on whether the reservoir they correspond to has been compromised or not. However, by doing so, we would undermine the simplicity of $V(G)$ as its currently defined.

Even if we were to limit our framework to the case where $q_1^{\text{ext}} > 0$ and $q_i^{\text{ext}} = 0$, for $i = 2, 3, \dots, k$, as seen in Figure 2.9a, it would be problematic to describe the system as a network prior to T_1^{mig} . At this time, the bottommost reservoir accumulates CO_2 so $v_1 \in V_{\text{sink}}(G)$, implying that $v_1 \notin V_{\text{source}}(G)$ as the sets are disjoint. This is not problematic in and of itself, as CO_2 does not migrate from the bottommost reservoir at this time. However, as $v_1 \notin V_{\text{source}}(G)$, $V_{\text{source}}(G)$ becomes an empty set, which contradicts the definition of $V_{\text{source}}(G)$ in Section 2.3.1. We would therefore need to implement a similar solution to that of the above paragraph, if we are to express the stratigraphic system as a network at this time, which again compromises the simplicity of $V(G)$ as its defined in Section 2.2.2. For the purposes of this thesis, we are therefore not going to entertain the idea of a family of networks on G any longer, although one could easily construct one by making the necessary changes to $V(G)$, as previously described. Instead, we may think of G along with $Q(G)$ as a weighted graph, which only requires the existence of a weight function similar to that of c in the definition of a network (Estrada 2011, ch. 1.3; Henning and van Vuuren 2022, ch. 4.2).

2.3.3. Effective Capillary Threshold Pressure

As part of a conference on greenhouse gas technology, Springer and Lindgren (2006) conducted laboratory tests on core samples taken from production well 15/9-A-11 at the Sleipner field. The tests were conducted in the pursuit of an estimate of the CTP of the Nordland shales, parts of which form the cap formations covering the reservoirs where CO_2 is injected. At a temperature of 37°C , believed to be the approximate in-situ temperature of CO_2 at the site (Cavanagh and Haszeldine 2014), the geologists reported a CTP of 1.7 MPa. Harrington et al. (2009) later substantiated this estimate by conducting additional experiments, concluding that the CTP of the Nordland shales is likely in the range

1.6 – 1.9 MPa. Applying IP theory, as outlined in Section 2.1.1, to the estimate reported by Springer and Lindgren (2006), we should expect the cap formations at the Sleipner field to contain CO₂ plumes with a height of several hundred metres. In particular, if we assume $\Delta\rho = 400 \text{ kg/m}^3$ as reported by Springer and Lindgren (2006), Eq. (2.1) implies $h_i(T_i^{\text{mig}}) \approx 433 \text{ m}$. Taking a single look at Figure 2.4 however, we notice that this is not the case in practice, as the CO₂ plumes rarely surpass 15 m in height prior to migration.

Cavanagh and Haszeldine (2014) noticed this as well, and through numerical simulations, estimated that a CO₂ column with a height of 10 – 15 m, exerting a pressure of approximately 50 kPa on the covering cap formations, are enough for inter-reservoir migration to commence at the Sleipner field. This is a 34-fold reduction in the CTP measured by Springer and Lindgren (2006), which Cavanagh and Haszeldine (2014) argued could only be the case if the cap formations contained existing fractures, as mentioned in Section 2.1.2. In particular, the width of the fractures decrease the effective CTP (eCTP) of the cap formations, enabling inter-reservoir migration of CO₂ at lower pressures (Cavanagh and Haszeldine 2014). As the composition of the Nordland shales, and cap formations in general, have local variations affecting the formation of fractures (Springer and Lindgren 2006; Wang et al. 2021), it seems only natural that some migration pathways have a lower eCTP than what is expected, given the CTP of the cap formations.

To account for this within our framework, we assign another set of weights to the migration pathways of our system. Specifically, we assign a weight $e_{i,j} \in [0, 1] \subset \mathbb{R}$ to each arc $a_{i,j} \in A(G)$ representing the relative eCTP of the migration pathway between the i th and j th reservoirs, and let $E(G) = \{e_{i,j}\}$ denote the set of all $e_{i,j}$. Determining when CO₂ starts to migrate along a particular pathway, is then achieved by generalising the approach of Section 2.1.2. In particular, we let CO₂ migrate along the pathway represented by $a_{i,j} \in A(G)$, only when $P_i(t)$ has surpassed the eCTP of the pathway, i.e. $P_i(t) > e_{i,j} \cdot C_i$. Similar to $\mathbf{Q}(G)$, we define a relative eCTP matrix $\mathbf{E}(G) = (e_{ij})$, by requiring each element of $\mathbf{E}(G)$ to satisfy,

$$e_{ij} = \begin{cases} e_{i,j}, & \text{if } a_{i,j} \in E(G). \\ 0, & \text{else.} \end{cases}$$

Or equivalently, using $\mathbf{A}(G)$,

$$e_{ij} = \begin{cases} e_{i,j}, & \text{if } a_{ij} = 1. \\ 0, & \text{else.} \end{cases}$$

As with $\mathbf{Q}(G)$, $\mathbf{E}(G)$ is $k \times k$ strictly upper-triangular, and so we only need to store the strictly upper-triangular part of $\mathbf{E}(G)$ on the computer. Defining $\mathbf{E}^{\text{sat}} = (e_{ij}^{\text{sat}})$ in the same way we defined $\mathbf{Q}^{\text{sat}}$ in Section 2.3.1, i.e. as the weights corresponding to a saturated system represented by G^{sat} , we can store $\mathbf{E}^{\text{sat}}$ on the computer and calculate $\mathbf{E}(G)$ only when needed, using the relation

$$\mathbf{E}(G) = \mathbf{E}^{\text{sat}} \odot \mathbf{A}(G).$$

Similar to when $\mathbf{Q}^{\text{sat}}$ is used to store the weights of $\mathbf{Q}(G)$ on the computer, it is here implicitly assumed that $e_{ij} = e_{ij}^{\text{sat}}$, for all $i, j : e_{i,j} \in E(G)$, when $\mathbf{E}^{\text{sat}}$ is used to store the weights of $\mathbf{E}(G)$.

Note that by allowing inter-reservoir migration of CO₂ at lower pressures, governed by the eCTP of each pathway as opposed to the CTP of each cap formation, Eq. (2.2) no longer provides a defining relation for T_i^{mig} , in the case where $i \neq k$. This follows from the fact that $e_{i,j} \leq 1$ for all $e_{i,j} \in E(G)$, which may allow CO₂ to migrate from reservoir $i \neq k$

prior to $P_i(t) > C_i$. As we still need a way to account for the time when CO₂ first migrates from a particular reservoir, we define the pathway specific migration time t_{ij}^{path} restricted to $A(G)$, as the time t when CO₂ first migrates from the i th to the j th reservoirs, along the pathway indicated by $a_{i,j} \in A(G)$. This can be expressed in terms of a minimum over t ,

$$t_{ij}^{\text{path}} = \min\{t > 0 : P_i(t) > e_{i,j}C_i\} = \operatorname{argmin}_{t \in (0, \infty)} \left\{ h_i(t) > \frac{e_{i,j}C_i}{\Delta\rho g} \right\}, \quad i, j : a_{i,j} \in A(G), \quad (2.8)$$

where the 2nd equality follows from a simple substitution with Eq. (2.1). As CO₂ cannot migrate (directly) from the i th to the j th reservoir if $a_{i,j} \notin A(G)$, we can extend t_{ij}^{path} to all $i, j = 1, 2, \dots, k$, by defining $t_{ij}^{\text{path}} = \infty$ for all $a_{i,j} \notin A(G)$. Recalling from Section 2.1.2, that T_i^{mig} is the time when CO₂ first migrates through ψ_i , we may now construct a new defining relation for T_i^{mig} using Eq. (2.8). In particular, for $i \neq k$, we have that

$$T_i^{\text{mig}} = \min_{j=1,2,\dots,k} t_{ij}^{\text{path}} = \operatorname{argmin}_{t \in (0, \infty)} \left\{ h_i(t) > \frac{(\min_{j=1,2,\dots,k} e_{i,j}) C_i}{\Delta\rho g} \right\}, \quad i = 1, 2, \dots, k-1. \quad (2.9)$$

Similar to the equation above, most of our results have thus far been stated exclusively in terms of scalar quantities, often having at least a single index that needed to be (re)specified whenever a result was given. To improve the readability of the thesis and avoid having to (re)specify indices unnecessarily, most results are from this point on stated in terms multidimensional vectors and vector-valued functions. We denote these quantities using their elements, by typesetting them in boldface and removing their index within the subscript, as outlined in International Organization for Standardization (2009, ch. 17). In particular, instead of having to specify a condition such as $P_i(t) > C_i$, for $i = 1, 2, \dots, k$, we simply write $\mathbf{P}(t) > \mathbf{C}$. In cases where the multidimensional vector is a matrix or a higher order tensor, standard conventions dictate that the underlying element is to be capitalised when writing it as a multidimensional vector (International Organization for Standardization 2009, ch. 17). Although we have not mentioned it explicitly until now, the matrices defined in previous sections, e.g. $\mathbf{A}(G)$ as defined in Section 2.2.2, have all been named using this particular convention. In the sections that follow, we may therefore denote e.g. the (collection of) pathway specific migration times using $\mathbf{T}^{\text{path}}$. Unless specifically stated, all quantities in the thesis whether scalar or multidimensional, adhere to the aforementioned convention.

2.4. Event-Driven Simulator for CO₂ Plume Development

If we are to avoid a situation in which the topmost reservoir of a stratigraphic system becomes compromised, thereby allowing CO₂ to (re)enter the atmosphere, it is crucial that we have a thorough understanding of the migration process at hand. Specifically, we have to be able to monitor the development of the CO₂ plume prior to T_k^{mig} , so that we can decommission the CCS project at a suitable time $t < T_k^{\text{mig}}$, thus preventing the CO₂ from (re)entering the atmosphere. In previous sections, namely Sections 2.1.2 and 2.2.1, we introduced quantities to describe various elements of the CO₂ plume at a particular time t , including the pressure head $\mathbf{P}(t)$, its height $\mathbf{h}(t)$ and the volume of accumulated CO₂ $\mathbf{V}(t)$. We also introduced the means by which we can convert between these quantities, i.e. Eqs. (2.1) and (2.6). In this section, we expand upon these results, developing a differentiable simulator $\mathcal{S} : t \mapsto \mathbf{h}(t)$ that to each $t < T_k^{\text{mig}}$ associates a

corresponding $\mathbf{h}(t)$. This enables us to describe the development of the CO₂ plume with greater accuracy, as we are no longer confined to using only the times represented by $\mathbf{T}^{\text{ent}}$, $\mathbf{T}^{\text{mig}}$ and $\mathbf{T}^{\text{path}}$. In particular, if we were to carry out some sort of data acquisition procedure that resulted in an estimate of $\mathbf{h}(t)$, see for instance Figure 2.4, we could use the simulator to assess how the plume heights are likely to develop in the time following the acquisition procedure. Determining when to decommission the CCS project can then be made on the basis of the likelihood of CO₂ migrating away from the topmost reservoir.

2.4.1. Constructing the Simulator

A core principle of IP theory, as described in Section 2.1.1, is the fact that the stratigraphic system changes intermittently in time, as opposed to continuously, when $\mathbf{P}(t)$ increases. For the purposes of this thesis, we refer to such changes as events, and recall from the previous section, that an event is triggered whenever the pressure build-up within a reservoir surpasses the eCTP of an associated migration pathway. Note that the number of events in total prior to T_k^{mig} , denoted by m , can be at most equal to $\varepsilon(G)$, i.e. the number of arcs in the digraph representing the stratigraphic system. This corresponds to the specific case where CO₂ is allowed to migrate along all $\varepsilon(G)$ pathways prior to T_k^{mig} , and where the parameters of the system are such that each pathway becomes available for the migrating CO₂ sequentially in time, meaning no 2 pathways become available simultaneously. When referring to a pathway along which CO₂ migrates at a particular time t , we say that the pathway is active at time t . Similarly, if CO₂ is not permitted to migrate along the pathway at the current time, the pathway is said to be inactive. A differentiable simulator that to each $t < T_k^{\text{mig}}$ associates a corresponding $\mathbf{h}(t)$, as outlined here, is then closely related to the state of the migration pathways prior to T_k^{mig} , as CO₂ cannot enter reservoirs through inactive pathways.

Determining the Net-Rate of Accumulation

As we continue to inject CO₂ into the system at constant rate $\mathbf{q}^{\text{ext}}$, the net-rate of accumulation in each reservoir, described by $\mathbf{q}^{\text{net}}(t)$, changes concurrently with the activation of each pathway. If we could determine the value of $\mathbf{q}^{\text{net}}(t)$ for $t < T_k^{\text{mig}}$, we could through Eqs. (2.4) and (2.6) obtain a corresponding value of $\mathbf{h}(t)$. Hence, an expression for the simulator could be obtained, if in addition $\mathbf{q}^{\text{net}}(t)$ is differentiable for $t < T_k^{\text{mig}}$. To that extent, we first define $\mathbf{A}^{\text{act}}(G, t)$ as the adjacency matrix of active migration pathways only, meaning that

$$a_{ij}^{\text{act}} = \begin{cases} a_{ij}, & \text{if } P_i(t) > e_{ij}C_i. \\ 0, & \text{if } P_i(t) \leq e_{ij}C_i. \end{cases} \quad i, j = 1, 2, \dots, k. \quad (2.10)$$

We note for the sake of clarity, that the inequalities in the above expression correspond to the generalised IP condition described in Section 2.3.3. The relative rate at which CO₂ migrates along active pathways at time t is then given by the relative flow matrix $\mathbf{F}(G, t)$, whose expression follows from $\mathbf{A}^{\text{act}}(G, t)$ and the relative weights in Section 2.3.1. In particular, we have that $\mathbf{F}(G, t) = \mathbf{A}^{\text{act}}(G, t) \odot \mathbf{Q}(G)$.

To compute $\mathbf{q}^{\text{net}}(t)$ at time $t < T_k^{\text{mig}}$, we first recall from its definition in Section 2.1.2, that $\mathbf{q}^{\text{net}}(t) = \mathbf{q}^{\text{ent}}(t) - \mathbf{q}^{\text{mig}}(t)$, where $\mathbf{q}^{\text{ent}}(t)$ and $\mathbf{q}^{\text{mig}}(t)$ are the rates at which CO₂ enters and migrates away from the reservoirs at time t , respectively. Note that $\mathbf{q}^{\text{ent}}(t)$ can

be expressed as the sum of $\mathbf{q}^{\text{ext}}$ and the fraction of $\mathbf{q}^{\text{ent}}(t)$ entering the reservoirs (from lower reservoirs) through active migration pathways at time t . Specifically, we have that

$$\mathbf{q}^{\text{ent}}(t) = \mathbf{q}^{\text{ext}} + \mathbf{F}^\top(G, t) \mathbf{q}^{\text{ent}}(t) \implies \mathbf{q}^{\text{ent}}(t) = (\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1} \mathbf{q}^{\text{ext}}, \quad t < T_k^{\text{mig}}, \quad (2.11)$$

where $\mathbf{I}_k$ denotes the identity on $\mathbb{R}^{k \times k}$, i.e. $\mathbf{I}_k = (\delta_{ij})$ using the Kronecker delta δ_{ij} , and where $\mathbf{F}^\top(G, t)$ denotes the transpose of $\mathbf{F}(G, t)$.

Note also that the matrix $\mathbf{I}_k - \mathbf{F}^\top(G, t)$ in the above equation always admits an inverse, as it is non-singular. Specifically, as $\mathbf{F}(G, t)$ is a Hadamard product of 2 strictly upper-triangular matrices, it is itself strictly upper-triangular. $\mathbf{I}_k - \mathbf{F}^\top(G, t)$ is therefore lower triangular with 1s on the diagonal, since we consider the transpose of $\mathbf{F}(G, t)$, resulting in a determinant that equals 1. In the literature, e.g. in Rapp (2017), such matrices are referred to as unit lower triangular. It follows, as shown in Appendix A, that the inverse of $\mathbf{I}_k - \mathbf{F}^\top(G, t)$ is a Neumann series (Suzuki 1976) truncated to $k - 1$,

$$(\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1} = \sum_{\ell=0}^{k-1} (\mathbf{F}^\top(G, t))^\ell = \mathbf{I}_k + \mathbf{F}^\top(G, t) + (\mathbf{F}^\top(G, t))^2 + \dots + (\mathbf{F}^\top(G, t))^{k-1}.$$

When solving for $\mathbf{q}^{\text{ent}}(t)$ in Eq. (2.11) on the computer, we do not compute the above inverse directly however, choosing instead to solve for $\mathbf{q}^{\text{ent}}(t)$ as a linear system of equations. As $\mathbf{I}_k - \mathbf{F}^\top(G, t)$ is lower triangular, this can be done using a single forward substitution, see Algorithm 2.1, thus reducing the number of elementary operations that need to be carried out on the computer. This reduces the potential of accumulating numerical errors and speeds up the routine significantly.

Algorithm 2.1 Compute $\mathbf{q}^{\text{ent}}(t)$ using forward substitution

Require: $\mathbf{q}^{\text{ext}}$ and $\mathbf{F}(G, t)$

Ensure: $\mathbf{q}^{\text{ent}}(t) = (\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1} \mathbf{q}^{\text{ext}}$

- 1: $q_1^{\text{ent}}(t) \leftarrow q_1^{\text{ext}}$
 - 2: **for** $i = 2, 3, \dots, k$ **do**
 - 3: $q_i^{\text{ent}} \leftarrow q_i^{\text{ext}} - \sum_{\ell=1}^{i-1} f_{\ell i} \cdot q_\ell^{\text{ent}}(t)$ $\triangleright$ Note that we can index $\mathbf{F}(G, t)$ directly.
 - 4: **end for**
-

Computing $\mathbf{q}^{\text{net}}(t)$ for $t < T_k^{\text{mig}}$ is then easily accomplished, if we in addition know the value of $\mathbf{q}^{\text{mig}}(t)$. Using the assumption on conservation of mass, $\mathbf{q}^{\text{mig}}(t)$ must equate to the fraction of $\mathbf{q}^{\text{ent}}(t)$ migrating (to higher order reservoirs) through active migration pathways at time t . Specifically,

$$\mathbf{q}^{\text{mig}}(t) = (\mathbf{F}(G, t) \mathbf{1}_k) \odot \mathbf{q}^{\text{ent}}(t), \quad t < T_k^{\text{mig}},$$

where $\mathbf{1}_k$ denotes the (real) vector in k dimensions whose elements are all equal to 1. It follows that,

$$\mathbf{q}^{\text{net}}(t) = (\mathbf{1}_k - \mathbf{F}(G, t) \mathbf{1}_k) \odot \mathbf{q}^{\text{ent}}(t) = \boldsymbol{\gamma}^{\text{ret}}(G, t) \odot \mathbf{q}^{\text{ent}}(t), \quad t < T_k^{\text{mig}}, \quad (2.12)$$

where we define the retention factor $\boldsymbol{\gamma}^{\text{ret}}(G, t) \in [0, 1]^k \subset \mathbb{R}$ as the (rate) factor of CO_2 that stays trapped within the system after entering the reservoirs at time t . In particular, note that $\gamma_i^{\text{ret}}(G, t)$, for $i = 1, 2, \dots, k$, is strictly decreasing and piecewise constant with respect to t , starting off with a value of 1 when no pathways from the i th reservoir is activated,

attaining the lower bound at 0 only when all the migration pathways are activated and the i th reservoir gets fully compromised.

By substituting Eq. (2.11) in Eq. (2.12), we obtain an expression for $\mathbf{q}^{\text{net}}(t)$ as a function of $\mathbf{q}^{\text{ext}}$, whenever $t < T_k^{\text{mig}}$. Unfortunately, this expression is not differentiable everywhere for $t < T_k^{\text{mig}}$, as the condition used to determine whether a migration pathway is active or inactive takes the form of an inequality, following our generalised approach to IP theory. If we try to construct the simulator as previously outlined, by substituting Eq. (2.12) in Eq. (2.4) and subsequently inverting Eq. (2.6) to obtain an expression for $\mathbf{h}(t)$, the resulting simulator would be non-differentiable at the event times, i.e. whenever the migration pathways activate.

Smoothing Pathway Activations to Ensure Differentiability

To circumvent this problem, we define a smooth differentiable approximation of $\mathbf{A}^{\text{act}}(G, t)$, denoted by $\mathbf{S}^{\text{act}}(G, t)$, that when substituted in the defining relation for $\mathbf{F}(G, t)$, yields a smooth approximation of $\mathbf{q}^{\text{net}}(t)$. To derive an expression for the particular approximation, we start by examining the values of $\mathbf{A}^{\text{act}}(G, t)$ as $\mathbf{P}(t)$ increases. If the pathway between the i th and j th reservoirs exists within the stratigraphic system, i.e. $a_{ij} = 1$, Eq. (2.10) implies that $a_{ij}^{\text{act}} = 1$ if $P_i(t) > e_{ij}C_i$ and 0 otherwise. Thus, as $P_i(t)$ increases, a_{ij}^{act} transitions from 0 to 1 whenever $P_i(t) - e_{ij}C_i > 0$. This type of function is usually referred to as the Heaviside (step) function, and as shown by e.g. Anguelov and Markov (2016), Martinelli et al. (2023) and Venetis (2025), the Heaviside function can typically be approximated by a sigmoid.

We choose therefore to approximate a_{ij}^{act} by a logistic sigmoid centred at $P_i(t) - e_{ij}C_i$, where we multiply the sigmoid by a_{ij} to ensure that $s_{ij}^{\text{act}} \neq 0$ only when $a_{ij} = 1$. Specifically, using the logistic function, we let

$$s_{ij}^{\text{act}} = a_{ij} \left[1 + \exp \left(-\frac{P_i(t) - e_{ij}C_i}{\tau^{\text{act}}} \right) \right]^{-1}, \quad i, j = 1, 2, \dots, k, \quad (2.13)$$

where $\exp()$ denotes the exponential function and where $\tau^{\text{act}} \in (0, \infty) \subset \mathbb{R}$ is a parameter controlling the smoothness of the approximation. In the limit, as we reduce the smoothness to 0, i.e. $\tau^{\text{act}} \rightarrow 0$ from above, we have that $\mathbf{S}^{\text{act}}(G, t) \rightarrow \mathbf{A}^{\text{act}}(G, t)$ almost uniformly, as shown in Appendix A.

In Figure 2.10, we illustrate how the value of τ^{act} affects the value of s_{ij}^{act} for migration pathways that are present within the system, i.e. where $a_{ij} = 1$. Notice that when we decrease the value of τ^{act} , the curves representing s_{ij}^{act} for different values of τ^{act} seemingly converge to the (black) discontinuous line annotated within the figure. As shown in Appendix A, this is indeed the line to which s_{ij}^{act} converges (pointwise) as $\tau^{\text{act}} \rightarrow 0$ from above. With the exception of the point at $(0, 1/2) \in \mathbb{R}^2$, we notice that this is also the line traced by a_{ij}^{act} for different values of $P_i(t) - e_{ij}C_i$. As previously mentioned, a_{ij}^{act} transitions from 0 to 1 at $P_i(t) - e_{ij}C_i > 0$, representing the activation of the migration pathway between the i th and j th reservoirs. We may therefore think of τ^{act} as a similarity parameter, in addition to a parameter controlling the smoothness of s_{ij}^{act} . For lower values of τ^{act} , the "degree of activation" indicated by s_{ij}^{act} more closely resembles the strict activation of a_{ij}^{act} . Decreasing the value of τ^{act} does however increase the slope of s_{ij}^{act} at the transition point $P_i(t) - e_{ij}C_i = 0$, which has the potential of introducing errors when differentiating the expression on a computer with finite memory. Care must therefore be taken when choosing an appropriate value of τ^{act} .

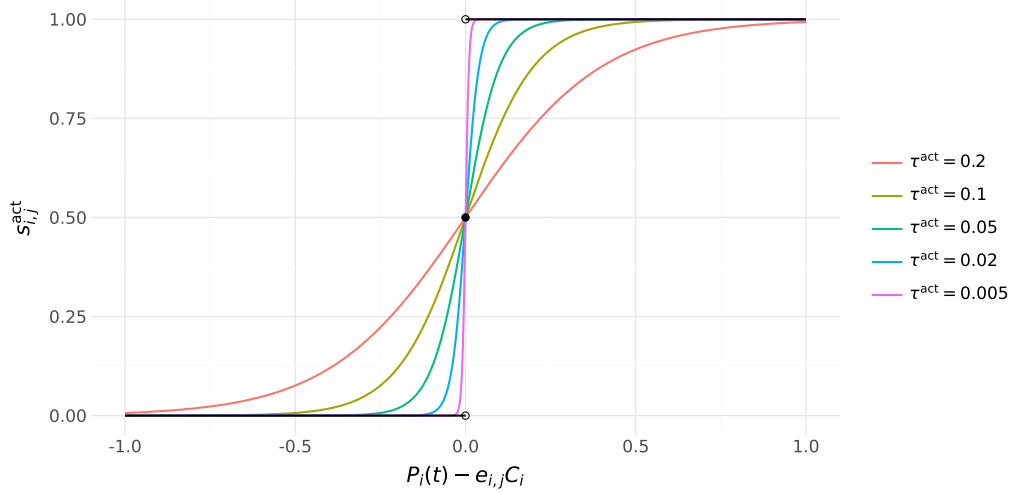


Figure 2.10.: Plot of s_{ij}^{act} against $P_i(t) - e_{i,j}C_i$ for 5 different values of τ^{act} , in the case where $a_{ij} = 1$. The black discontinuous line that takes on the value of $1/2$ at $P_i(t) - e_{i,j}C_i = 0$ represents the (pointwise) limit of s_{ij}^{act} as $\tau^{\text{act}} \rightarrow 0$.

2.4.2. Optimising the Simulator

We have now technically constructed the differentiable simulator that we set out to construct in the previous section. For each $t < T_k^{\text{mig}}$, we construct $\mathcal{S}$ by computing $\mathbf{q}^{\text{net}}(t)$ using Eqs. (2.11) and (2.12), which yields $\mathbf{h}(t)$ by substitution in Eq. (2.4) and subsequent inversion of Eq. (2.6), noting in addition that we have substituted $\mathbf{A}^{\text{act}}(G, t)$ for $\mathbf{S}^{\text{act}}(G, t)$ in $\mathbf{F}(G, t)$. This is a rather inefficient approach however, as we need to recompute $\mathbf{q}^{\text{net}}(t)$ for each $t < T_k^{\text{mig}}$ we consider. Depending on the value of τ^{act} , $\mathbf{F}(G, t)$ (and by extension $\mathbf{q}^{\text{net}}(t)$) might barely change in the time between pathway activations. As $\tau^{\text{act}} \rightarrow 0$ from above, $\mathbf{q}^{\text{net}}(t)$ only changes intermittently in time, concurrent with the activation of the migration pathways. An efficient algorithm would account for this, only computing $\mathbf{q}^{\text{net}}(t)$ when absolutely necessary.

Reducing the Number of Evaluations of $\mathbf{q}^{\text{net}}(t)$

We accomplish this by exploiting the approximate piecewise constant nature of $\mathbf{q}^{\text{net}}(t)$, which implies by Eq. (2.4), that $\mathbf{V}(t)$ is approximate piecewise linear for $t < T_k^{\text{mig}}$. In particular, letting $t^{(\ell)}$ denote the time of occurrence (or event time) of the ℓ th event, we compute $\mathbf{q}^{\text{net}}(t)$ and $\mathbf{V}(t)$ only at the times $t^{(0)}, t^{(1)}, \dots, t^{(m+1)}$ leading up to and including T_k^{mig} , where we define $t^{(0)} = 0$ and $t^{(m+1)} = T_k^{\text{mig}}$. Linear interpolation can then be used to approximate $\mathbf{V}(t)$ for other values of $t < T_k^{\text{mig}}$, i.e.

$$\mathbf{V}(t) \approx \mathbf{V}(t^{(\ell)}) + \frac{t - t^{(\ell)}}{t^{(\ell+1)} - t^{(\ell)}} \left[\mathbf{V}(t^{(\ell+1)}) - \mathbf{V}(t^{(\ell)}) \right], \quad \ell = 0, 1, \dots, m. \quad (2.14)$$

As previously mentioned, when $\tau^{\text{act}} \rightarrow 0$ from above, $\mathbf{q}^{\text{net}}(t)$ only changes concurrently with the activation of each migration pathway. As a result, $\mathbf{q}^{\text{net}}(t)$ becomes piecewise constant with respect to t in the limit, making $\mathbf{V}(t)$ piecewise linear. It follows that the above approximation of $\mathbf{V}(t)$ is exact, when we let $\tau^{\text{act}} \rightarrow 0$ from above.

Moreover, using Eq. (2.4), we have that

$$t^{(\ell+1)} \approx t^{(\ell)} + \frac{V(t^{(\ell+1)}) - V(t^{(\ell)})}{q^{\text{net}}(t^{(\ell)})}, \quad \ell = 0, 1, \dots, m,$$

where the approximation once again becomes exact in the limit, as $\tau^{\text{act}} \rightarrow 0$ from above. Note that if we can express $V(t^{(\ell+1)})$ in some other way, so as to make the expression explicit, we can compute the event times iteratively, starting at $t^{(0)}$. Using Eqs. (2.1) and (2.6), the condition for inter-reservoir migration developed in Section 2.3.3 implies that the pathway between the i th and j th reservoirs, if present within the system, activate whenever $V_i(t)$ surpasses the pathway's critical volume v_{ij}^{crit} , defined by

$$v_{ij}^{\text{crit}} = \frac{\pi \alpha_i^2}{3\beta_i^2} \left(\frac{e_{i,j} C_i}{\Delta \rho g} \right)^2 \left[3\beta_i - \frac{e_{i,j} C_i}{\Delta \rho g} \right], \quad i, j : a_{i,j} \in A(G).$$

This allows us to approximate $t^{(\ell+1)}$ by a minimum over positive fractions, i.e.

$$t^{(\ell+1)} \approx t^{(\ell)} + \min_{\substack{i,j=1,2,\dots,k \\ q_i^{\text{net}}(t^{(\ell)}) \neq 0}} \left\{ \frac{v_{ij}^{\text{crit}} - V_i(t^{(\ell)})}{q_i^{\text{net}}(t^{(\ell)})} > 0 \right\}, \quad \ell = 0, 1, \dots, m-1, \quad (2.15)$$

where we extend v_{ij}^{crit} to all $i, j = 1, 2, \dots, k$, by defining $v_{ij}^{\text{crit}} = \infty$ for all $a_{i,j} \notin A(G)$.

In relation to the above expression, we immediately make the following notes. In particular, if $q_i^{\text{net}}(t^{(\ell)}) = 0$, then CO_2 does not accumulate within the i th reservoir at the current time. As a result, CO_2 has either not migrated into the i th reservoir at the time or the reservoir has been fully compromised, following the activation of every migration pathway emanating from it. In either case, there are no pathways emanating from the i th reservoir that are a candidate for activation, and so we should not include their corresponding fraction in the minimum. A similar conclusion can be drawn from fractions that are non-positive, as $V_i(t^{(\ell)}) > v_{ij}^{\text{crit}}$ in this case, following the fact that $q_i^{\text{net}}(t) \geq 0$. If the volume within a reservoir has surpassed the critical volume of a migration pathway, the pathway in question must have already been activated during an event prior to or including the ℓ th event. The pathway can therefore not be part of a future event, and so it should not be included in the minimum. Lastly, we also note that the running index ℓ can be at most equal to $m-1$ in this particular case, as opposed to m in the earlier approximations, a direct result of us defining $t^{(m+1)} = T_k^{\text{mig}}$. In particular, we might have a case in which some of the system's $\varepsilon(G)$ migration pathways have not activated prior to T_k^{mig} . If we let $\ell = m$ in this case, Eq. (2.15) would provide an approximation of the first event occurring after T_k^{mig} , as opposed to T_k^{mig} itself.

To ensure that we are able to approximate $t^{(m+1)}$ in such a way that the approximation tends to the exact value of T_k^{mig} in the limit as $\tau^{\text{act}} \rightarrow 0$ from above, we nest Eq. (2.15) in an outer minimum, comparing its value to an estimate of the remaining time until the topmost reservoir gets compromised. In particular, we use the pathways that are active at $t^{(\ell)}$ and compute the time until CO_2 starts to migrate from the topmost reservoir, assuming no other pathways activate in the meantime. Denoting this estimate by $T^{\text{remain}}(t^{(\ell)})$, we have that, for $\ell = 0, 1, \dots, m$,

$$T^{\text{remain}}(t^{(\ell)}) = \begin{cases} \frac{1}{q_k^{\text{net}}(t^{(\ell)})} \left(\frac{\pi \alpha_k^2}{3\beta_k^2} \left(\frac{C_k}{\Delta \rho g} \right)^2 \left[3\beta_k - \frac{C_k}{\Delta \rho g} \right] - V_k(t^{(\ell)}) \right), & \text{if } q_k^{\text{net}}(t^{(\ell)}) \neq 0. \\ \infty, & \text{if } q_k^{\text{net}}(t^{(\ell)}) = 0. \end{cases}$$

This follows from the approximate piecewise linear nature of $\mathbf{V}(t)$, by applying Eqs. (2.1) and (2.6). Note in particular, that the condition under which CO_2 migration from the topmost reservoir can occur, is governed by the IP condition in Section 2.1.2, as there are no migration pathways emanating from the topmost reservoir. If we denote by $T^{\text{inner}}(t^{(\ell)})$ the minimum in Eq. (2.15), or the time between events ℓ and $\ell + 1$ assuming that $\ell \neq m$ is not the final event prior to T_k^{mig} , we have that

$$t^{(\ell+1)} \approx t^{(\ell)} + \min \left\{ T^{\text{remain}}(t^{(\ell)}), T^{\text{inner}}(t^{(\ell)}) \right\}, \quad \ell = 0, 1, \dots, m. \quad (2.16)$$

Using the above equation to iteratively approximate $t^{(1)}, t^{(2)}, \dots, t^{(m+1)}$ has a detrimental flaw however, as we cannot differentiate a minimum at the points where its arguments coincide. Since $\mathcal{S}$ has to be differentiable for all $t < T_k^{\text{mig}}$, this becomes a major problem. Similar to how we softened the hard activation of $\mathbf{A}^{\text{act}}(G, t)$ in Section 2.4.1, we mediate this problem by introducing a 2nd smoothness parameter $\tau^{\text{min}} \in (0, \infty) \subset \mathbb{R}$ that softens the minimum in Eqs. (2.15) and (2.16). Specifically, we approximate the minimum of n real variables by a shifted and scaled logarithm-sum-exponential function (Blanchard et al. 2021; Bonnot et al. 2024),

$$\text{LSE}(x_1, x_2, \dots, x_n) = \tau^{\text{min}} \left[\ln n - \ln \sum_{i=1}^n \exp \left(-\frac{x_i}{\tau^{\text{min}}} \right) \right], \quad (2.17)$$

where in the case of the aforementioned equations, $n = k^2$ and $n = 2$ respectively. By shifting and scaling the function in this particular way, it is possible to show that it is bounded from below by the minimum it estimates, i.e. $\min\{x_1, x_2, \dots, x_n\}$. Moreover, as $\tau^{\text{min}} \rightarrow 0$ from above, the function converges uniformly to $\min\{x_1, x_2, \dots, x_n\}$ for all $x_1, x_2, \dots, x_n \in \mathbb{R}$. Note that a proof of both of these statements have been included in Appendix A.

We can therefore reduce the number of evaluations of Eq. (2.12) significantly, by instead approximating $t^{(1)}, t^{(2)}, \dots, t^{(m)}$ iteratively using Eq. (2.16), where the minimum has been replaced by Eq. (2.17), only evaluating $\mathbf{q}^{\text{net}}(t)$ at each approximate event time. Eq. (2.14) can then be used to obtain an approximation of $\mathbf{V}(t)$ for all $t < T_k^{\text{mig}}$, which upon inversion of Eq. (2.6), yields a corresponding approximation of $\mathbf{h}(t)$. We can then control the smoothness of the resulting differentiable simulator by varying the parameters τ^{act} and τ^{min} , noting that in the limit as $\tau^{\text{act}}, \tau^{\text{min}} \rightarrow 0$ jointly from above, $\mathcal{S}$ becomes exact at the expense of not being differentiable at the event times. When deciding on a value of τ^{min} , care must be taken to ensure that the value is not too small compared with τ^{act} . Ideally, the value of τ^{min} is such that when we evaluate $S^{\text{act}}(G, t^{(\ell)})$ at the event times, $s_{ij}^{\text{act}} \approx 1$ if the pathway between the i th and j th reservoirs are active following the event, as this reduces the approximation error of having non-zero smoothing parameters.

To see why this is the case, we first reiterate the fact that Eq. (2.17) is bounded from below by the minimum it approximates. Hence, when we replace the minimum in Eq. (2.16) by Eq. (2.17), Eq. (2.16) provides an approximation of the event time that is greater than or equal to the true event time, where we have equality only in the case where $\tau^{\text{min}} \rightarrow 0$ from above. Unless we are in the limit, it follows that $P_i(t) - e_{i,j}C_i > 0$ at the approximated event time, where the migration pathway between the i th and j th reservoirs activate, as $P_i(t)$ increases whenever $q_i^{\text{net}}(t) > 0$. If the value of τ^{min} is too small compared to that of τ^{act} , the difference or delay between the true- and approximated event times would be too small as well. As a result, s_{ij}^{act} would not have had enough time to increase from its value of $1/2$ at $P_i(t) - e_{i,j}C_i = 0$. If e.g. $\tau^{\text{act}} = 0.2$ and we choose a value of τ^{min}

resulting in $P_i(t) - e_{i,j}C_i = 0.25$ at the approximated event time, we see from Figure 2.10 that $s_{ij}^{\text{act}} \approx 0.8$, compared with $s_{ij}^{\text{act}} \approx 1$ for $\tau^{\text{act}} < 0.05$. When evaluating $q_j^{\text{net}}(t)$ at the approximated event time, the resulting rate would be significantly smaller, as only 0.8 of the rate of CO₂ that should have migrated along the pathway between the i th and j th reservoirs is allowed to do so. As $\mathbf{q}^{\text{net}}(t)$ is included in the denominator of Eq. (2.15), the approximation provided by the equation has the potential of blowing up over successive iterations. Our approximation of the event times and $t^{(m+1)} = T_k^{\text{mig}}$ in particular, would likely be much higher than the true event times. The resulting $\mathcal{S}$, although differentiable for all $t < T_k^{\text{mig}}$, would likely not provide an accurate approximation of $\mathbf{h}(t)$.

Scaling of Physical Quantities

As is recommended when modelling physical processes using mathematical equations, we further optimise the simulator by scaling some of the quantities used in its construction. This produces equations that are often more numerically stable compared to their non-scaled counterparts, as shown in e.g. Casella and Braun (2017), and can highlight certain features of the model that would otherwise be lost. Researchers such as Dym (2004, ch. 3) have treated this in great detail, showing how proper scaling of underlying quantities can expose limits to the validity of the model in question. We proceed, as outlined in Dym (2004, ch. 3.5), by introducing characteristic quantities, such that when we divide the physical quantities by their respective characteristic quantities, we obtain a dimensionless variable in the range between 0 and 1. The equations used to construct $\mathcal{S}$ can then be adapted so that they apply to these new dimensionless variables, noting that we can at any time reconstruct the actual physical quantities through multiplication by the characteristic quantities.

We recall from Section 2.2.1 that β denotes the length of the vertical semi-axis of the reservoirs of the stratigraphic system. Hence, if we divide $\mathbf{h}(t)$ by β , we obtain a scaled quantity that is an element on $[0, 1]^k \in \mathbb{R}^k$ for all $t \geq 0$, as the height of the CO₂ plume cannot possibly grow larger than the vertical extent of the reservoirs (in which case the CO₂ would overflow and enter the next reservoir in higher topological order). Denoting the scaled height of the CO₂ plume within the i th reservoir at time t by $x_i(t) = h_i(t)/\beta_i$, we have by substitution in Eq. (2.6),

$$V_i(t) = \frac{\pi\alpha_i^2\beta_i}{3}x_i^2(t)[3 - x_i(t)], \quad i = 1, 2, \dots, k.$$

Noting that $x_i(t) \leq 1$ for all $i = 1, 2, \dots, k$, it follows that $V_i(t) \leq 2\pi\alpha_i^2\beta_i/3$. This is not immediately obvious without scaling the quantities, and serves therefore to illustrate one of the benefits of scaling, namely the fact that a characteristic quantity, such as $2\pi\alpha_i^2\beta_i/3$ in the case of $V_i(t)$, often have a physical meaning apart from just providing a necessary scale (Dym 2004, ch. 3). In this case, we note that $2\pi\alpha_i^2\beta_i/3$ is the maximum amount of CO₂ that can accumulate within the i th reservoir (before the CO₂ starts to overflow). By defining the scaled volume of accumulated CO₂ within the i th reservoir at time t using $y_i(t) = 3V_i(t)/(2\pi\alpha_i^2\beta_i)$, we can express Eq. (2.6) as

$$\mathbf{y}(t) = \frac{1}{2}\mathbf{x}^2(t) \odot [3(\mathbf{1}_k) - \mathbf{x}(t)], \quad (2.18)$$

where the exponentiation is taken element-wise, i.e. $\mathbf{x}^2(t) = \mathbf{x}(t) \odot \mathbf{x}(t)$.

Notice in particular the lack of reservoir parameters in the above equation, as these have all been absorbed by the scaling factors. For each $\mathbf{y}(t)$, we can therefore obtain a corresponding $\mathbf{x}(t)$ without having knowledge of the reservoir parameters. We do however need

the value of β to retrieve $\mathbf{h}(t)$ from $\mathbf{x}(t)$, noting that the lateral extent of the reservoirs, as parametrised by α , are irrelevant in this regard. This might at first come as a surprise, as it means that we can retrieve the height of the CO₂ plume within the reservoirs, which obviously has a lateral extent, without actually considering the extent so long as we have the associated scaled volume.

Moreover, as $V_i(t) \leq 2\pi\alpha_i^2\beta_i/3$ for all $i = 1, 2, \dots, k$, the characteristic volume $2\pi\alpha_i^2\beta_i/3$ also provides a natural scaling factor for the various CO₂ rates defined thus far. In particular, by letting $z_i(t) = 3q_i^{\text{net}}(t)/(2\pi\alpha_i^2\beta_i)$ denote the scaled net rate of CO₂ accumulation within the i th reservoir at time t , $\mathbf{z}(t)$ now directly corresponds to the fraction of the total capacity of CO₂ accumulating within the reservoirs at time t . If for instance $z_k(t) = 1/2$, then 1/2 of the total CO₂ capacity of the topmost reservoir accumulates within the reservoir at time t .

We are now able to express the event times in terms of an iterative expression using scaled physical quantities. Using the expression for $\mathbf{x}(t)$, Eq. (2.1) and the migration condition developed in Section 2.3.3, we first define the scaled critical volume of a migration pathway between the i th and j th reservoirs as follows,

$$y_{ij}^{\text{crit}} = \begin{cases} \frac{1}{2} \left(\frac{e_{ij}C_i}{\Delta\rho g\beta_i} \right)^2 \left[3 - \frac{e_{ij}C_i}{\Delta\rho g\beta_i} \right], & \text{if } a_{i,j} \in A(G). \\ \infty, & \text{if } a_{i,j} \notin A(G). \end{cases} \quad (2.19)$$

Using the above expression, we have that

$$t^{(\ell+1)} \approx t^{(\ell)} + \text{LSE} \left(\tilde{T}^{\text{remain}}(t^{(\ell)}), \tilde{T}^{\text{inner}}(t^{(\ell)}) \right), \quad \ell = 0, 1, \dots, m, \quad (2.20)$$

where

$$\tilde{T}^{\text{inner}}(t^{(\ell)}) = \text{LSE} \left(\left\{ \frac{y_{ij}^{\text{crit}} - y_i(t^{(\ell)})}{z_i(t^{(\ell)})} > 0 : i, j = 1, 2, \dots, k \wedge z_i(t^{(\ell)}) \neq 0 \right\} \right)$$

and

$$\tilde{T}^{\text{remain}}(t^{(\ell)}) = \begin{cases} \frac{1}{2 \cdot z_k(t^{(\ell)})} \left(\frac{C_k}{\Delta\rho g\beta_k} \right)^2 \left[3 - \frac{C_k}{\Delta\rho g\beta_k} \right] - y_k(t^{(\ell)}), & \text{if } z_k(t^{(\ell)}) \neq 0. \\ \infty, & \text{if } z_k(t^{(\ell)}) = 0. \end{cases}$$

In particular, we have that $\tilde{T}^{\text{remain}}$ and $\tilde{T}^{\text{inner}}$ denotes the scaled versions of T^{remain} and T^{inner} in Eq. (2.16), respectively. Ideally, we would provide a scale for t as well, using e.g. T_k^{mig} so that the event times $t^{(1)}, t^{(2)}, \dots, t^{(m)}$ are all bounded by 1. As we compute $t^{(m+1)} = T_k^{\text{mig}}$ iteratively using Eq. (2.20), this will not work however, as the value of T_k^{mig} remains unknown prior to the computation of the earlier event times. For the purposes of this thesis, we therefore choose to let t remain unscaled.

By approximating the event times using the aforementioned equation and noting that the approximation is exact as $\tau^{\text{act}}, \tau^{\text{min}} \rightarrow 0$ jointly from above, we obtain yet another way of constructing the simulator in Section 2.4.1, the details of which are found in Algorithm 2.2. Note in particular that if we are to obtain a value of $\mathbf{h}(t)$ for some $t < T_k^{\text{mig}}$, we have to account for the activation of every migration pathway in the time leading up to t . It is not enough to account for the state of the system at time t only, as $\mathbf{h}(t)$ is dependent on the history of $\mathbf{q}^{\text{net}}(t)$ through Eqs. (2.4) and (2.6), and as demonstrated in Section 2.4.1, the activation of the system's migration pathways. We have therefore made a conscious

decision to refer to $\mathcal{S}$ as a simulator in the preceding sections, rather than a map of $[0, T_k^{\text{mig}}) \subset \mathbb{R}$ onto $[0, \beta_1] \times [0, \beta_2] \times \cdots \times [0, \beta_k] \subset \mathbb{R}^k$ that sends $t \mapsto \mathbf{h}(t)$, as this term fails to convey the importance of the history of $\mathbf{q}^{\text{net}}(t)$ when evaluating $\mathbf{h}(t)$.

Algorithm 2.2 $\mathcal{S}$ - Differentiable simulator for $\mathbf{h}(t)$

Require: $t, \tau^{\text{act}}, \tau^{\text{min}}, \Delta\rho, \mathbf{q}^{\text{ext}}, \boldsymbol{\alpha}, \beta, \mathbf{C}, \mathbf{Q}^{\text{sat}}, \mathbf{E}^{\text{sat}}$ and $\mathbf{A}(G)$

Ensure: $\mathbf{h}(t)$

```

1: procedure GETNETRATE( $t$ )                                ▷ Compute  $\mathbf{z}(t)$ .
2:    $\mathbf{S}^{\text{act}}(G, t) \stackrel{(2.13)}{\leftarrow} \tau^{\text{act}}, \mathbf{Y}^{\text{crit}}, \mathbf{y}(t), \mathbf{A}(G)$     ▷ Replace  $P_i(t) - e_{i,j}C_i$  by  $y_{ij}^{\text{crit}} - y_i(t)$ .
3:    $\mathbf{F}(G, t) \leftarrow \mathbf{S}^{\text{act}}(G, t) \odot \mathbf{Q}^{\text{sat}} \odot \mathbf{A}(G)$ 
4:    $\mathbf{q}^{\text{ent}}(t) \stackrel{(2.11)}{\leftarrow} \mathbf{F}(G, t), \mathbf{q}^{\text{ext}}$                                 ▷ Use Algorithm 2.1.
5:    $\mathbf{q}^{\text{net}}(t) \stackrel{(2.12)}{\leftarrow} \mathbf{F}(G, t), \mathbf{q}^{\text{ent}}(t)$ 
6:    $\mathbf{z}(t) \leftarrow \frac{3}{2\pi} \boldsymbol{\alpha}^{-2} \odot \beta^{-1} \odot \mathbf{q}^{\text{net}}(t)$     ▷ Exponentiation is taken element-wise.
7: end procedure

8: procedure SIMULATEEVENTS                                ▷ Compute  $t^{(\ell)}$  and  $\mathbf{y}(t^{(\ell)})$  for  $\ell = 0, 1, \dots, m+1$ .
9:    $\mathbf{Y}^{\text{crit}} \stackrel{(2.19)}{\leftarrow} \Delta\rho, \beta, \mathbf{C}, \mathbf{E}^{\text{sat}}, \mathbf{A}(G)$     ▷ Recall that  $\mathbf{E}(G) = \mathbf{E}^{\text{sat}} \odot \mathbf{A}(G)$ .
10:   $t^{(0)} \leftarrow 0$ 
11:   $\mathbf{y}(t^{(0)}) \leftarrow \mathbf{0}_k$                                 ▷  $\mathbf{0}_k \in \mathbb{R}^k$  is the 0-element on  $\mathbb{R}^k$ .
12:  for  $\ell = 0, 1, \dots, m$  do
13:    GETNETRATE( $t^{(\ell)}$ )
14:     $t^{(\ell+1)} \stackrel{(2.20)}{\leftarrow} t^{(\ell)}, \tau^{\text{min}}, \mathbf{Y}^{\text{crit}}, \mathbf{y}(t^{(\ell)}), \mathbf{C}, \beta, \Delta\rho, \mathbf{z}(t^{(\ell)})$ 
15:     $\mathbf{y}(t^{(\ell+1)}) \leftarrow \mathbf{y}(t^{(\ell)}) + \mathbf{z}(t^{(\ell)})[t^{(\ell+1)} - t^{(\ell)}]$ 
16:  end for
17: end procedure

18: procedure SIMULATEHEIGHT( $t$ )                                ▷ Compute  $\mathbf{h}(t)$ .
19:   SIMULATEEVENTS
20:    $\mathbf{y}(t) \stackrel{(2.14)}{\leftarrow} t, t^{(\ell)}, \mathbf{y}(t^{(\ell)})$  for  $\ell = 0, 1, \dots, m+1$     ▷ Replace  $\mathbf{V}(t)$  by  $\mathbf{y}(t)$ .
21:    $\mathbf{x}(t) \stackrel{(2.18)}{\leftarrow} \mathbf{y}(t)$                                 ▷ Use e.g. Newton-Raphson to invert the equation.
22:    $\mathbf{h}(t) \leftarrow \beta \odot \mathbf{x}(t)$ 
23: end procedure

```

Using the stratigraphic system depicted in Figure 2.9a, Figure 2.11 illustrates the process in which $\mathcal{S}$ obtains $\mathbf{h}(t)$ for a particular $t \leq T_k^{\text{mig}}$. It might at first seem wasteful that the simulator computes the state of the system at every event time leading up to and including T_k^{mig} , if for instance $t^{(0)} < t < t^{(1)}$. This is done with intension however, since a computer implementation will be able to store the state of the system at each event time. If we need to compute the value of $\mathbf{h}(t)$ for another $t \leq T_k^{\text{mig}}$ within the same system, e.g. for the purposes of plotting $\mathbf{h}(t)$ against t , we can do so directly, without having to recompute the event times or test whether the necessary event times are already computed. We also note that the number of events in total prior to T_k^{mig} , which we denoted by m , can be at most equal to $\varepsilon(G)$, as shown in Section 2.4.1. The added complexity of having to evaluate the state of the system at $m+1$ time points (accounting for T_k^{mig}), is therefore somewhat limited.

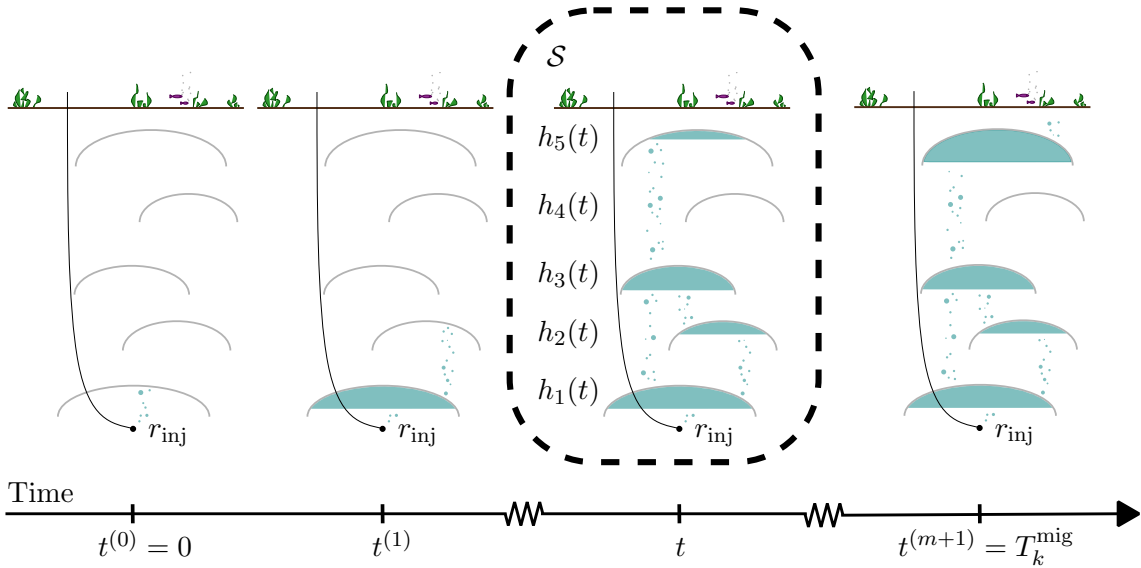


Figure 2.11.: Representation of the differentiable simulator $\mathcal{S}$, based on the stratigraphic system depicted in Figure 2.9a. $\mathcal{S}$ simulates the development of the CO_2 plume at each event time $t^{(0)}, t^{(1)}, \dots, t^{(m+1)}$, and uses linear interpolation to obtain the (scaled) volume of accumulated CO_2 within the reservoirs at time t . Eq. (2.18) is then used to obtain the (scaled) height of the CO_2 plume at time t . Note that some variables have been suppressed in the figure, so as to make it easier to interpret.

2.4.3. Connection to Existing Research

Before we continue developing our framework further, it is worth mentioning how the use of a simulator ties into existing research on CO_2 migration in multi-layered stratigraphic systems, and why the introduction of a graph based framework might prove superior in some situations. Earlier in the thesis, namely in Section 2.1.2, we mentioned that Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear) all considered inter-reservoir CO_2 migration as part of their research. To make inference on the migration process, the researchers simulated the migration of CO_2 throughout the system based on the original IP criteria presented in Section 2.1.2, i.e. they let CO_2 migrate from the i th reservoir only when $P_i(t) > C_i$. As they held a Markovian assumption on the migration of CO_2 , their simulator essentially reduced to a 2 step process, in which $\mathbf{T}^{\text{mig}}$ was computed iteratively and $\mathbf{h}(t)$ obtained through a piecewise expression involving $\mathbf{T}^{\text{mig}}$ (Dutta et al. 2026; Eidsvik et al. to appear). In this regard, we note that the researchers made an additional assumption of constant injection in the bottommost reservoir only.

We loosen the Markovian assumption and allow constant injection in all reservoirs, not just the bottom one. A direct consequence of this is the fact that $\mathbf{q}^{\text{net}}(t)$ is no longer trivially tied to $\mathbf{q}^{\text{ext}}$. Using the assumptions of the earlier researchers, we would have that for all $i = 2, 3, \dots, k$,

$$q_i^{\text{net}} = \begin{cases} 0, & \text{if } t < T_{i-1}^{\text{mig}}. \\ r_{\text{inj}}, & \text{if } T_{i-1}^{\text{mig}} \leq t < T_i^{\text{mig}}. \\ 0, & \text{if } t > T_i^{\text{mig}}. \end{cases}$$

This is obviously not the case when loosening the Markovian assumption, as seen in Section 2.3.1. Consequently, our simulator has to account for the changes in $\mathbf{q}^{\text{net}}(t)$ in a

more complex manner, making an iterative algorithm on $\mathbf{T}^{\text{mig}}$ unfeasible. Our simulator does however borrow on the design of Dutta et al. (2026) and Eidsvik et al. (to appear), when it comes to the computation of $\mathbf{V}(t)$ using linear interpolation. Although we cannot use $\mathbf{T}^{\text{mig}}$ for this purpose, Eq. (2.14) shows that we can compute $\mathbf{V}(t)$ by interpolation over $t^{(0)}, t^{(1)}, \dots, t^{(m+1)}$.

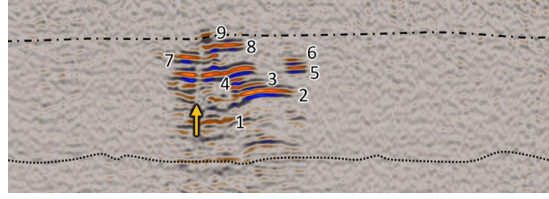
In between each event, our simulator has to recompute $\mathbf{q}^{\text{net}}(t)$ using Eqs. (2.11) and (2.12), which illustrates the advantage of representing the migration pathways as a DAG. In particular, it facilitates easy computation of complex quantities that are a result of propagating simpler quantities along the arcs of the digraph. As the corresponding adjacency matrix is triangular, such quantities can often be written implicitly in terms of a system of equations, as shown in Eq. (2.11), whose solution can be obtained through a single forward substitution. As far as the author is aware, the report written by Grødeland (2026) is the first example of someone representing the migration pathways of a multi-layered stratigraphic system in terms of a DAG. Note however that in Grødeland (2026), a recursion based algorithm was used to recompute $\mathbf{q}^{\text{net}}(t)$ for $t < T_k^{\text{mig}}$, as opposed to a forward substitution followed by a Hadamard product. The number of possible DAGs considered for a system comprising k reservoirs was also significantly lower in Grødeland (2026), compared to the number of possible DAGs considered in this thesis. Note that we elaborate on this particular fact in the sections that follow.

For the purposes of being complete, we also note that the practice of using DAGs to represent dependencies within complex systems is already well established, even though its use to represent migration pathways in multi-layered stratigraphic systems is seemingly lacking. Pearl and Verma (1987) outlined many of the benefits of using DAGs, in particular their ability to efficiently communicate conditional independence between underlying quantities, thus making them particularly well suited for use in probabilistic models. In the years that followed, researchers from multiple scientific fields, e.g. Berrie et al. (2025) and Zhang et al. (2019), used DAGs to represent dependencies in complex systems, ranging from epidemiology to machine learning and artificial intelligence.

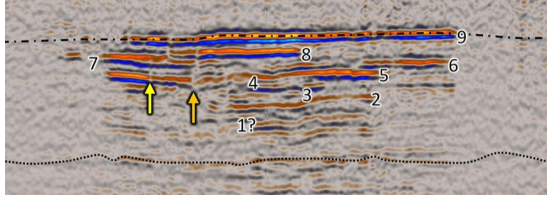
3. Bayesian Analysis



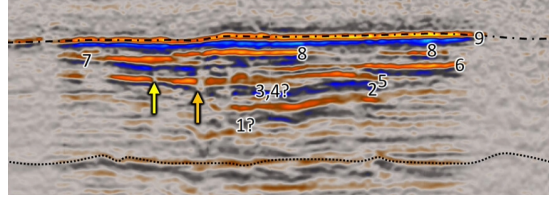
(a) Seismic baseline from the 1994 pre-injection survey. The white line indicates the location of the single injection well (15/9-A-16) used for CO₂ sequestration at the Sleipner field, and the star indicates the location where CO₂ is injected into the strata.



(b) Seismic difference between the 1994 baseline and the 1999 seismic survey, conducted 3 years after the injection process had commenced.



(c) Seismic difference between the 1994 baseline and the 2010 seismic survey.



(d) Seismic difference between the 1994 baseline and the 2020 seismic survey.

Figure 3.1.: Seismic sections of the stratigraphic system at the Sleipner field, along the main axis of the plume (see Figure 2.4). Decreases in acoustic impedance are shown as red/yellow returns, indicating the presence of CO₂ within the strata. The black numbers within the figure correspond to the $k = 9$ reservoirs making up the stratigraphic system. The figure has been taken from Furre et al. (2024), and both coloured arrows and black dashed/dotted lines correspond to geological anomalies/features discussed in the article.

3.1. Prior Distributions

The quantities of previous sections have till now all been considered deterministic, the value of which known to the researcher prior to the start of the injection process. In reality, quantities such as α and β , affecting the geometry of the CO₂ plume, vary significantly, even amongst reservoirs that are part of the same stratigraphic system. This is illustrated particularly well in Figure 2.4, showing the profile of the stratigraphic system at the Sleipner field after the injection of approximately 5 Mt CO₂. Notice how the lateral extent, in particular, vary significantly between the different reservoirs. In Figure 3.1, seismic sections along the main axis of the CO₂ plume (see Figure 2.4) are shown at different points in time, corresponding to 4 separate surveys conducted in the time between 1994 and 2020. The later surveys conducted in 1999, 2010 and 2020 have all been compared with the 1994 baseline (Display a), to show the location of sequestered CO₂ at the time of the later surveys (Display b) to (Display d). Notice how we are unable to determine

the extent of the reservoirs using only the 1994 and 1999 surveys, and that the location of sequestered CO₂ changes ever so slightly between the 2010 and 2020 survey. Furre et al. (2024) specifically states that frequent surveys were needed in the early 2000s to increase confidence regarding the storage safety of the stratigraphic system.

As mentioned in Section 2.3.3, the CTP of the cap formations also tend to exhibit local variation, with values in the range 1.6 – 1.9 MPa being considered normal for the Nordland shales. Similar to Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear), we account for this variation within our framework by adding uncertainty to the parameters of our model. Specifically, we assign prior distributions to the parameters in question, as shown in the following subsections, where we have chosen to consider continuous- and discrete parameters separately for added clarity.

3.1.1. Prior on Continuous Parameters

Prior on CTPs and the Topology of the Reservoirs

Similar to Dutta et al. (2026), we assume that the uncertainty in the CTPs of our system can be modelled using a joint log-normal distribution with log-mean vector $\boldsymbol{\mu}_{\tilde{\mathbf{C}}}$ and log-covariance matrix $\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}$. If we denote by $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ the (multivariate) normal distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, this can be stated as

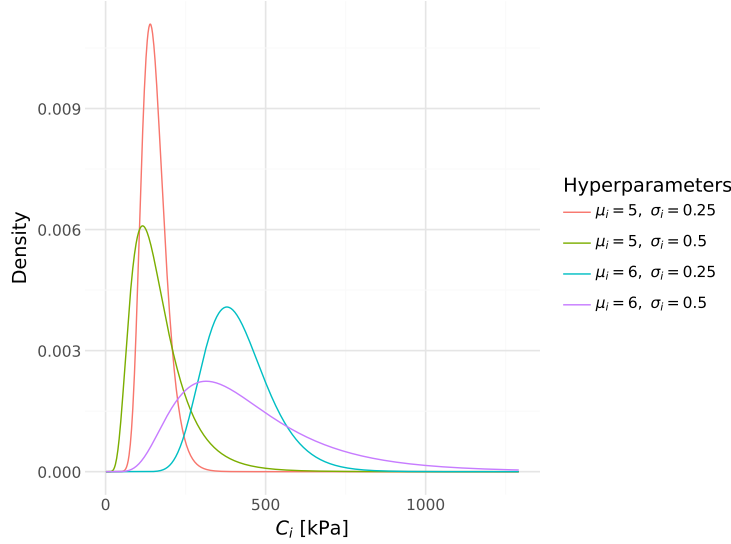
$$\ln \mathbf{C} \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{C}}}, \boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}).$$

Note in particular that we loosen the assumption of mutual independence held by Dutta et al. (2026), allowing the CTPs of our cap formations to be correlated. We make this generalisation so that we can account for any spatial features effecting the integrity of the cap formations. This could for instance be the local composition of impermeable rock found at the site of the stratigraphic system. As for the Sleipner field, the composition of the cap formations at the site are exclusively shale (Cavanagh and Haszeldine 2014), a type of mudstone, and so we should expect the properties of the formations to be similar to each other, as their composition are exactly the same. In other words, we should expect some systems to exhibit a high degree of correlation, when it comes to the CTPs of their cap formations.

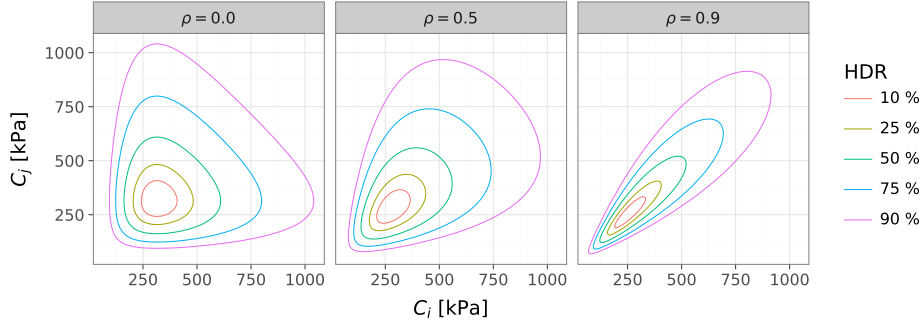
Figure 3.2 shows how the log-normal prior on $\mathbf{C}$ changes for different parameter settings. In particular, we illustrate how the marginal distribution of a single component C_i is affected by the choice of hyperparameters (Display a), and how the joint distribution of C_i and C_j , where we consider 2 separate reservoirs i and j , changes based on the log-correlation between the CTPs (Display b). The joint distribution is here visualised in terms of 5 Highest Density Regions (HDRs), representing a total probability mass ranging from 10 % to 90 %. If we for the moment let $f(C_i, C_j)$ denote the joint density function of C_i and C_j , then we can describe the $100(1 - \alpha)$ % HDR, denoted $R(f_\alpha)$, as the specific subset of the sample space of C_i and C_j defined by

$$R(f_\alpha) = \{C_i, C_j : f(C_i, C_j) \geq f_\alpha\},$$

where we define f_α as the largest possible constant such that $\Pr(C_i, C_j \in R(f_\alpha)) \geq 1 - \alpha$ (Hyndman 1996). In other words, the $100(1 - \alpha)$ % HDR represents the smallest possible subset of the sample space of the 2 CTPs, such that the probability of any sample being an element of the HDR is $1 - \alpha$.



(a) Marginal distribution of a single component C_i , where μ_i is the i th component of $\boldsymbol{\mu}_{\tilde{\mathbf{C}}}$ and σ_i^2 the i th diagonal component of $\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}$.



(b) Joint distribution of 2 components C_i and C_j , where the marginal distribution of each component corresponds to the purple density in (Display a). Each contour corresponds to the (approximate) boundary of a Highest Density Region (HDR), see for instance Hyndman (1996), and the log-correlation of C_i and C_j are denoted ρ .

Figure 3.2.: Illustration of the log-normal prior on the CTPs $\mathbf{C}$ for different parameter settings.

We note that by increasing the component of $\boldsymbol{\mu}_{\tilde{\mathbf{C}}}$ corresponding to the i th reservoir (denoted μ_i in the figure) the mode of the marginal prior distribution is shifted to the right. A similar shift is also visible when we increase the diagonal component of $\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}$ corresponding to the i th reservoir (denoted σ_i^2 in the figure), in which case the skewness of the marginal distribution increases and the mode is shifted to the left. The components of $\boldsymbol{\mu}_{\tilde{\mathbf{C}}}$ and diagonal elements of $\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}$ can therefore be used to express prior knowledge of the range of values typically taken by the CTPs. If our prior knowledge dictate that the CTPs are correlated, as we argued were likely in the case of the Sleipner field, we can model this by increasing the prior log-correlation between the CTPs in question. As for Figure 3.2b, we clearly notice that by increasing the log-correlation between the 2 CTPs (denoted ρ in the figure) the joint prior distribution of C_i and C_j gets concentrated in regions where C_i and C_j are similarly valued. An expression for both the marginal distribution in (Display a) and the joint distribution in (Display b) is available in Appendix A.

To account for the uncertainty in the topology of each reservoir, Dutta et al. (2026) used a

non-linear least-squares solver to fit multiple ellipsoids to different subsets of each reservoir within the stratigraphic system at the Sleipner field. The authors then used the resulting empirical distribution on the semi-axes of the ellipsoid, to account for the uncertainty in the geometric parameters of their model. We choose instead to assign joint log-normal priors to the lateral- and vertical extent of our reservoirs, similar to the prior we assigned to the CTPs of the cap formations. In particular, we assume that

$$\ln \alpha \sim \mathcal{N}(\mu_{\tilde{\alpha}}, \Sigma_{\tilde{\alpha}}) \quad \text{and} \quad \ln \beta \sim \mathcal{N}(\mu_{\tilde{\beta}}, \Sigma_{\tilde{\beta}}),$$

where $\mu_{\tilde{\alpha}}$ and $\Sigma_{\tilde{\alpha}}$ are the log-mean vector and log-covariance matrix of α , and $\mu_{\tilde{\beta}}$ and $\Sigma_{\tilde{\beta}}$ the log-mean vector and log-covariance matrix of β , respectively. We can then express our prior knowledge of the range of values typically taken by the lateral- and vertical extent of our reservoirs, by choosing suitable values for the hyperparameters. Since we use the same underlying prior for the lateral- and vertical extent as we did for the CTPs, the effect of the hyperparameters on the prior distributions of α and β are identical to the effects observed in Figure 3.2. The benefit of using log-normal priors as opposed to a least-squares solver in this case, is both its simplicity as well as a reduced risk of over-fitting. If we were to impose an empirical distribution based on the reservoirs of the stratigraphic system at the Sleipner field, the prior would not necessarily reflect the general uncertainty found in other stratigraphic systems.

This is similar to the approach taken by Eidsvik et al. (to appear), who themselves assumed a lower-truncated normal distribution on the geometric parameters of their stratigraphic system. As the truncated normal is locally symmetric around its mean, the prior used by Eidsvik et al. (to appear) might actually provide a more realistic distribution on the geometric parameters of the model. We choose however to use log-normal priors in this case, as it is easier to transform a log-normal variable such that the resulting transformation is unbounded on $\mathbb{R}$, which becomes important in the sections that follow. By using a prior with positive skewness, e.g. the log-normal, we are essentially assuming that it is more likely for the extent of the reservoirs to be smaller than what is indicated by the mean extent. In other words, we are assuming that the median of the distribution is smaller than the mean. We note for the sake of clarity, that we in this case have to ensure that α , β and C are non-negative, as it would be unphysical to consider the contrary.

Prior on Pathway Specific Weights

Both Dutta et al. (2026) and Eidsvik et al. (to appear) stopped assigning prior distributions at this point, as they had no more parameters to assign priors to. As for our framework, we introduced the notion of pathway specific weights in Sections 2.3.1 and 2.3.3, represented by the matrices $Q(G)$ and $E(G)$, both of which need to have prior densities associated with them. In the report by Grødeland (2026), a possible set of such priors were outlined, utilising independent and identically distributed (iid) Dirichlet- and Beta distributions, so as to ensure that the constraints placed on the weights in the aforementioned sections were satisfied. As part of this thesis, we take a different approach, assigning normal distributions to the weights and then applying a transformation, so as to ensure that the constraints are satisfied.

We can think of the prior distributions on α , β and C in a similar manner, where we first assign a normal distribution to their unconstrained counterparts, denoted by $\tilde{\alpha}$, $\tilde{\beta}$ and $\tilde{C}$ respectively, and then apply an exponential transformation to obtain the original quantities. Using α as an example, we have that $\alpha = \exp \tilde{\alpha}$, where $\tilde{\alpha} \sim \mathcal{N}(\mu_{\tilde{\alpha}}, \Sigma_{\tilde{\alpha}})$. As $\tilde{\alpha}$, $\tilde{\beta}$ and $\tilde{C}$ are all unconstrained elements on $\mathbb{R}^k$, this further illustrates how we can

transform our original quantities in such a way that they become unconstrained, namely by taking the (natural) logarithm of the quantities. As previously stated, the reason for wanting to do so should become apparent in the sections that follow.

We recall from Sections 2.3.1 and 2.3.3, that $\mathbf{Q}(G)$ and $\mathbf{E}(G)$ can be computed by a Hadamard product involving the saturated matrices $\mathbf{Q}^{\text{sat}}$ and $\mathbf{E}^{\text{sat}}$, hence we may state the priors on the weights of the migration pathways in terms of $\mathbf{Q}^{\text{sat}}$ and $\mathbf{E}^{\text{sat}}$. Moreover, as $\mathbf{Q}^{\text{sat}}$ and $\mathbf{E}^{\text{sat}}$ are both strictly upper-triangular, we only need to define priors on the strictly upper-triangular elements of each matrix. In practice, this is done by reducing the matrices to row-major form, and then assigning priors to the resulting vectors. To that extent, we define $\mathbf{q}^{\text{sat}} = (q_{12}^{\text{sat}}, q_{13}^{\text{sat}}, \dots, q_{1k}^{\text{sat}}, q_{23}^{\text{sat}}, \dots, q_{(k-1)k}^{\text{sat}})$ and $\mathbf{e}^{\text{sat}} = (e_{12}^{\text{sat}}, e_{13}^{\text{sat}}, \dots, e_{1k}^{\text{sat}}, e_{23}^{\text{sat}}, \dots, e_{(k-1)k}^{\text{sat}})$ as the vectors resulting from reducing $\mathbf{Q}^{\text{sat}}$ and $\mathbf{E}^{\text{sat}}$ to row-major form, noting that the lower diagonal elements of the matrices (all of which are equal to 0) have been discarded in the process. Letting $\tilde{\mathbf{q}}^{\text{sat}}$ and $\tilde{\mathbf{e}}^{\text{sat}}$ denote the unconstrained counterparts of $\mathbf{q}^{\text{sat}}$ and $\mathbf{e}^{\text{sat}}$ respectively, our earlier assumption implies that

$$\tilde{\mathbf{q}}^{\text{sat}} \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{q}}^{\text{sat}}}, \boldsymbol{\Sigma}_{\tilde{\mathbf{q}}^{\text{sat}}}) \quad \text{and} \quad \tilde{\mathbf{e}}^{\text{sat}} \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{e}}^{\text{sat}}}, \boldsymbol{\Sigma}_{\tilde{\mathbf{e}}^{\text{sat}}}),$$

where $\boldsymbol{\mu}_{\tilde{\mathbf{q}}^{\text{sat}}}$ and $\boldsymbol{\Sigma}_{\tilde{\mathbf{q}}^{\text{sat}}}$ are the log-mean vector and log-covariance matrix of $\tilde{\mathbf{q}}^{\text{sat}}$, and $\boldsymbol{\mu}_{\tilde{\mathbf{e}}^{\text{sat}}}$ and $\boldsymbol{\Sigma}_{\tilde{\mathbf{e}}^{\text{sat}}}$ the log-mean vector and log-covariance matrix of $\tilde{\mathbf{e}}^{\text{sat}}$, respectively.

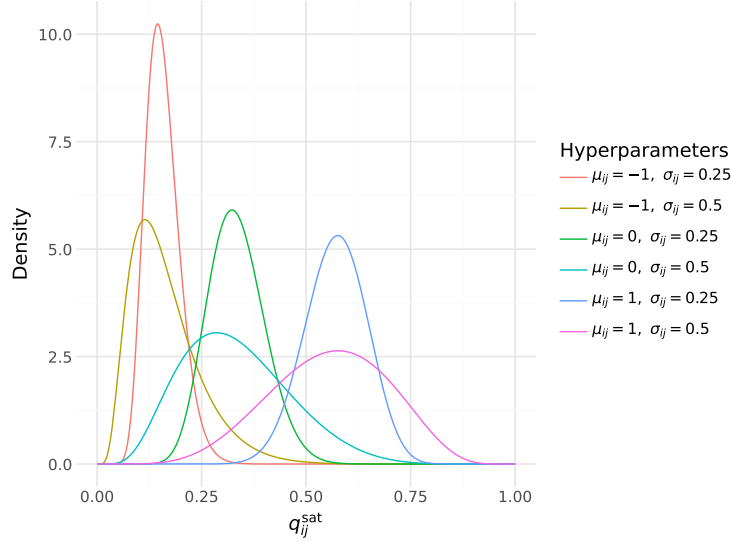
To obtain a transformation of $\tilde{\mathbf{q}}^{\text{sat}}$ that respects the constraints of Section 2.3.1, we note per the definition of $q_{i,j} \in Q(G)$, that $q_{i,j} \in [0, 1] \subset \mathbb{R}$ and $\sum_{j: a_{i,j} \in A(g)} q_{i,j} = 1$. In terms of $\tilde{\mathbf{q}}^{\text{sat}}$, this means that the elements of the vector have to be in the range $[0, 1] \subset \mathbb{R}$, such that when we sum over the columns of $\mathbf{Q}^{\text{sat}}$, the resulting sum equals 1 for every row. Note that if we exponentiate $\tilde{\mathbf{q}}^{\text{sat}}$, the resulting transformation would satisfy some of the constraints imposed by the definition, as the exponential function maps real values to the non-negative real line. Hence, the resulting transformation would be bounded from below by 0. If we then scale each element of the transformation by the sum of the terms appearing on the same "row", the remaining constraints would be satisfied as well. Defining the transformation $\tilde{\mathbf{q}}^{\text{sat}} \mapsto \mathbf{q}^{\text{sat}}$ by the relation,

$$q_{ij}^{\text{sat}} = \frac{\exp \tilde{q}_{ij}^{\text{sat}}}{\sum_{\ell=i+1}^k \exp \tilde{q}_{i\ell}^{\text{sat}}}, \quad (3.1)$$

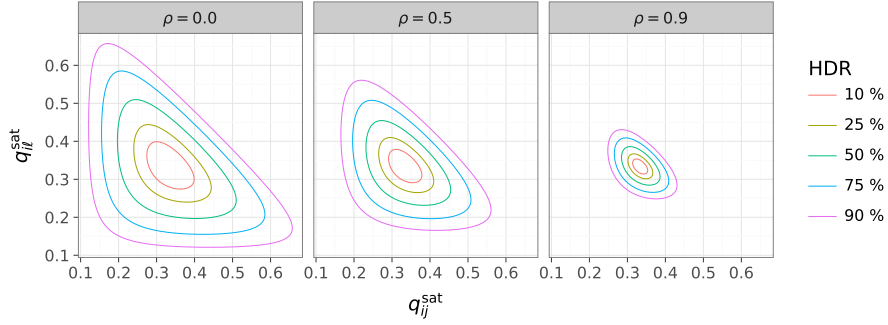
for each pair of indices (i, j) , such that $i = 1, 2, \dots, k-1$ and $j = i+1, i+2, \dots, k$, we obtain a transformation that respects the given constraints.

This function is usually referred to as the soft-max function within scientific literature, see for instance Blanchard et al. (2021). Especially within the field of computer science, where it is often used to normalise the output of a neural network, so that the output corresponds to a probability distribution on the labels of the final layer (Martínez-Ramón et al. 2024, ch. 1.5.2.2). As shown in Blanchard et al. (2021), the soft-max function also corresponds to the derivative of the ordinary logarithm-sum-exponential function, a scaled and shifted version of which we used in Section 2.4.2 to approximate the minimum in Eqs. (2.15) and (2.16).

Similar to Figure 3.2, Figure 3.3 shows how the prior distribution on $\mathbf{q}^{\text{sat}}$ changes for different parameter settings. In particular, we illustrate how the marginal distribution of a single component q_{ij}^{sat} is affected by the choice of hyperparameters (Display a), and how the joint distribution of q_{ij}^{sat} and $q_{i\ell}^{\text{sat}}$ changes based on the correlation between their unconstrained counterparts $\tilde{q}_{ij}^{\text{sat}}$ and $\tilde{q}_{i\ell}^{\text{sat}}$ (Display b). Note that we have here assumed a total of



(a) Marginal distribution of a single component q_{ij}^{sat} , whose corresponding unconstrained weight $\tilde{q}_{ij}^{\text{sat}}$ has mean μ_{ij} and standard deviation σ_{ij} .



(b) Joint distribution of 2 components q_{ij}^{sat} and q_{il}^{sat} , where the marginal distribution of each component corresponds to the cyan density in (Display a). Each contour corresponds to the (approximate) boundary of a Highest Density Region (HDR), see for instance Hyndman (1996), and the correlation between the unconstrained counterparts $\tilde{q}_{ij}^{\text{sat}}$ and $\tilde{q}_{il}^{\text{sat}}$ are denoted ρ .

Figure 3.3.: Illustration of the prior density on the weights $\mathbf{q}^{\text{sat}}$ for different parameter settings.

3 migration pathways emanating from the i th reservoir when evaluating the prior distributions in Figure 3.3, which technically means that we are considering pathways emanating from the $(k - 3)$ rd reservoir¹. This is important to be aware of, as the transformation in Eq. (3.1) does not factor over individual components. When marginalising over the joint distribution, the resulting expression therefore depends on the components of the hyperparameters corresponding to every unconstrained weight in Eq. (3.1), see Appendix A for additional details.

For the marginal distributions in Figure 3.3a, we assumed that the 3 unconstrained weights are uncorrelated with equal standard deviation (denoted σ_{ij} in the figure). We also fixed the means of the other 2 unconstrained weights at 0, only varying the mean of the weight whose marginal distribution is shown in the figure, i.e. q_{ij}^{sat} . This is done following the translational invariance of the soft-max function, in particular we have that for any shift

¹For a saturated system with migration pathways to every reservoir of higher topological order, the only reservoir with 3 emanating pathways is reservoir $k - 3$.

$s \in \mathbb{R}$,

$$q_{ij}^{\text{sat}} = \frac{\exp \tilde{q}_{ij}^{\text{sat}}}{\sum_{\ell=i+1}^k \exp \tilde{q}_{i\ell}^{\text{sat}}} = \left(\frac{\exp s}{\exp s} \right) \frac{\exp \tilde{q}_{ij}^{\text{sat}}}{\sum_{\ell=i+1}^k \exp \tilde{q}_{i\ell}^{\text{sat}}} = \frac{\exp (\tilde{q}_{ij}^{\text{sat}} + s)}{\sum_{\ell=i+1}^k \exp (\tilde{q}_{i\ell}^{\text{sat}} + s)}.$$

Hence, if we were to assume equal means as well, we would not observe any change in the marginal prior distribution of q_{ij}^{sat} . By increasing the unconstrained mean of q_{ij}^{sat} (denoted μ_{ij} in the figure), whilst keeping the other unconstrained means fixed, we notice from the figure that the mode of the marginal distribution shifts to the right. If on the other hand, we increase the common standard deviation of the unconstrained weights, the spread of the distribution increases, consistent with a higher prior uncertainty on the relative migration rates of our model.

For the joint distribution in Figure 3.3b, we assumed both a common standard deviation of 0.5 and a common mean² for all 3 unconstrained weights, choosing to vary only the pairwise correlation between the unconstrained weights (denoted ρ in the figure). Similar to Figure 3.2, we notice that by increasing the pairwise correlation, the joint distribution concentrates in regions where the weights are similarly valued. It follows from Eq. (3.1) that the 3rd weight is deterministic when we condition on the other weights, in particular, we have that the 3rd weight is given by the expression $1 - (q_{ij}^{\text{sat}} + q_{il}^{\text{sat}})$. When we increase the pairwise correlation between the weights, the prior density therefore concentrates in the region around the point $(1/3, 1/3)$ in the figure. Notice also that the HDRs are all subsets of $\{q_{ij}^{\text{sat}}, q_{il}^{\text{sat}} \in [0, 1] \subset \mathbb{R} : q_{ij}^{\text{sat}} + q_{il}^{\text{sat}} \leq 1\}$. Mathematically, this set is referred to as the 2-dimensional simplex on $\mathbb{R}^3$, see for instance Hatcher (2002, ch. 0), and we use Δ^{n-1} to denote the general $(n-1)$ -dimensional simplex on $\mathbb{R}^n$, i.e.

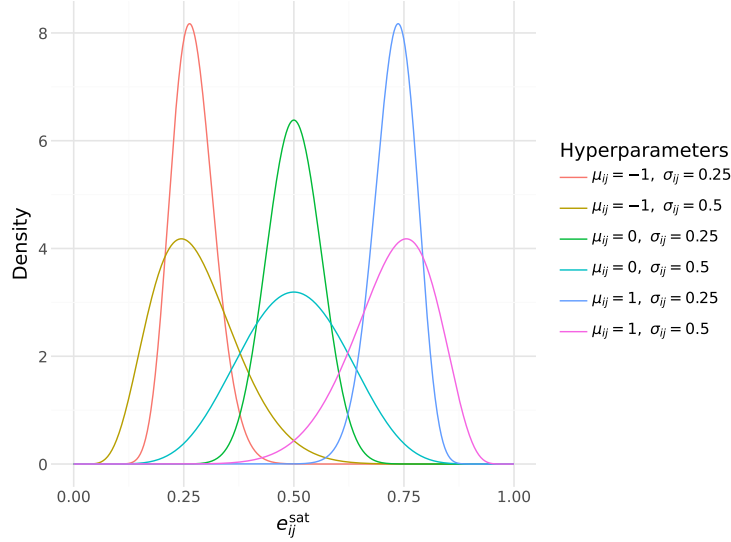
$$\Delta^{n-1} = \left\{ (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1 \wedge x_i \geq 0 \text{ for } i = 1, 2, \dots, n \right\}.$$

This illustrates the main reason as to why we chose to consider 3 weights when evaluating the prior distribution in Figure 3.3, i.e. the weights corresponding to the pathways emanating from the $(k-3)$ rd reservoir of the saturated system. If we had chosen to consider only 2 weights (similar to what we did in Figure 3.2), then the sample space of the joint distribution in Figure 3.3b would be a 1-dimensional simplex, i.e. the line $q_{ij}^{\text{sat}} + q_{il}^{\text{sat}} = 1$, and so the figure would not be as interesting. Note also that in the case of a single migration pathway emanating from the i th reservoir, which is the case for the $(k-1)$ st reservoir, the sample space of its corresponding relative flow rate would equal $\Delta^0 = \{1\}$. The marginal prior distribution of the relative flow rate therefore becomes degenerate at $q_{ij}^{\text{sat}} = 1$, which is expected following our assumption on conservation of mass.

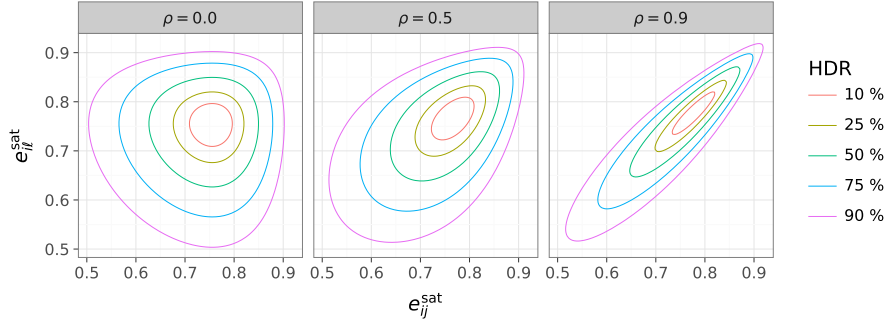
To obtain a transformation of the unconstrained eCTPs $\tilde{e}^{\text{sat}}$ that respects the constraints of Section 2.3.3, we note per the definition of $e_{i,j} \in E(G)$, that $e_{i,j} \in [0, 1] \subset \mathbb{R}$. Hence, we only need the transformation of $\tilde{e}^{\text{sat}}$ to map each of its elements from $\mathbb{R}$ onto $[0, 1] \subset \mathbb{R}$, as the elements of e^{sat} must all be in the range between 0 and 1. Recall that we have already introduced such a transformation earlier, that being the logistic transformation in Eq. (2.13), where $(P_i(t) - e_{i,j}C_i)/\tau^{\text{act}}$ is mapped onto $[0, 1] \subset \mathbb{R}$. Using the same transformation on the unconstrained eCTPs, we get that $\tilde{e}^{\text{sat}} \mapsto e^{\text{sat}}$ through the relation

$$e^{\text{sat}} = \left[1 + \exp(-\tilde{e}^{\text{sat}}) \right]^{-1}. \quad (3.2)$$

²Due to translational invariance, the exact value of the common mean does not affect the joint distribution. That being said, a common mean of 0 was assumed in this case.



(a) Marginal distribution of a single component e_{ij}^{sat} , whose corresponding unconstrained weight $\tilde{e}_{ij}^{\text{sat}}$ has mean μ_{ij} and standard deviation σ_{ij} .



(b) Joint distribution of 2 components e_{ij}^{sat} and e_{il}^{sat} , where the marginal distribution of each component corresponds to the magenta density in (Display a). Each contour corresponds to the (approximate) boundary of a Highest Density Region (HDR), see for instance Hyndman (1996), and the correlation between the unconstrained counterparts $\tilde{e}_{ij}^{\text{sat}}$ and $\tilde{e}_{il}^{\text{sat}}$ are denoted ρ .

Figure 3.4.: Illustration of the prior density on the weights $\mathbf{e}^{\text{sat}}$ for different parameter settings.

This transformation has many use cases within both computer science and statistics. In the case of the former, the transformation might be used to normalise the output of a neural network used for binary classification, such that the output corresponds to a probability mass on the label of the final layer (Martínez-Ramón et al. 2024, ch. 1.5.2.1). Within statistics, we might use the same transformation as the response function of a binary (logistic) regression model, connecting the model’s linear predictor to the expectation of the binary outcome (Fahrmeir et al. 2022, ch. 2.3).

Similar to Figures 3.2 and 3.3, we illustrate in Figure 3.4 how the prior distribution on $\mathbf{e}^{\text{sat}}$ changes for different parameter settings. As Eq. (3.2) factorises over individual components, the marginal distribution of a single component (Display a) is affected only by the choice of hyperparameters corresponding to the unconstrained weight $\tilde{e}_{ij}^{\text{sat}}$. In the figure, we denote by μ_{ij} and σ_{ij} the mean and standard deviation associated with this particular unconstrained weight. Note that by increasing the value of μ_{ij} , the mode of the marginal prior distribution on e_{ij}^{sat} is shifted to the right. Moreover, if we decrease

the value of μ_{ij} by a similar amount, the marginal prior distribution gets mirrored around $e_{ij}^{\text{sat}} = 1/2$. This is a result of the rotational symmetry that is inherent to the logistic transformation. If we for the moment denote by σ the logistic transformation in Eq. (3.2), we have that $\sigma(\tilde{e}_{ij}^{\text{sat}}) = 1 - \sigma(-\tilde{e}_{ij}^{\text{sat}})$, and so the transformation is rotationally symmetric around the point $(0, \sigma(0)) = (0, 1/2)$, as seen in Figure 2.10. If we instead increase the value of σ_{ij} , we notice that the spread of the distribution increases. As for the joint prior of e_{ij}^{sat} and e_{il}^{sat} in (Display b), we notice that the results are similar to the ones in Figures 3.2b and 3.3b. In particular, we notice that by increasing the correlation of the unconstrained weights $\tilde{e}_{ij}^{\text{sat}}$ and $\tilde{e}_{il}^{\text{sat}}$, the joint prior distribution on the corresponding weights gets concentrated in regions where e_{ij}^{sat} and e_{il}^{sat} are similarly valued. Note also that the transformation restricts the support of the distribution to the unit interval, or unit cube in case of the joint distribution in (Display b), similarly to how the transformation in Eq. (3.1) restricts the support of the prior on $\mathbf{q}^{\text{sat}}$ to simplices, whose dimension depends on the number of weights under consideration.

Sampling of the Prior Distribution

In the sections that follow, we often refer to the continuous parameters as a whole, and so to simplify our notation, we use $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}, \mathbf{C}, \mathbf{q}^{\text{sat}}, \mathbf{e}^{\text{sat}})$ to denote the vector of continuous parameters. The sample space of $\boldsymbol{\theta}$, which we denote by $\Omega_{\boldsymbol{\theta}}$, is then given by the following Cartesian product,

$$\Omega_{\boldsymbol{\theta}} = [0, \infty)^{3k} \times (\Delta^{k-2} \times \Delta^{k-3} \times \dots \times \Delta^0) \times [0, 1]^{\mathcal{T}_{k-1}} \subset \mathbb{R}^{k(k+2)},$$

where we use $\mathcal{T}_n = \sum_{i=1}^n i = n(n+1)/2$ to denote the n th triangular number (Beldon and Gardiner 2002), and where Δ^{n-1} denotes the $(n-1)$ -dimensional simplex on $\mathbb{R}^n$, as previously defined. Similar to the earlier parameters, we use $\tilde{\boldsymbol{\theta}} = (\tilde{\boldsymbol{\alpha}}, \tilde{\boldsymbol{\beta}}, \tilde{\mathbf{C}}, \tilde{\mathbf{q}}^{\text{sat}}, \tilde{\mathbf{e}}^{\text{sat}})$ to denote the unconstrained vector of continuous parameters, whose sample space is given by $\Omega_{\tilde{\boldsymbol{\theta}}} = \mathbb{R}^{k(k+2)}$.

Algorithm 3.1 then provides an overview of the process used to sample $\boldsymbol{\theta}$ from its prior distribution, denoted $p(\boldsymbol{\theta})$, by first sampling $\tilde{\boldsymbol{\theta}}$ from the corresponding unconstrained prior $p(\tilde{\boldsymbol{\theta}})$. Generally, we use $p(\cdot)$ to denote the probability density function (pdf) of its argument within the context of the thesis, or equivalently, the probability mass function (pmf) in cases where the argument is discrete. We have here made the assumption of mutual independence between the different parameter vectors, i.e. $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\mathbf{C}$, $\mathbf{q}^{\text{sat}}$ and $\mathbf{e}^{\text{sat}}$, noting that the elements of each vector might still be correlated with other elements of the same vector. The assumption of independence between different parameter vectors are made following Dutta et al. (2026) and Eidsvik et al. (to appear), who themselves made similar assumptions when developing their own models. Note however, that we are easily able to modify $p(\boldsymbol{\theta})$ in the future, if the assumption of independence were to fail. We can for instance replace the 1st procedure in Algorithm 3.1 with a procedure that samples $\tilde{\boldsymbol{\theta}}$ from a joint normal that does not factorise over individual component vectors. This illustrates one of the benefits of modelling the uncertainty in $\mathbf{q}^{\text{sat}}$ and $\mathbf{e}^{\text{sat}}$ through normal distributions and subsequent transformations, in that it becomes easy to account for any correlation between the parameter vectors. If we were to follow the outline proposed by Grødeland (2026), and instead model the uncertainty in terms of a Dirichlet distribution on $\mathbf{q}^{\text{sat}}$ and a Beta distribution on $\mathbf{e}^{\text{sat}}$, this would be noticeably harder.

Algorithm 3.1 Sample $\theta \sim p(\theta)$

Require: $\mu_{\tilde{v}}$ and $\Sigma_{\tilde{v}}$ for all $\tilde{v} \in \{\tilde{\alpha}, \tilde{\beta}, \tilde{C}, \tilde{q}^{\text{sat}}, \tilde{e}^{\text{sat}}\}$ **Ensure:** $\theta \sim p(\theta)$

```
1: procedure SAMPLEUNCONSTRAINED ▷ Sample  $\tilde{\theta} \sim p(\tilde{\theta})$ .
2:   for all  $\tilde{v} \in \{\tilde{\alpha}, \tilde{\beta}, \tilde{C}, \tilde{q}^{\text{sat}}, \tilde{e}^{\text{sat}}\}$  do
3:     sample  $\tilde{v} \sim \mathcal{N}(\mu_{\tilde{v}}, \Sigma_{\tilde{v}})$  ▷ Sample each component vector  $\tilde{v}$ .
4:   end for
5: end procedure

6: procedure TRANSFORMSAMPLE ▷ Obtain  $\theta$  through transformation of  $\tilde{\theta}$ .
7:   SAMPLEUNCONSTRAINED
8:   for all  $\tilde{v} \in \{\tilde{\alpha}, \tilde{\beta}, \tilde{C}\}$  do
9:      $\tilde{v} \leftarrow \ln \tilde{v}$ 
10:  end for
11:   $q^{\text{sat}} \xleftarrow{(3.1)} \tilde{q}^{\text{sat}}$ 
12:   $e^{\text{sat}} \xleftarrow{(3.2)} \tilde{e}^{\text{sat}}$ 
13: end procedure
```

3.1.2. Prior on Discrete Parameters

We have thus far considered the migration pathways of our system to be fixed, in the same manner as the continuous parameters of the previous sections. In reality, the migration pathways, which we model using a strict connected DAG G , is latent and need to be inferred from the migrating CO₂. Taking a look at the various sketches of the stratigraphic systems presented thus far, e.g. Figures 2.9a and 2.11, one might be led to believe that the process of determining where CO₂ migrates once a reservoir is compromised is trivial. In reality, modern acquisition procedures often result in images more similar to Figures 2.4 and 3.1, where we are able to detect the accumulation of CO₂ within each reservoir, but where the pathways of the migrating CO₂ is impossible to distinguish from the background. As we loosen the Markovian assumption held by previous researchers, including Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear), we introduce an uncertainty on the migration pathways of our system.

Constructing a Prior on G

We model this by assigning a prior distribution to G , assuming that the number of vertices $\nu(G)$ in G is fixed. In practice, nothing is stopping us from including the vertices in the prior as well, thus making the number of reservoirs in our system, denoted by k , stochastic. We choose not to do this however, for a couple of reasons. Firstly, obtaining an accurate estimate of the number of reservoirs in the system can often be done using a single repeat measurement, as seen in Figure 3.1b in the case of the Sleipner field. This essentially collapses the probability distribution around the estimate, and so we gain very little by allowing our framework to account for the uncertainty in the number of reservoirs k . Secondly, the sample space of G , which we denote by Ω_G , grows rapidly even for small changes in k , as we demonstrate later. When making inference based on a subset of Ω_G , we might therefore need to limit the value of k , so as to ensure that the subset is sufficiently representative for Ω_G . If we fail to do so, our results might be biased and we likely run

the risk of severely underestimating the true uncertainty associated with G . We also note that both Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear) chose to fix the value of k in their models.

Our prior on G therefore reduces to a prior on the arcs only. In particular, it has to account for the uncertainty in both the number of arcs, represented by $\varepsilon(G)$, and the location of each arc. As G is a strict connected DAG with vertices representing reservoirs and arcs representing migration pathways, following its construction in Section 2.2.2, we know that the number of arcs joining the i th vertex to vertices of higher topological order must be between 1 and $k - i$. Moreover, as the migration pattern of a particular stratigraphic system likely depends on the amount of pre-existing fractures within the cap formations, as mentioned in Sections 2.1.2 and 2.3.3, our prior distribution should reflect this by accounting for the condition of the cap formations in the system.

We choose therefore to model the number of migration pathways emanating from the i th reservoir, or equivalently, the number of arcs having the i th vertex as its tail, using a truncated Poisson distribution. Denoting this number by κ_i , we have that

$$\kappa_i \sim \text{Poisson}(\lambda_i) \mid 1 \leq \kappa_i \leq k - i, \quad i = 1, 2, \dots, k - 2, \quad (3.3)$$

where $\lambda_i \in (0, \infty) \subset \mathbb{R}$ is a sparsity parameter tied to the condition of the i th cap formation ψ_i . By increasing the value of λ_i we are essentially increasing the amount of pre-existing fractures within ψ_i , and so we should expect an increase in the number of migration pathways emanating from the i th reservoir. We have in Appendix A included a derivation of the expectation of κ_i , denoted $\mathbb{E}[\kappa_i]$, that shows that this is indeed the case. In particular, we have that

$$\mathbb{E}[\kappa_i] = \lambda_i \left(\sum_{\ell=0}^{k-i-1} \frac{\lambda_i^\ell}{\ell!} \right) / \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right), \quad i = 1, 2, \dots, k - 2,$$

which is strictly increasing in λ_i . Moreover, in the limit as $\lambda_i \rightarrow 0$ from above and $\lambda_i \rightarrow \infty$, $\mathbb{E}[\kappa_i] \rightarrow 1$ and $\mathbb{E}[\kappa_i] \rightarrow k - i$, respectively. In other words, if ψ_i is essentially void of any pre-existing fractures, we only expect a single migration pathway to emanate from the i th reservoir. Similarly, if ψ_i comprises an extensive network of pre-existing fractures, then we expect to have migration pathways to all reservoirs of higher topological order, of which there are $k - i$ in total.

Note in particular that we only consider κ_i for $i \leq k - 2$ to be stochastic. This follows for the structure of G , being a connected DAG. Specifically, there will always be a migration pathway between the 2 topmost reservoirs, and since this is the only pathway admissible for these 2 reservoirs, we have that $\kappa_{k-1} = 1$ and $\kappa_k = 0$. We can then represent the uncertainty in the number of pathways in total, by summation over the indices of $\boldsymbol{\kappa}$. In particular, we have that

$$\varepsilon(G) = \sum_{i=1}^k \kappa_i = 1 + \sum_{i=1}^{k-2} \kappa_i.$$

Note that by assigning a truncated Poisson distribution to the number of pathways emanating from the i th reservoir, we can also quite easily account for the uncertainty in their location, through a hierarchical prior on the reservoirs of higher topological order. In particular, if we condition on κ_i , then we can represent the uncertainty through a probability distribution on the subsets of size κ_i of unique indices in $\{i + 1, i + 2, \dots, k\}$, representing the reservoirs to which the κ_i migration pathways point. In other words, if we know how

many migration pathways are emanating from the i th reservoir, we only need to determine which of the $k - i$ reservoirs of higher topological order the pathways point to. For a single pathway to bypass multiple reservoirs, we would need subsequent cap formations to exhibit large networks of pre-existing fractures. Even in the cases where this is true, as for the Sleipner field, previous studies have only indicated the presence of a single large chimney bypassing multiple reservoirs (Callioli Santi et al. 2025; Vinje et al. 2025; Williams and Chadwick 2021). The other chimneys only bypass a single reservoir each, and so we should expect our framework to favour shorter migration pathways, bypassing at most a few reservoirs.

We allow the prior on G to account for this by associating each valid pathway with a weight that decreases exponentially, as a function of the distance between the reservoirs it connects. Specifically, using a distance penalty parameter $\eta_i \in (0, \infty) \subset \mathbb{R}$, we associate the migration pathway between the i th and j th reservoirs with a weight of the form

$$w_{i,j} = \exp \left[-\eta_i(j - i) \right], \quad j = i + 1, i + 2, \dots, k,$$

where the 2nd index has to be strictly larger than the 1st for the pathway to be valid, see Section 2.2.2. Note that by varying the penalty parameter η_i , we can control how much we penalise migration pathways that bypass multiple reservoirs. If for instance $\eta_i \rightarrow 0$ from above, then $w_{i,j} \rightarrow 1$ for all j , and so we do not introduce any penalty at all. If $\eta_i \gg 0$ on the other hand, then the weights corresponding to longer migration pathways are heavily penalised, resulting in $w_{i,k} \ll w_{i,k-1} \ll \dots \ll w_{i,i+1}$.

By conditioning on κ_i , we can then use these weights to create a probability distribution on the indices of the reservoirs to which the migration pathways point. In particular, if we let $\mathfrak{J}_i(\kappa_i)$ denote the collection of all possible such index sets for the i th reservoir, i.e.

$$\mathfrak{J}_i(\kappa_i) = \{ \mathcal{I} \subseteq \{i + 1, i + 2, \dots, k\} : |\mathcal{I}| = \kappa_i \}, \quad (3.4)$$

we can to any one set $\mathcal{J}_i \in \mathfrak{J}_i(\kappa_i)$ associate a probability that is proportional to the product of the weights of the corresponding pathways. In other words, we have that

$$p(\mathcal{J}_i \mid \kappa_i) \propto \prod_{j \in \mathcal{J}_i} w_{i,j}, \quad i = 1, 2, \dots, k - 2.$$

Note in particular that as the weights are exponential in the distance between reservoirs, index sets corresponding to short migration pathways receive a higher prior probability compared to sets corresponding to longer pathways.

The normalising constant of this pmf is then given by a sum over all sets in $\mathfrak{J}_i(\kappa_i)$, i.e.

$$e_{\kappa_i}(w_{i,i+1}, w_{i,i+2}, \dots, w_{i,k}) = \sum_{\mathcal{I} \in \mathfrak{J}_i(\kappa_i)} \prod_{j \in \mathcal{I}} w_{i,j}, \quad i = 1, 2, \dots, k - 2,$$

where, in the field of algebraic combinatorics, e_{κ_i} is usually referred to as the elementary symmetric polynomial (ESP) of degree κ_i in the weights $w_{i,i+1}, w_{i,i+2}, \dots, w_{i,k}$ (Alatawi 2022). We mention this, as ESPs satisfy a recursive relation of the form

$$e_\ell^{(n)} = e_\ell^{(n-1)} + w_{i,i+n} e_{\ell-1}^{(n-1)}, \quad (3.5)$$

where $e_0^{(n)} = 1$, $e_\ell^{(n)} = 0$ for $\ell > n$ or $\ell < 0$, and where $e_\ell^{(n)} = e_\ell(w_{i,i+1}, w_{i,i+2}, \dots, w_{i,i+n})$ denotes the ESP of ℓ th degree where we only consider the first n weights, i.e.

$$e_\ell^{(n)} = \sum_{\substack{\mathcal{I} \subseteq \{i+1, i+2, \dots, i+n\} \\ |\mathcal{I}| = \ell}} \prod_{j \in \mathcal{I}} w_{i,j}.$$

Using this relation, it is therefore possible to compute the normalising constant explicitly, and so we get that for $i = 1, 2, \dots, k - 2$,

$$p(\mathcal{J}_i \mid \kappa_i) = \frac{\prod_{j \in \mathcal{J}_i} w_{i,j}}{e_{\kappa_i}^{(k-i)}}, \quad \mathcal{J}_i \in \mathfrak{J}_i(\kappa_i). \quad (3.6)$$

We can then construct a prior on G by making the additional assumption of mutual independence between the migration pathways emanating from different reservoirs. In particular, we have that

$$p(G) = \prod_{i=1}^{k-2} p(\mathcal{J}_i \mid \kappa_i) p(\kappa_i), \quad G \in \Omega_G, \quad (3.7)$$

where we once again note that there will always be a migration pathway between the 2 topmost reservoirs in G , following its construction in Section 2.2.2. We therefore only need to consider reservoirs whose index is less than or equal to $k - 2$ when computing the above product.

It is worth mentioning that the assumption of mutual independence between pathways that emanate from different reservoirs has the potential to be rather unrealistic. To see why this is the case, we can for instance imagine a situation in which $\kappa_1 = k - 1$, i.e. where there exists a migration pathway from the bottommost reservoir to all other reservoirs within the stratigraphic system. For this to be possible, an extensive network of fractures is likely present within every cap formation. Hence, one would expect the other reservoirs to also have multiple migration pathways emanating from them. Using the above formulation, we neglect to account for this when evaluating the prior probability of G .

It is however relatively easy to account for this, by slightly modifying the pmfs of Eq. (3.7). In particular, if we were to introduce a (joint) prior on the hyper-parameters $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$, denoted by $p(\boldsymbol{\lambda}, \boldsymbol{\eta})$, we could assume that both the number of migration pathways within the system and their locations, as represented by $\boldsymbol{\kappa}$ and $\boldsymbol{\mathcal{J}}$, are conditionally independent given $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$. To obtain a prior on G , we would then marginalise over $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$, i.e. by computing

$$p(G) = \int \left[\prod_{i=1}^{k-2} p(\mathcal{J}_i \mid \kappa_i, \eta_i) p(\kappa_i \mid \lambda_i) \right] p(\boldsymbol{\lambda}, \boldsymbol{\eta}) \, d\boldsymbol{\lambda} \, d\boldsymbol{\eta}$$

where the integral is taken over the (joint) sample space of $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$. To ensure a more realistic prior, we would define $p(\boldsymbol{\lambda}, \boldsymbol{\eta})$ in such a way that large values of $\boldsymbol{\lambda}$ and small values of $\boldsymbol{\eta}$ have high prior probability in systems where the cap formations are in poor condition, e.g. by exhibiting large interconnected fracture networks. By ensuring that the components of $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$ are highly correlated, we obtain a more realistic prior on G . In particular, for stratigraphic systems with poor cap formations, there would be a high prior probability of having multiple migration pathways per reservoir. In addition, the penalty associated with longer pathways would be reduced, and so systems where the migration pathways bypass multiple reservoirs would be assigned a higher prior probability.

For the purposes of this thesis, we are not going to introduce such a prior on the hyper-parameters however, as it makes the prior on G unnecessarily complicated. We instead assume a prior of the form given in Eq. (3.7), which easily factorises over each reservoir and where we can use the (fixed) hyper-parameters $\boldsymbol{\lambda}$ and $\boldsymbol{\eta}$ to control how the prior behaves. In particular, by controlling the degree to which we favour sparse systems, i.e. systems in which there are few migration pathways emanating from each reservoir, and the degree to which we want to penalise migration pathways that bypass multiple reservoirs.

Efficient Sampling of the Prior Distribution

To sample a DAG G from its prior distribution, as defined in Eq. (3.7), we first sample κ_i , the number of migration pathways emanating from the i th reservoir, from its prior distribution, as defined in Eq. (3.3). Specifically, we evaluate the probability of each element in the sample space $\Omega_{\kappa_i} = \{1, 2, \dots, k-i\}$ and then sample κ_i from the resulting categorical distribution. When deriving the expectation of κ_i in Appendix A, we specifically found that

$$p(\kappa_i) = \frac{\lambda_i^{\kappa_i}}{\kappa_i!} \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right), \quad \kappa_i \in \Omega_{\kappa_i},$$

and so this is not problematic.

If we used a similar naive approach to obtain $\mathcal{J}_i \in \mathfrak{J}_i(\kappa_i)$, the indices to which the migration pathways emanating from the i th reservoir point, we would likely run into a problem however, especially for larger values of k . This follows from the fact that we would have to compute the probability of every set in $\mathfrak{J}_i(\kappa_i)$ to obtain the categorical prior we are after. Computing the probabilities themselves is not problematic, as we can use Eq. (3.6) for that purpose, the problem first arises when we consider just how many probabilities we actually have to compute. From its definition in Eq. (3.4), we see that each set in $\mathfrak{J}_i(\kappa_i)$ is essentially a way in which κ_i indices can be chosen from amongst $k-i$ indices. The number of sets in $\mathfrak{J}_i(\kappa_i)$ is therefore equal to the binomial coefficient

$$\binom{k-i}{\kappa_i}.$$

If we consider for instance a system comprising $k = 20$ reservoirs, and are interested in sampling the reservoirs to which the pathways of the 1st reservoir point, knowing that we have $\kappa_1 = 9$ pathways in total, we would need to compute over 92,000 probabilities to construct the associated categorical distribution. Even for a more modest stratigraphic system like the one at the Sleipner field, where we have $k = 9$ reservoirs in total, we would potentially need to evaluate up to 70 probabilities to construct the categorical distribution associated with the migration pathways of a single reservoir³. This is computationally demanding, especially for larger values of k , and so we should ideally use a different approach to sample $\mathcal{J}_i$ from its prior distribution.

To understand how this can be accomplished, we first need to take a closer look at the ESPs that appear in Eq. (3.5). In particular, we need to understand what they represent in terms of probabilities on the index sets of $\mathfrak{J}_i(\kappa_i)$. As $e_{\kappa_i}^{(k-i)} = e_{\kappa_i}(w_{i,i+1}, w_{i,i+2}, \dots, w_{i,k})$ represents the normalising constant of the prior probability associated with index sets in $\mathfrak{J}_i(\kappa_i)$, as seen in Eq. (3.6), it corresponds to the total unnormalised probability mass associated with these sets. In general, if we restrict the ESP of ℓ th degree to the first n weights only, the resulting expression $e_\ell^{(n)} = e_\ell(w_{i,i+1}, w_{i,i+2}, \dots, w_{i,i+n})$ corresponds to the total unnormalised probability mass associated with distributing ℓ migration pathways amongst the first n reservoirs of higher topological order. Using Eq. (3.5), we can then partition this probability mass according to whether reservoir $i+n$ is pointed to by one of the ℓ migration pathways. If this is the case, then we must include its corresponding weight $w_{i,i+n}$ in the partition and still distribute $\ell-1$ migration pathways amongst the remaining reservoirs. If not, then we have to distribute all ℓ migration pathways amongst the remaining $n-1$ reservoirs of higher topological order.

³If $\kappa_1 = 4$, we must consider $\binom{9-1}{4} = 70$ probabilities to sample the migration pathways emanating from the 1st reservoir.

The terms on the right hand side of Eq. (3.5) therefore correspond to 2 mutually exclusive and collectively exhaustive events, which we can use to compute the conditional probability of our system having a migration pathway between the i th and $(i+n)$ th reservoirs. In particular, we have that

$$\pi_{i,i+n}^{(\ell)} = \frac{w_{i,i+n} e_{\ell-1}^{(n-1)}}{e_{\ell}^{(n)}} = 1 - \frac{e_{\ell}^{(n-1)}}{e_{\ell}^{(n)}} \quad (3.8)$$

denotes the prior probability of our system having a migration pathway between the i th and $(i+n)$ th reservoirs, conditional on there being ℓ migration pathways left to distribute amongst the n reservoirs we currently consider, i.e. reservoirs $i+1, i+2, \dots, i+n$.

By conditioning on κ_i , we can therefore construct a sample $\mathcal{J}_i \in \mathfrak{J}_i(\kappa_i)$ from its conditional prior in Eq. (3.6), by considering sequentially whether each index $k, k-1, \dots, i+1$ is an element of $\mathcal{J}_i$. In other words, we can determine which reservoirs are pointed to by the κ_i pathways one after the other, instead of having to consider them all at once. Algorithm 3.2 provides an overview of this procedure, assuming we have already precomputed the necessary weights and ESPs, as outlined by the algorithm.

Note in particular that the algorithm only produces sets of size κ_i , even though the number of individual Bernoulli trials conducted as part of the procedure is equal to $k-i \geq \kappa_i$, corresponding to the number of reservoir of higher topological order. To see why this is the case, we note that as soon as we have included all κ_i indices in $\mathcal{J}_i$, the probability of including any additional indices drops to 0. This follows directly from Eq. (3.8), as $\pi_{i,i+n}^{(0)} \propto e_{-1}^{n-1} = 0$, following the base case $e_{\ell}^{(n)} = 0$ for all $\ell < 0$ (see Eq. (3.5)). Similarly, if we have to distribute a total of $\ell \leq \kappa_i$ pathways amongst the remaining ℓ reservoirs we consider, we have that $\pi_{i,i+\ell}^{(\ell)} = 1 - e_{\ell}^{(\ell-1)}/e_{\ell}^{(\ell)} = 1$, following the base case $e_{\ell}^{(n)} = 0$ for all $\ell > n$. As a result, we include $i+\ell \in \mathcal{J}_i$ almost surely, and so we have to distribute $\ell-1$ additional pathways amongst $\ell-1$ reservoirs. By induction, it then follows that the indices of the remaining reservoirs are all included in $\mathcal{J}_i$.

Algorithm 3.2 Sample $\mathcal{J}_i \sim p(\mathcal{J}_i \mid \kappa_i)$

Require: $k, i, \kappa_i, \{w_{i,i+n}\}$ and $\{e_{\ell}^{(n)}\}$ for $n = 1, 2, \dots, k-i$ and $\ell = 1, 2, \dots, \kappa_i$

Ensure: $\mathcal{J}_i \sim p(\mathcal{J}_i \mid \kappa_i)$

```

1:  $\mathcal{J}_i \leftarrow \emptyset$  ▷ Initialise sampled indices.
2:  $\ell \leftarrow \kappa_i$  ▷ Initialise number of remaining pathways to distribute.
3: for  $n = k-i, k-i-1, \dots, i+1$  do ▷ For each possible reservoir (index)...
4:    $\pi_{i,i+n}^{(\ell)} \stackrel{(3.8)}{\leftarrow} w_{i,i+n} e_{\ell-1}^{(n-1)} / e_{\ell}^{(n)}$  ▷ Compute probability of including the index in  $\mathcal{J}_i$ .
5:   sample  $u \sim \text{Uniform}(0, 1)$ 
6:   if  $u < \pi_{i,i+n}^{(\ell)}$  then ▷ If Bernoulli trial is a success...
7:      $\mathcal{J}_i \leftarrow \mathcal{J}_i \cup \{i+n\}$  ▷ Include the index in  $\mathcal{J}_i$ .
8:      $\ell \leftarrow \ell - 1$  ▷ Decrement number of remaining pathways to distribute.
9:   end if
10: end for
```

It might at first seem tempting to introduce conditional logic inside Algorithm 3.2 to return $\mathcal{J}_i$ immediately once all κ_i elements have been added, or alternatively, add the remaining indices in bulk, in cases where the number of such indices match the number of migration pathways still to be distributed. We choose not to do this however, as it would negatively effect computer implementations that vectorise over multiple samples. In particular, if multiple samples are to be drawn from the conditional prior $p(\mathcal{J}_i \mid \kappa_i)$,

an efficient computer implementation of Algorithm 3.2 would construct all samples simultaneously in terms of a large vector, instead of constructing each sample sequentially. If some of the samples suddenly were to terminate before the others, the computer implementation would have to account for this with additional logic. This in turn would likely be detrimental to the efficiency of the algorithm, and so we would gain very little by vectorising over the samples.

To obtain a sample from the prior on G in Eq. (3.7), we then repeat the sampling of κ_i and $\mathcal{J}_i$ for each of the $k - 2$ lowermost reservoirs, noting that we have to pre-compute all necessary weights and ESPs beforehand, as seen in Algorithm 3.3. It is then natural to question whether we have gained anything by sampling the indices to which the migration pathways point using Algorithm 3.2, as we still have to compute a number of weights and ESPs beforehand. It is however straight forward to illustrate the computational benefit of the algorithm. In particular, if we examine the procedure in Algorithm 3.3 that pre-computes the ESPs, we notice that we have to store a total of $(k - i)^2$ ESPs and an additional $k - i$ weights $\{w_{i,i+n}\}$ to construct a sample $\mathcal{J}_i \in \mathfrak{J}_i(\kappa_i)$, according to its conditional prior. The computational complexity of the algorithm is therefore quadratic in the number of higher order reservoirs that we have to consider, which we can express as $\mathcal{O}((k - i)^2)$ using asymptotic notation⁴. Moreover, when sampling from the conditional prior $p(\mathcal{J}_i \mid \kappa_i)$ multiple times, we can pre-compute the weights once for all the samples. The amortized complexity of Algorithm 3.2 is therefore equal to $\mathcal{O}(k - i)$, as we can average the cost of computing both the weights and ESPs over every sample. For the sake of clarity, we note that the linear complexity of the algorithm is a result of having to consider whether the index of each reservoir of higher topological order is an element of the sample $\mathcal{J}_i$, of which there are $k - i$ in total.

This is in stark contrast to the naive approach, in which we have to evaluate a total of $\binom{k-i}{\kappa_i}$ probabilities to obtain a single sample from $p(\mathcal{J}_i \mid \kappa_i)$. In terms of computational complexity, we prove in Appendix A that if κ_i is fixed, such an algorithm has complexity $\mathcal{O}((k - i)^{\kappa_i})$. That is, the naive approach has polynomial complexity given a fixed κ_i , and so it only becomes linear in the case where $\kappa_i = 1$, corresponding to the specific case in which we only have a single migration pathway to distribute. Generally however, unless λ_i is sufficiently small, κ_i will take on values in the whole range $[1, k - i] \in \mathbb{N}$. If we instead consider κ_i a function of $k - i$, the computational complexity of the naive approach increases significantly. If for instance $\kappa_i = \lfloor (k - i)/2 \rfloor$, where $\lfloor \cdot \rfloor$ denotes the floor operator, the complexity is now given by $\mathcal{O}(2^{k-i}/\sqrt{k-i})$, as seen in Appendix A. We choose to illustrate the complexity for this particular value, as the binomial coefficient attains its maximum here⁵. The computational complexity of the naive approach can in other words become exponential in $k - i$, unless λ_i is sufficiently small.

We have already seen this in practice in one of the previous examples, in which we considered a system comprising $k = 20$ reservoirs and had to distribute $\kappa_1 = \lfloor (20 - 1)/2 \rfloor = 9$ migration pathways emanating from the 1st reservoir. Using the naive approach, we would have had to compute over 92,000 probabilities to construct the required categorical prior. Using our alternative approach however, we only need to compute $(20 - 1)^2 = 361$ ESPs once, of which we evaluate a total of 19 each time we sample from the conditional prior $p(\mathcal{J}_1 \mid \kappa_1 = 9)$. The difference becomes less pronounced for more modest systems, but still, in the case of the Sleipner field where $k = 9$, we will at most have to pre-compute 64 ESPs corresponding to the 1st reservoir, of which no more than 8 have to be evalu-

⁴Often referred to as big O notation within computer science.

⁵ $\binom{n}{k}$ has a single maximum at $k = n/2 = \lfloor n/2 \rfloor = \lceil n/2 \rceil$ if n is even and 2 maximums at $k = \lfloor n/2 \rfloor$ and $k = \lceil n/2 \rceil$ if n is odd, where $\lceil \cdot \rceil$ denotes the ceiling operator.

Algorithm 3.3 Sample $G \sim p(G)$

Require: k, λ and η **Ensure:** $G \sim p(G)$

```
1: procedure PRECOMPUTEESP                                ▷ Pre-compute necessary weights and ESPs.
2:    $e_0^{(n)} \leftarrow 1$  for  $n = 1, 2, \dots, k - 1$           ▷ Initialise base cases, max number of path-
                                                                ways to consider is  $k - 1$ .
3:    $e_\ell^{(0)} \leftarrow 0$  for  $\ell = 1, 2, \dots, k - 1$       ▷ Initialise base cases, max number of path-
                                                                ways to consider is  $k - 1$ .
4:   for  $i = 1, 2, \dots, k - 2$  do                                ▷ For each reservoir...
5:     for  $n = 1, 2, \dots, k - i$  do                                ▷ Consider all possible pathways...
6:        $w_{i,i+n} \leftarrow \exp(-\eta_i \cdot n)$                 ▷ Compute its weight.
7:       for  $\ell = 1, 2, \dots, n$  do
8:          $e_\ell^{(n)} \leftarrow e_\ell^{(n-1)} + w_{i,i+n}e_{\ell-1}^{(n-1)}$     ▷ Compute ESPs using Eq. (3.5).
9:       end for
10:    end for
11:  end for
12: end procedure

13: procedure SAMPLEGRAPH                                    ▷ Sample  $G \sim p(G)$ .
14:   PRECOMPUTEESP
15:    $a_{ij} \leftarrow 0$  for  $i, j = 1, 2, \dots, k$           ▷ Initialise adjacency matrix with 0s.
16:   for  $i = 1, 2, \dots, k - 2$  do                                ▷ For each reservoir...
17:     sample  $\kappa_i \sim \text{Poisson}(\lambda_i) \mid 1 \leq \kappa_i \leq k - i$     ▷ Determine number of pathways.
18:     sample  $\mathcal{J}_i \sim p(\mathcal{J}_i \mid \kappa_i)$                 ▷ Use Algorithm 3.2.
19:     for all  $j \in \mathcal{J}_i$  do
20:        $a_{ij} \leftarrow 1$                                 ▷ Update adjacency matrix based on sampled indices.
21:     end for
22:   end for
23:    $a_{(k-1)k} \leftarrow 1$                                 ▷ Include topmost pathway, which is deterministic.
24: end procedure
```

ated per sample. With the naive approach, we would potentially be forced to evaluate 70 probabilities per sample, as previously mentioned.

On the Sample Space of G

In the beginning of this section, we denoted by Ω_G the sample space of G and argued that it grows rapidly in k . We later found that the cardinality of $\mathfrak{J}_i(\kappa_i)$ is equal to the number of ways in which κ_i migration pathways can be distributed amongst $k - i$ reservoirs of higher topological order, i.e. $\binom{k-i}{\kappa_i}$. As κ_i takes on values in the whole range $[1, k - i] \in \mathbb{N}$, the total number of DAGs generated by considering the i th reservoir only, becomes

$$\sum_{\ell=1}^{k-i} |\mathfrak{J}_i(\ell)| = \sum_{\ell=1}^{k-i} \binom{k-i}{\ell}.$$

To determine the cardinality of Ω_G , we must then consider the product of all the reservoirs whose migration pathways are stochastic, resulting in

$$|\Omega_G| = \begin{cases} \prod_{i=1}^{k-2} \sum_{\ell=1}^{k-i} \binom{k-i}{\ell}, & \text{if } k > 2. \\ 1, & \text{if } 1 \leq k \leq 2. \end{cases}$$

Note in particular, that if the number of reservoirs k is equal to either 1 or 2, then the migration pathways of the system are all deterministic. In the case of the former, we would have no migration pathways at all, and so the corresponding arc set would be $A(G) = \emptyset$. In the case of the latter, we would have a single migration pathway only, i.e. $A(G) = \{a_{1,2}\}$, and since it is the pathway joining the 2 topmost reservoirs, it is present within the DAG deterministically. Hence, we only consider a single DAG in both cases, and so $|\Omega_G| = 1$.

For higher values of k on the other hand, we notice that the cardinality of Ω_G is a product of sums of binomial coefficients. Using the binomial theorem, we have that

$$2^{k-i} = (1+1)^{k-i} = \sum_{\ell=0}^{k-i} \binom{k-i}{\ell} 1^{k-i-\ell} 1^\ell = 1 + \sum_{\ell=1}^{k-i} \binom{k-i}{\ell},$$

by factoring out the first coefficient at $\ell = 0$. We may therefore express the cardinality of Ω_G in terms of a simpler product, i.e.

$$|\Omega_G| = \prod_{\ell=2}^{k-1} (2^\ell - 1), \quad k = 3, 4, \dots,$$

where we made the change of variables $\ell = k - i$ and reversed the order in which we compute the product. Notice in particular that this is a product over $k - 2$ exponentials, and so it tends to increase extremely fast in k .

For reference, Grødeland (2026) considered a similar situation, in which DAGs were used to model the migration of CO₂ in multi-layered stratigraphic systems. By only allowing a single migration pathway per reservoir, i.e. $\kappa_i = 1$ for all $i = 1, 2, \dots, k-1$, the author only needed to consider $(k-1)!$ distinct DAGs. As mentioned by Grødeland (2026), this follows from the fact that each pathway can point to $k - i$ different reservoirs. When accounting for the total number of distinct DAGs, a product over each migration pathway, or rather the reservoirs where they emanate, is taken. Hence the corresponding sample space, which we here denote using $\Omega_{G'}$, has cardinality

$$|\Omega_{G'}| = \prod_{i=1}^{k-1} (k-i) = (k-1)!, \quad k = 1, 2, \dots$$

If we tabulate the cardinalities of Ω_G and $\Omega_{G'}$ for the first few values of k , as seen in Table 3.1, the rapid rate at which our sample space Ω_G grows as a function of k becomes apparent. For systems with as few as $k = 4$ reservoirs, we already notice that we have to consider 3.5 times the number of graphs considered by Grødeland (2026). In Figure 3.5, this is illustrated by showing the diagram of every graph in $\Omega_{G'}$ for $k = 4$ (Display a), and the diagram of every additional graph present in Ω_G that is not present in $\Omega_{G'}$ (Display b). Notice also that for the system we once considered to be modest, i.e. the one corresponding to the system at the Sleipner field having $k = 9$, the number of possible graphs explodes being approximately equal to 20 billion. It is therefore not feasible to

k	1	2	3	4	5	6	7	8	9
$ \Omega_{G'} $	1	1	2	6	24	120	720	5,040	40,320
$ \Omega_G $	1	1	3	21	315	9,765	615,195	78,129,765	19,923,090,075

Table 3.1.: Cardinality of Ω_G and $\Omega_{G'}$ for stratigraphic systems comprising $k = 1, 2, \dots, 9$ reservoirs in total.

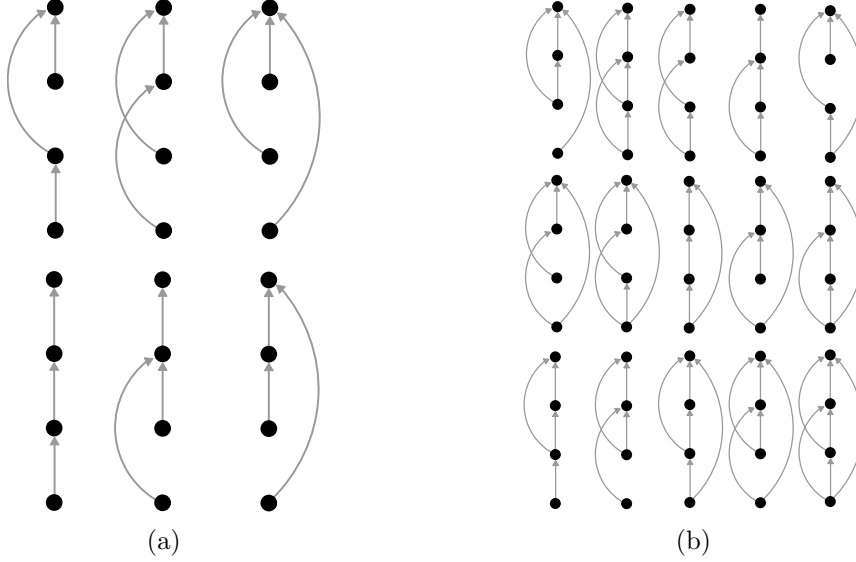


Figure 3.5.: Diagram of all DAGs in $\Omega_{G'}$ for a system comprising $k = 4$ reservoirs (Display a), and the elements of Ω_G that is not present in $\Omega_{G'}$ (Display b). The labels on the vertices and arcs of each DAG have been suppressed in the figure, so as to make it easier to interpret.

make inference on the whole sample space. One could argue that with a strong enough dedication and the budget to match it, it would technically be possible to make inference based on the whole sample space, even for $k = 9$. This quickly breaks down for larger values of k however, as it would be impossible even for modern supercomputers to evaluate each possible DAG in a reasonable amount of time.

For reference, we may consider the stratigraphic system comprising $k = 20$ reservoirs, which we previously used as an example in this section. For this particular system, the cardinality of Ω_G is approximately equal to $4.532 \cdot 10^{56}$, an astronomically large number that is difficult to wrap one's head around. If we for instance let each DAG be represented by a single water molecule, then it's possible to show, see Appendix A, that the resulting body of water (consisting of $4.532 \cdot 10^{56}$ molecules) would at standard density occupy a sphere whose diameter is 4.252 times the size of the diameter of the sun. Following recent advancements within computer science, some supercomputers have been able to compute on the order of 10^{18} floating point operations (FLOP) per second (Rashmi et al. 2024). Even if we could evaluate each DAG using a single FLOP, which is generally impossible, it would take today's most powerful supercomputers on the order of 10^{31} years to evaluate each DAG in a system comprising $k = 20$ reservoirs.⁶ Recent estimates on the age of the universe, indicates that the universe is approximately $13.8196 \cdot 10^9$ years old (Costa et al. 2023). In other words, if we started evaluating each possible DAG the instant the universe was created using today's most powerful supercomputers, assuming we could do so using

⁶ Assuming we can evaluate $\sim 10^{18}$ DAGs per second, it would take $\sim 4.532 \cdot 10^{38}$ seconds or $\sim 1.437 \cdot 10^{31}$ years to evaluate all $4.532 \cdot 10^{56}$ DAGs, as there are $4.0176 \cdot 10^7$ seconds in a year.

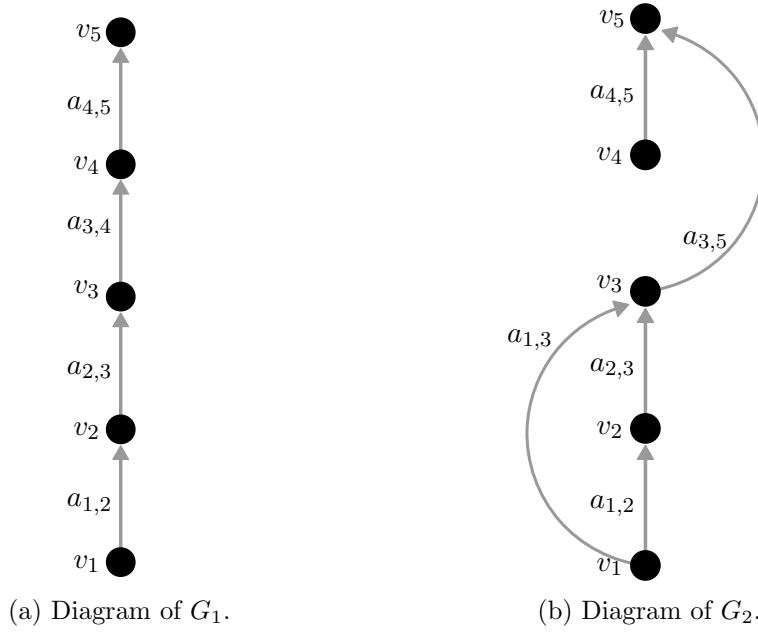


Figure 3.6.: Diagram of a Markovian DAG (Display a) and a non-Markovian DAG (Display b) used for uncertainty propagation. We use v_i , for $i = 1, 2, \dots, 5$, to denote vertices in the DAGs and label each arc according to the vertices it joins. In particular, $a_{i,j}$ denotes the arc joining the i th and j th vertices.

a single FLOP per DAG, we would still only have evaluated a tiny fraction $\sim 1/10^{21}$ of every possible DAG.

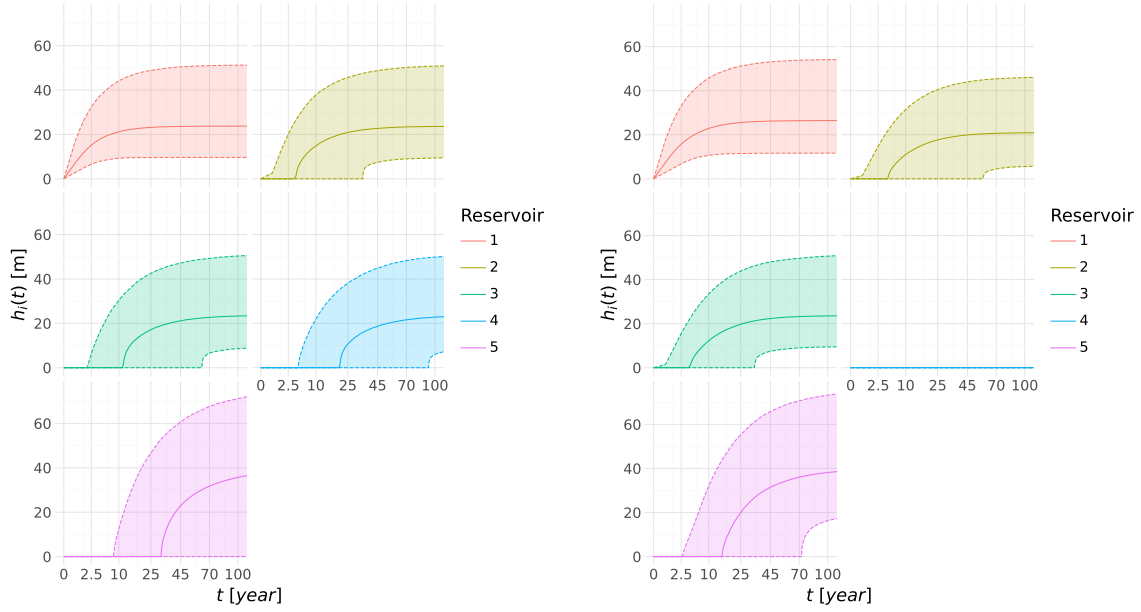
3.2. Propagation of Prior Uncertainty

Having assigned priors to both the continuous- and the discrete parameters θ and G in the preceding sections, it is only natural to question how the prior uncertainty affects the CO₂ plumes within the stratigraphic system at a particular time $t < T_k^{\text{mig}}$. Using the differentiable simulator $\mathcal{S}$ that we developed in Section 2.4, we can propagate the prior uncertainty in θ and G through the simulator to characterise the resulting uncertainty in the CO₂ plumes. Specifically, if we sample N realisations from both priors,

$$\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)} \sim p(\theta) \quad \text{and} \quad G^{(1)}, G^{(2)}, \dots, G^{(N)} \sim p(G),$$

and evaluate the simulator conditional on each realisation, i.e. $\mathbf{h}^{(\ell)}(t) = \mathcal{S}(t \mid \theta^{(\ell)}, G^{(\ell)})$ for $\ell = 1, 2, \dots, N$, we obtain N plume heights $\mathbf{h}^{(1)}(t), \mathbf{h}^{(2)}(t), \dots, \mathbf{h}^{(N)}(t)$ that are distributed according to the induced prior on $\mathbf{h}(t)$. Monte Carlo integration can then be used to approximate various quantities of interest.

In this section, we choose to fix the DAG G for each realisation of θ and subsequent propagation through the simulator $\mathcal{S}$, so as to isolate the prior uncertainty that is induced by the continuous parameters θ . To illustrate how the DAGs affect the development of the CO₂ plumes, we instead repeat the sampling for 2 distinct DAGs G_1 and G_2 , whose diagram is depicted in Figure 3.6. Note in particular, that G_1 is the DAG corresponding to a Markovian assumption on the migration of CO₂, whilst G_2 is the non-Markovian DAG corresponding to the stratigraphic system in Figure 2.9a. By conditioning on G_1 , it is possible to compare the results of the uncertainty propagation with those of previous



(a) Height distribution for the stratigraphic system corresponding to G_1 , the diagram of which is shown in Figure 3.6a. (b) Height distribution for the stratigraphic system corresponding to G_2 , the diagram of which is shown in Figure 3.6b.

Figure 3.7.: Empirical percentiles of the prior induced distribution on plume heights $\mathbf{h}(t)$ as a function of time t , for 2 stratigraphic systems comprising $k = 5$ reservoirs each. Solid lines correspond to the median (50th percentile) of $N = 75,000$ samples, with dotted lines indicating the 5th and 95th percentiles. The shaded regions therefore cover 90 % of the simulated plume heights $\mathbf{h}(t)$ at each time t . The parameter settings used to obtain the samples are listed in Appendix B.

studies, where the authors assumed the migration of CO_2 to be a Markovian process. Hence, the use of a differentiable simulator $\mathcal{S}$ can be substantiated, in addition to providing insight into the differences between Markovian and non-Markovian stratigraphic systems.

Figure 3.7 provides an illustration of the induced uncertainty on $\mathbf{h}(t)$ as a function of time, for the specific parameter settings shown in Appendix B. In particular, it shows the empirical median of $N = 75,000$ plume heights for both G_1 and G_2 as a function of time, along with the 5th and 95th empirical percentiles. We immediately notice that when conditioning on G_2 , the plume height within the 4th reservoir as a function of time $h_4(t) = 0$ for all $N = 75,000$ samples. This is expected when we only inject CO_2 in the bottommost reservoir, as there are no migration pathways pointing to the 4th reservoir in G_2 , see Figure 3.6b. Hence, there is no way for the migrating CO_2 to enter the 4th reservoir once the lower reservoirs are compromised. We also notice that the median time until $h_5(t) > 0$, i.e. the median entry time of the 5th reservoir T_5^{ent} or time until CO_2 starts to accumulate in the reservoir, is much higher for G_1 than for G_2 . In particular, the median entry time for the topmost reservoir conditional on G_1 is ~ 31 years, whilst the median entry time for same reservoir is only ~ 17 years when conditioning on G_2 , noting that the horizontal axis in the figure has been scaled in terms of the square root of its labels⁷. This is easily explained by the structure of the 2 DAGs. Specifically, when

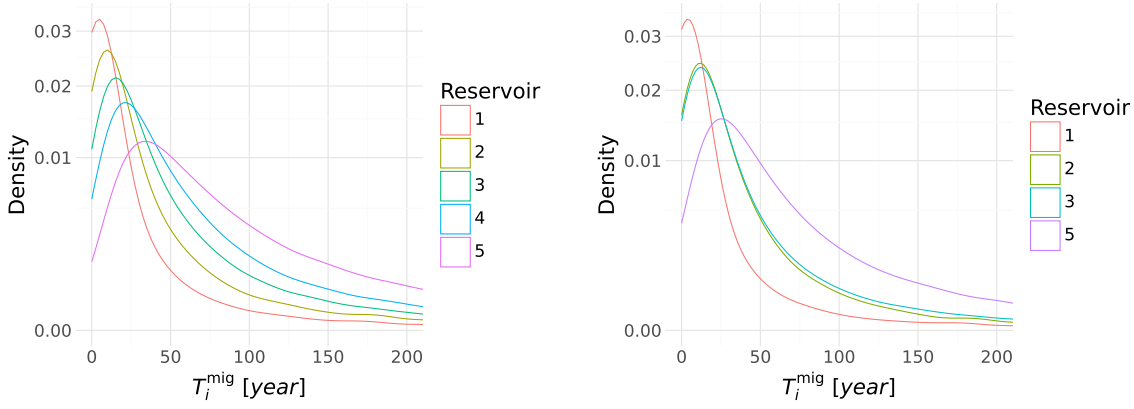
⁷Hence, if the median plume height within the topmost reservoir in Figure 3.7b is non-zero only after the approximate midpoint between 10 and 25 years, then $t = (\sqrt{10}/2 + \sqrt{25}/2)^2 = 16.65 \dots$ years. Similarly, if this holds only after 1/3 of the way between 25 and 45 years, as we see in Figure 3.7a, then $t = (2\sqrt{25}/3 + \sqrt{45}/3)^2 = 31.01 \dots$ years.

conditioning on G_2 , we could, depending on the sampled continuous parameters θ , end up in a situation where CO_2 only enters a single intermediate reservoir (reservoir 3) before it enters the topmost reservoir. When conditioning on G_1 however, this is not possible, as the CO_2 must enter reservoir $i - 1$ before it can enter the i th reservoir. As a result, each and every reservoir below the topmost reservoir must be compromised before CO_2 can migrate into it, and so the median entry times of reservoirs higher up in the strata increase.

This mirrors what we know from previous studies on the Sleipner field in particular, where the observed propagation of CO_2 is not compatible with cap formations that are completely sealing. As mentioned in Section 2.1.2, multiple studies, including Cavanagh and Haszeldine (2014), Vinje et al. (2025) and Williams and Chadwick (2021), argue that the observed migration of CO_2 at the Sleipner field can only be explained by the presence of chimney-like structures within the stratigraphic system. Figure 3.7 then illustrate an important result of the presence of such structures within a stratigraphic system, namely that shallow reservoirs, and the topmost reservoir in particular, have smaller median entry times compared with similar Markovian systems. We also note that both Dutta et al. (2026) and Eidsvik et al. (to appear) obtained similar results to those illustrated in Figure 3.7a, when plotting the plume heights $h(t)$ as a function of time t . By assuming a Markovian process on the migration of CO_2 , their models only predicted an increase in the plume height $h_i(t)$ for a particular reservoir i once $h_{i-1}(t)$ was sufficiently large, corresponding to the CTP needed to break through the cap formation of the $(i - 1)$ st reservoir.

Note also that for the 2nd reservoir, the width of the interval between the 5th and 95th percentiles differ significantly between the 2 DAGs, particularly when $h_2(t) \approx 0$. This suggests that the induced uncertainty in the entry times of the 2nd reservoir is higher when conditioning on the non-Markovian DAG G_2 , as opposed to the Markovian DAG G_1 . To understand why this is the case, we once again turn to the diagrams of Figure 3.6. Depending on the value of the continuous parameters θ , we might be in a situation where the pathway between the 1st and 3rd reservoirs activate prior to the pathway between the 1st and 2nd reservoirs. If this the case, some of the injected CO_2 will start to migrate to the 3rd reservoir, thus reducing the net rate of accumulation in the 1st reservoir $q_1^{\text{net}}(t)$. As a result, the pressure head of the 1st reservoir $P_1(t)$ will increase more slowly, and so it takes longer for $P_1(t)$ to surpass the eCTP of the remaining migration pathway. According to the generalised IP condition in Section 2.3.3, the time until the pathway between the 1st and 2nd reservoirs activate will therefore increase as well. As for the Markovian DAG G_1 , we only have to consider a single migration pathway emanating from the 1st reservoir, and so there is no additional uncertainty as to whether $q_1^{\text{net}}(t)$ decreases prior to the activation of the pathway between the 1st and 2nd reservoirs. It follows that the width of the interval between the 2 percentiles of $h_2(t)$ is smaller when we condition on G_1 as opposed to G_2 .

By using Eqs. (2.2) and (2.9) in combination with the $N = 75,000$ plume heights needed to produce Figure 3.7, we can examine how the prior distribution on the continuous parameters $p(\theta)$ induces uncertainty in the migration times $\mathbf{T}^{\text{mig}}$ of a stratigraphic system. Figure 3.8 shows the estimated kernel density of the induced distribution on $\mathbf{T}^{\text{mig}}$ for the 2 stratigraphic systems represented by G_1 and G_2 . Note that by conditioning on G_2 , the distribution of the migration times for the 2nd and 3rd reservoirs, T_2^{mig} and T_3^{mig} , are seemingly identical. This need not be the case in general, but since the hyperparameters corresponding to the 2nd and 3rd reservoirs are equal, see Appendix B, the prior distribution does not differentiate between the geometries of the 2 reservoirs or the weights placed on the pathways pointing to them. As each reservoir is directly reachable from the 1st



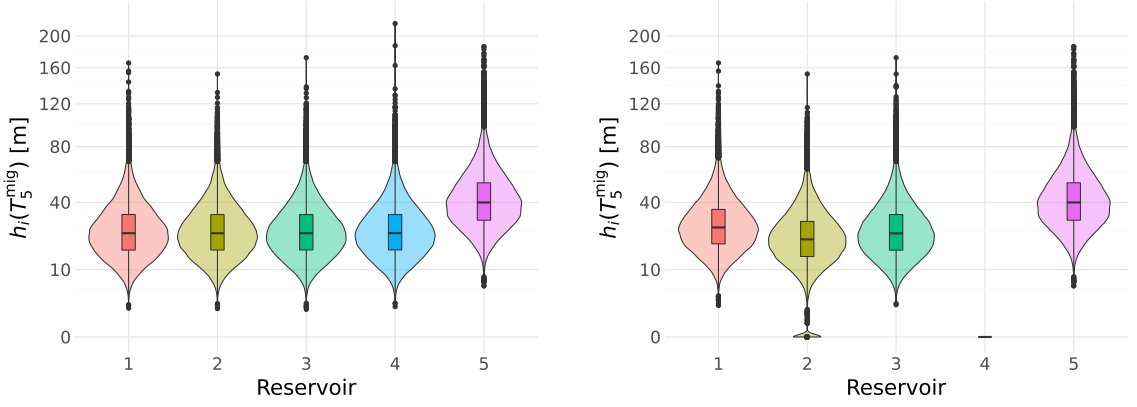
(a) Estimated kernel density corresponding to the stratigraphic system represented by G_1 , a diagram of which is shown in Figure 3.6a. (b) Estimated kernel density corresponding to the stratigraphic system represented by G_2 , a diagram of which is shown in Figure 3.6b.

Figure 3.8.: Distribution of the migration times T_i^{mig} of 2 stratigraphic systems comprising $k = 5$ reservoirs each, induced by the prior on the continuous parameters $p(\theta)$. The kernel estimate uses $N = 75,000$ samples in total, and the parameter settings used to obtain each sample is listed in Appendix B.

reservoir, it follows that the time needed for CO_2 to migrate away from either reservoir is similar. We also notice the missing density corresponding to the 4th reservoir in Figure 3.8b, which due to the lack of any migration pathways pointing to it, is degenerate at $T_4^{\text{mig}} = \infty$ in the case where we only inject CO_2 in the bottommost reservoir.

Moreover, we notice that the marginal density of the migration time of each reservoir T_i^{mig} becomes more spread out as we move higher into the strata. This suggests that there is a higher uncertainty associated with the migration times of shallow reservoirs, which seems only natural, as the migration pathways induce dependence between reservoirs. Specifically, if a reservoir is pointed to by a migration pathway emanating from a 2nd reservoir, then the 1st reservoir is dependent on this 2nd reservoir. To see why this is the case, we recall from Section 2.4.1 and Eq. (2.11) in particular, that the rate at which CO_2 enters a reservoir $q_i^{\text{ent}}(t)$ depends in part on the rate at which CO_2 migrates through active pathways from lower reservoirs at the current time. The net rate of accumulation in the i th reservoir $q_i^{\text{net}}(t)$ is therefore dependent on the parameters of the reservoirs pointing to it, and the weights of the migration pathways along which the CO_2 migrates. As the uncertainty accumulates, the the marginal density of the migration times become more spread out. We note that Dutta et al. (2026) obtained qualitative similar results when applying their model to a case inspired by the stratigraphic system at the Sleipner field. In particular, the mode of the induced distribution on the migration times shifted to the right for each subsequent reservoir, in addition to becoming more spread out.

Figure 3.9 then provides an illustration of the induced uncertainty on the plume heights of the stratigraphic systems represented by G_1 and G_2 , at the time of migration from the topmost reservoir, i.e. $h_i(T_5^{\text{mig}})$ for $i = 1, 2, \dots, 5$. It is particularly interesting to evaluate the plume heights at this specific time, as it represents the time at which the stratigraphic system is no longer capable of providing a seal for the migrating CO_2 . In other words, the CO_2 will start to (re)enter the atmosphere immediately after T_5^{mig} , and so the height of the CO_2 plumes at this particular instant represents an upper limit on the plume heights that our stratigraphic system can handle, while still ensuring that the CO_2 remains sequestered.



(a) Plots corresponding to the stratigraphic system represented by G_1 , a diagram of which is shown in Figure 3.6a. (b) Plots corresponding to the stratigraphic system represented by G_2 , a diagram of which is shown in Figure 3.6b.

Figure 3.9.: Box- and violin plots of the prior induced distribution of the plume heights of 2 stratigraphic systems comprising $k = 5$ reservoirs each, at the time of migration from the topmost reservoir T_5^{mig} . The plots are based on $N = 75,000$ samples in total, and the parameter settings used to obtain each sample is listed in Appendix B.

We notice that in both sub-figures the CO_2 plume within the topmost reservoir tends to be considerably higher compared to the other reservoirs. This is a result of the specific parameter settings we used for the prior on the continuous parameter $p(\theta)$ in the examples of this section, see Appendix B. In particular, by not introducing any differences between the reservoirs and assuming that the unconstrained mean of the relative eCTPs is equal to the 0-vector, we notice from Figure 3.4 that the relative eCTPs all have mean 0.5. The pathways of the stratigraphic system will therefore tend to activate once the pressure head $P(t)$ reaches half the value of the CTP, as per the generalised IP condition in Section 2.3.3. Migration from the topmost reservoir only occurs when $P_5(t) > C_5$ however, and so the plume within the topmost reservoir will tend to be 2 times the height of the plumes within the lower reservoirs at the time of migration, noting that the vertical axis is here scaled in terms of the square root of its labels.

Similar to Figures 3.7 and 3.8, we note that when conditioning on G_2 , the height of the CO_2 plume within 4th reservoir will be 0 for all t . Interestingly enough, the height of the CO_2 plume within the 2nd reservoir also tends to be 0 for some values of the prior parameters θ when conditioning on G_2 , as seen in Figure 3.9b. Similar to how the interval between the 5th and 95th percentiles were particularly wide in the case of the 2nd reservoir in Figure 3.7b when $h_2(t) \approx 0$, this is an indication of injected CO_2 bypassing the 2nd reservoir entirely. It shows that for some values of θ , CO_2 is able to migrate from the 1st to the 5th reservoir, without ever entering the 2nd reservoir. This is of course not possible under a Markovian paradigm, and so when we condition on G_1 in Figure 3.9a, the plume heights are all strictly positive at the time of migration from the topmost reservoir.

3.3. Acquisition Model

When we in Section 2.4 introduced the notion of a differentiable simulator $\mathcal{S}$, that computes the plume heights $\mathbf{h}(t)$ at a particular time $t < T_k^{\text{mig}}$, we specifically mentioned

that the simulator could be used in combination with a suitable acquisition procedure. We have since seen an example of how we can implement such a procedure in practice, namely through the use of repeated seismic surveys, as illustrated in Figure 3.1 and further elaborated on in Section 3.1. The term 4D seismic analysis is often used to describe this process, in which reflected amplitudes and their travel-times are compared between surveys to infer where CO₂ accumulates within the strata (Eidsvik et al. to appear; He et al. 2016; Martinez et al. 2024). Note in particular, that we do not measure $h(t)$ directly when carrying out such a survey, but rather the effect it has on seismic waves travelling through the strata.

Physically, as CO₂ accumulates within a reservoir, the acoustic impedance of the reservoir decreases, consistent with a reduction in the velocity of seismic waves travelling through the reservoir, and a decrease in the reservoir’s density (Arts et al. 2003; Furre et al. 2024; Onajite 2014). As we transmit seismic waves into the strata, parts of the transmitted energy will be reflected at the interface between regions that have different acoustic impedances (Onajite 2014). If we then compare the received signals with a baseline survey, we can determine which signals likely correspond to the accumulation of CO₂ within the strata. The time it takes for a seismic wave to be transmitted and then reflected back to a receiver, often referred to as two-way time (TWT) within the literature (Arts et al. 2003; Furre et al. 2024), can then be used to estimate both the depth and height of a particular CO₂ plume, in combination with the amplitude of the reflected wave.

There are however a couple of caveats associated with this procedure, when it comes to obtaining accurate estimates. Firstly, as the velocity of seismic waves decrease when travelling through CO₂ saturated rock, the reflections corresponding to lower reservoirs have a tendency to appear deeper in the strata than they are in reality, an effect known as velocity push-down (Arts et al. 2003). Secondly, the seismic waves that are transmitted may reflect back and forth between the reservoirs multiple times before they reach a receiver. The signal we receive is therefore a superposition of many reflections, not just the primary reflection from each reservoir. As waves interfere constructively or destructively depending on their relative phase shift, this affects the amplitude of the reflections corresponding to each reservoir, and may additionally create apparent reflections deeper in the strata where there are no reservoirs in reality. Arts et al. (2003) refer to this as the tuning effect, the results of which are actually quite noticeable in Figure 3.1, especially for the 2020 survey indicated in (Display d). Although we have not mentioned it earlier, the figure actually shows the returns as function of the TWT, with the y -axis ranging from approximately 1,200 to 800 ms. The faint horizontal returns visible in the lower half of image are therefore likely a result of secondary or tertiary reflections from the above reservoirs, rather than primary reflections from some other anomaly within the strata.

Note also that as the depth of the reservoirs decrease, the pressure of the overburden and temperature of the surrounding rock decrease as well. As a result, the phase properties of the injected CO₂ is expected to change, and the CO₂ may even undergo a phase change. In the case of the Sleipner field, multiple studies agree that the pressure-temperature conditions of the reservoirs are such that the CO₂ is expected to stay in a supercritical state for the whole duration of the CCS project, see for instance Alnes et al. (2011), Arts et al. (2003) and Callioli Santi et al. (2025). It is however mentioned that the conditions are not far off from the supercritical point of CO₂, and so small changes in either pressure or temperature will likely result in large changes in the density of the injected CO₂. Using precision gravity measurements from 30 seafloor stations atop the Utsira formation, Alnes et al. (2011) estimated the density of CO₂ at the Sleipner field to range from approximately 710 kg/m³ in the lowermost reservoir to 425 kg/m³ in the topmost reservoir. This decrease

in density, results in a larger density difference between CO₂ and brine saturated rock, which in turn increases the difference in acoustic impedance between CO₂ filled reservoirs and the surrounding strata. This makes it easier to detect the accumulation of CO₂ in shallow reservoirs, which currently has a detection limit of ~ 1 m (Martinez et al. 2024).

It should be evident from the preceding description of 4D seismic analysis, that an estimate of $\mathbf{h}(t)$ is likely not exact. To account for the uncertainty in the acquisition procedure, in addition to the aforementioned detection limit, we assume that any observations of $\mathbf{h}(t)$ are left censored with Gaussian noise. In particular, if $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\zeta}})$ is centred Gaussian noise with covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}}$, and $\mathbf{h}^{\text{thres}}$ denotes the minimum detectable plume heights using 4D seismic analysis, we assume that for each reservoir $i = 1, 2, \dots, k$, the estimated or observed apex plume height at time t is given by

$$d_i(t) = \begin{cases} h_i(t) + \zeta_i, & \text{if } h_i(t) + \zeta_i \geq h_i^{\text{thres}}. \\ 0, & \text{if } h_i(t) + \zeta_i < h_i^{\text{thres}}. \end{cases} \quad (3.9)$$

We further assume that $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_k^2)$ is diagonal, which results in an observation vector $\mathbf{d}(t)$ whose elements are mutually independent conditional on $\mathbf{h}(t)$. This assumption is made following Dutta et al. (2026) and Eidsvik et al. (to appear), who themselves used essentially the same observation model, although they assumed common noise terms and detection limits for all reservoirs. We generalise this somewhat for a more realistic model, allowing our reservoirs to have individual noise terms and detection limits. As previously mentioned, it is easier to detect the accumulation of CO₂ in shallow reservoirs, and so we should expect both the detection limit and the amount of noise to increase as we move deeper into the strata. This is essentially a state space model, in which observations $\mathbf{d}(t)$ at time $t < T_k^{\text{mig}}$ are conditionally independent given the underlying latent state, which in this case consists of $\mathbf{h}(t)$.

Note that Eq. (3.9) induces a likelihood model on the plume heights $\mathbf{h}(t)$ of the stratigraphic system. In particular, we have that

$$p(\mathbf{d}(t) | \mathbf{h}(t)) = \prod_{i=1}^k p(d_i(t) | h_i(t)), \quad (3.10)$$

where, if we let $\Phi(\cdot)$ and $\varphi(\cdot)$ denote the cumulative distribution function (cdf) and pdf of the standard normal respectively,

$$p(d_i(t) | h_i(t)) = \begin{cases} \Phi\left(\frac{h_i^{\text{thres}} - h_i(t)}{\sigma_i}\right), & \text{if } d_i(t) = 0. \\ \frac{1}{\sigma_i} \varphi\left(\frac{d_i(t) - h_i(t)}{\sigma_i}\right), & \text{if } d_i(t) \geq h_i^{\text{thres}}. \end{cases}$$

In other words, the likelihood model becomes a mixture model that factorises over each reservoir. For a particular reservoir i , the likelihood model consists of a point mass at $d_i(t) = 0$, representing the probability of a censored measurement, and a normal density that is centred at the actual plume height $h_i(t)$, for observations above (or at) the detection threshold h_i^{thres} .

Figure 3.10 then illustrates the effect of the detection threshold h_i^{thres} and associated measurement noise σ_i on the likelihood of $h_i(t)$. In particular, by fixing the actual plume height $h_i(t)$ and letting the observed height $d_i(t)$ vary, we notice how the probability mass corresponding to a censored measurement changes as the plume height $h_i(t)$ increases. If

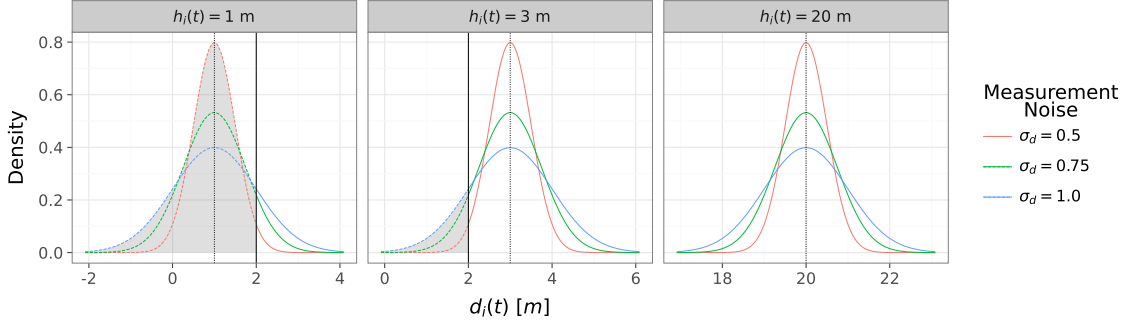


Figure 3.10.: Likelihood of the plume height in the i th reservoir $p(d_i(t) | h_i(t))$, as a function of the observed height $d_i(t)$ for 3 distinct values of $h_i(t)$. The standard error of the measurement noise has been indicated in the figure, and we assume an observation threshold $h_i^{\text{thres}} = 2$ m for each panel. The solid vertical lines correspond to this threshold and the dotted line corresponds to the particular values of $h_i(t)$. The greyed out area in the leftmost- and centre panels corresponds to the density that collapses at $d_i(t) = 0$, according to the likelihood model in Eq. (3.10).

$h_i(t) < h_i^{\text{thres}}$ as seen in the leftmost panel, then there is a high probability of $d_i(t)$ being censored. The density that falls below the observation threshold, which has been greyed out in the figure, then corresponds to the point mass at $d_i(t) = 0$. Even if $h_i(t) > h_i^{\text{thres}}$, the probability of obtaining a censored measurement $d_i(t)$ might be relatively high, if for instance the true height $h_i(t)$ is only barely larger than h_i^{thres} or the measurement noise is substantial, in which case σ_i is high. This is illustrated in the centre panel. Note that if $h_i(t) \gg h_i^{\text{thres}}$, as seen in the rightmost panel, then the probability mass at $d_i(t) = 0$ is essentially 0, and we are highly likely to obtain an uncensored measurement, even if the measurement noise is moderately high.

3.4. Sampling of the Posterior Distribution

Assuming that the acquisition procedure of the preceding section results in a series of $D \in \mathbb{N}$ measurements, all of which are acquired prior to the time of migration from the topmost reservoir T_k^{mig} , i.e.

$$\mathbf{d}^{(1)}(t_1), \mathbf{d}^{(2)}(t_2), \dots, \mathbf{d}^{(D)}(t_D) \quad \text{for} \quad 0 < t_1 < t_2 < \dots < t_D < T_k^{\text{mig}},$$

one might question whether we can use these measurements to make inference on the model parameters. In particular, one might ask whether it is possible to reduce the prior uncertainty in the parameters of our model, by accounting for the information provided by the measurements. In Bayesian statistics, this is accomplished by constructing the associated posterior distribution, using Bayes' theorem. Specifically, we have that

$$p(\boldsymbol{\theta}, G | \mathcal{D}) = \frac{p(\mathcal{D} | \boldsymbol{\theta}, G) \cdot p(\boldsymbol{\theta}) \cdot p(G)}{p(\mathcal{D})}, \quad (3.11)$$

where we define the set of observations $\mathcal{D} = \{\mathbf{d}^{(\ell)}(t_\ell) : \ell = 1, 2, \dots, D\}$, assuming, in addition, that the joint prior on the continuous- and discrete parameters $p(\boldsymbol{\theta}, G)$ factorises over each parameter type. $p(\mathcal{D} | \boldsymbol{\theta}, G)$ then denotes the joint-likelihood of $\boldsymbol{\theta}$ and G , given the entire set of measured plume heights $\mathcal{D}$, and $p(\mathcal{D})$ denotes the marginal likelihood of $\mathcal{D}$.

From Section 3.3 we know that the acquisition procedure induces a likelihood model on the plume heights $\mathbf{h}(t)$ of the form $p(\mathbf{d}(t) | \mathbf{h}(t))$, as seen in Eq. (3.10). By substituting $\mathbf{h}(t)$ in the likelihood model with the plume heights obtained by evaluating our differentiable simulator $\mathcal{S}$, conditional on the parameters of our model $\boldsymbol{\theta}$ and G , i.e. $\mathcal{S}(t | \boldsymbol{\theta}, G)$, we obtain a likelihood model on the parameters as a function of a single measurement $\mathbf{d}(t)$. If we denote this particular likelihood by $p(\mathbf{d}(t) | \boldsymbol{\theta}, G)$, and recall from the previous section that $\mathbf{d}(t)$ is conditionally independent given the underlying plume heights $\mathbf{h}(t)$, it follows that the joint-likelihood in Eq. (3.11) can be written as a product over observations in $\mathcal{D}$,

$$p(\mathcal{D} | \boldsymbol{\theta}, G) = \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \boldsymbol{\theta}, G) = \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \mathbf{h}(t)), \quad \mathbf{h}(t) = \mathcal{S}(t | \boldsymbol{\theta}, G).$$

Direct evaluation of the posterior distribution in Eq. (3.11) would then require the evaluation of the marginal likelihood of all observations $p(\mathcal{D})$, which takes the form of a sum over the sample space Ω_G of integrals over the sample space $\Omega_{\boldsymbol{\theta}}$. Specifically, we have that

$$p(\mathcal{D}) = \sum_{G \in \Omega_G} p(G) \int_{\Omega_{\boldsymbol{\theta}}} \left(\prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \boldsymbol{\theta}, G) \right) p(\boldsymbol{\theta}) d\boldsymbol{\theta},$$

which is generally unfeasible to compute, noting for instance that the cardinality of Ω_G grows extremely fast with the number of reservoirs k , as seen in Table 3.1. We therefore choose to approximate the posterior distribution using Markov chain Monte Carlo (McMC), which only requires the evaluation of the numerator in Eq. (3.11) (Gilks et al. 1995). In particular, by constructing a Markov chain whose transition kernel has Eq. (3.11) as its stationary distribution, samples from the Markov chain are approximately distributed according to the posterior distribution. Monte Carlo integration using the samples of the chain can therefore be used to approximate various posterior quantities of interest.

Since we have both continuous and discrete parameters in our model, we choose to construct separate transition kernels for each parameter type. This is accomplished by letting the Markov chain on $\boldsymbol{\theta}$ and G each target their corresponding full conditional distribution. In terms of $\boldsymbol{\theta}$, the full conditional distribution is given by

$$p(\boldsymbol{\theta} | \mathcal{D}, G) \propto p(\boldsymbol{\theta}) \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \mathbf{h}(t)), \quad \boldsymbol{\theta} \in \Omega_{\boldsymbol{\theta}}, \quad (3.12)$$

and similarly for G , we have that its full conditional distribution is given by

$$p(G | \mathcal{D}, \boldsymbol{\theta}) \propto p(G) \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \mathbf{h}(t)), \quad G \in \Omega_G, \quad (3.13)$$

where in both cases, $\mathbf{h}(t) = \mathcal{S}(t | \boldsymbol{\theta}, G)$. This type of construction, where we partition the model parameters in smaller blocks and update each block one at a time using separate transition kernels, is often referred to as Metropolis-within-Gibbs or component-wise McMC in the literature, see for instance Ascolani et al. (2026). As we use different methods to construct each transition kernel, we have decided to treat them separately in the subsections that follow.

3.4.1. Hamiltonian Monte Carlo on Continuous Parameters

Recall from Section 3.1.1, that the sample space of continuous parameters $\Omega_{\boldsymbol{\theta}} \subset \mathbb{R}^{k(k+2)}$ grows quadratically with the number of reservoirs k . For the purposes of constructing a

Markov chain whose stationary distribution is that of Eq. (3.12), this can quickly become problematic due to a phenomena typically referred to as the "the curse of dimensionality". In particular, when working with distributions in high-dimensional spaces, the neighbourhood around the mode of the distribution, where the probability density is high, occupies an increasingly small volume of the sample space, relative to the neighbourhoods far from the mode (Betancourt 2018). When evaluating posterior expectations, the increasingly small volume of this neighbourhood results in it having a negligible effect on the expectations in question, even though the probability density associated with the parameters in this neighbourhood is high. A similar result exists for neighbourhoods far from the mode of the distribution (Betancourt 2018). Even though the volume of such neighbourhoods are increasingly large, compared to that of the neighbourhood around the mode of the distribution, the probability density of the parameters in these neighbourhoods are vanishingly small.

For samples of the Markov chain to contribute meaningfully to posterior expectations, they therefore have to lie in an increasingly narrow band between these 2 extremes, in a neighbourhood whose volume is not too small and where the density of the parameters are such that the integral of Eq. (3.12) over the neighbourhood in question is of meaningful size. Betancourt (2018) refer to this particular neighbourhood as the typical set, a term we adopt for the remainder of the thesis. Efficient sampling from the full conditional distribution in Eq. (3.12) therefore requires us to consistently draw parameters that lie in the typical set of Ω_{θ} . This becomes increasingly difficult to accomplish using ordinary random-walk Metropolis-Hastings (Betancourt 2018; Yamada et al. 2022), which was the type of transition kernel considered by Grødeland (2026), for a stratigraphic system whose continuous parameters only consisted of α , β and C .

To see why this is the case, we first note that the transition kernel of random-walk Metropolis-Hastings can be characterised in terms of a 2-step process. In particular, if $\theta^{(\ell)}$ denotes the current state of the chain, random-walk Metropolis-Hastings first generates a candidate sample $\theta' \sim \mathcal{N}(\theta^{(\ell)}, \Sigma)$ according to a normal density centred in $\theta^{(\ell)}$, having covariance matrix Σ . The candidate sample is then accepted ($\theta^{(\ell+1)} = \theta'$) or rejected ($\theta^{(\ell+1)} = \theta^{(\ell)}$) based on a specific probability α_{accept} of accepting the sample. α_{accept} has to ensure that Eq. (3.12) is the stationary distribution of the chain, and since the normal distribution is symmetric, the probability is given by a ratio of posterior probabilities (Betancourt 2018; Yamada et al. 2022). In particular, we have that

$$\alpha_{\text{accept}} = \min \left\{ 1, \frac{p(\theta' | \mathcal{D}, G)}{p(\theta^{(M)} | \mathcal{D}, G)} \right\}.$$

The difficulty of high-dimensional random-walk Metropolis-Hastings is then to choose a covariance matrix Σ , so that the Markov chain on θ efficiently explores the whole typical set of Ω_{θ} (Betancourt 2018; Yamada et al. 2022). If the components of Σ are too large, then the proposed samples θ' regularly fall outside the typical set in regions where the posterior density is particularly small. The above acceptance probability therefore becomes close to 0, and so we reject most of the proposed samples. If instead the components of Σ are too small, then the proposed samples θ' , although likely part of the typical set, only ever result in small jumps from the previous states. As a result, the Markov chain is likely highly autocorrelated, and so we need increasingly many samples to adequately explore the whole typical set. In both cases, random-walk Metropolis-Hastings fails to explore the typical set efficiently, as the method is unable to account for the geometry of the set (Betancourt 2018).

We instead turn to a sampling technique known as Hamiltonian Monte Carlo (HMC), to sample the continuous parameters $\boldsymbol{\theta}$ according to their full conditional distribution in Eq. (3.12). In simple terms, HMC manages to efficiently explore the typical set of well behaved distributions, even in high-dimensional spaces, by leveraging the geometric structure encoded in the log-gradient of the full conditional distribution. In other words, it computes the value of

$$\nabla_{\boldsymbol{\theta}} \ln p(\boldsymbol{\theta} \mid \mathcal{D}, G) = \nabla_{\boldsymbol{\theta}} \ln p(\boldsymbol{\theta}) + \sum_{\mathbf{d}(t) \in \mathcal{D}} \nabla_{\boldsymbol{\theta}} \ln p(\mathbf{d}(t) \mid \mathbf{h}(t)), \quad \mathbf{h}(t) = \mathcal{S}(t \mid \boldsymbol{\theta}, G),$$

which it uses to make informed decisions as to where to propose new states $\boldsymbol{\theta}'$, such that the proposals lie in the typical set of $\Omega_{\boldsymbol{\theta}}$.⁸ The method draws inspiration from physics and the field of Hamiltonian dynamics in particular, where the Hamiltonian $H(\boldsymbol{\theta}, \mathbf{p})$ represents the total energy of a system, as a function of both position $\boldsymbol{\theta}$ and momentum $\mathbf{p}$ (Yamada et al. 2022). By treating the continuous parameters $\boldsymbol{\theta}$ as position variables and introducing auxiliary momentum variables $\mathbf{p}$, thus doubling the size of our parameter space $\Omega_{\boldsymbol{\theta}}$, we can evolve the variables over time according to Hamilton’s equations,

$$\frac{d\boldsymbol{\theta}}{dt} = \nabla_{\mathbf{p}} H(\boldsymbol{\theta}, \mathbf{p}), \quad \frac{d\mathbf{p}}{dt} = -\nabla_{\boldsymbol{\theta}} H(\boldsymbol{\theta}, \mathbf{p}).$$

A proposal $\boldsymbol{\theta}'$ is then made by first sampling the auxiliary momentum $\mathbf{p}^{(\ell)} \sim p(\mathbf{p}^{(\ell)} \mid \boldsymbol{\theta}^{(\ell)})$ according to some conditional distribution, where $\boldsymbol{\theta}^{(\ell)}$ is the current state of the Markov chain. A common choice here is to use a normal distribution centred at 0, where the covariance matrix is either dependent or independent of $\boldsymbol{\theta}^{(\ell)}$ (Betancourt 2018; Yamada et al. 2022). In case of the latter, the distribution simplifies by no longer being dependent on the position variables. We then evolve the joint state $(\boldsymbol{\theta}^{(\ell)}, \mathbf{p}^{(\ell)})$ for some time t using Hamilton’s equations, to obtain a proposal $(\boldsymbol{\theta}', \mathbf{p}')$. For most practical purposes, we are unable to integrate these equations analytically however, and so we have to use some form of numerical integration scheme to obtain the proposals $(\boldsymbol{\theta}', \mathbf{p}')$. A popular choice here is to use the symplectic leapfrog integrator, as it is volume preserving (Betancourt 2018; Yamada et al. 2022). This does however introduce a numerical error, such that the Hamiltonian $H(\boldsymbol{\theta}, \mathbf{p})$ is no longer constant with respect to time. A Metropolis correction step is therefore needed in practice to ensure that the Markov chain targets the full conditional distribution in Eq. (3.12). The resulting acceptance probability is then a function of $H(\boldsymbol{\theta}, \mathbf{p})$. Specifically, we have that

$$\alpha_{\text{accept}} = \min \left\{ 1, \exp \left[H(\boldsymbol{\theta}^{(\ell)}, \mathbf{p}^{(\ell)}) - H(\boldsymbol{\theta}', \mathbf{p}') \right] \right\}.$$

For a more detailed description of the motivation behind HMC, the aforementioned procedure or the use of a symplectic integrator such as the leapfrog method, the reader may refer to Betancourt (2018), Neal (2011) and Yamada et al. (2022), who covered these concepts in great detail. It is worth mentioning that even though HMC is more computationally demanding relative to random-walk Metropolis-Hastings, the method usually performs better when it comes to high-dimensional sample spaces, since it is able to efficiently explore the whole typical set. The amount of samples needed to perform reliable inference is therefore considerably lower when sampled using HMC, and so the method often results in considerably better over all run-times (Yamada et al. 2022).

⁸The normalising constant in Eq. (3.12), i.e. $p(\mathcal{D} \mid G) = \int_{\Omega_{\boldsymbol{\theta}}} \left(\prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G) \right) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$, does not depend on $\boldsymbol{\theta}$, and so its contribution to the log-gradient is 0.

Note however that for HMC to work in our case, we need to be able to compute the log-gradient of the likelihood of the simulator $\mathcal{S}$ at time t , conditional on the model parameters θ and G . This is precisely the reason why we had to make the simulator differentiable for every $t < T_k^{\text{mig}}$, in Section 2.4. To compute the aforementioned log-gradient we use a technique called Automatic Differentiation (AD), which is treated in depth by authors such as Bartholomew-Biggs et al. (2000), Margossian (2019) and Rall and Corliss (1996). In simple terms, AD works by repeatedly applying the chain rule of differentiation to simple elementary operations, appearing in the function to be differentiated. If we consider for instance the simple scalar function $f(x) = x^2 + 2$, AD works by first creating a tape of the elementary operations needed to produce $f(x)$. Starting with the identity operations $f_1 = x$ and $f_2 = 2$, we can define $f(x)$ in terms of the operations $f_3 = (f_1)^2$ and $f_4 = f_3 + f_2$, in which case $f(x) = f_4$. To compute the derivative $\partial f / \partial x = 2x$, AD applies the chain rule recursively to obtain

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial f_4} \cdot \frac{\partial f_4}{\partial x} = 1 \cdot \left(\frac{\partial f_4}{\partial f_3} \frac{\partial f_3}{\partial x} + \frac{\partial f_4}{\partial f_2} \frac{\partial f_2}{\partial x} \right) = 1 \cdot \frac{\partial f_3}{\partial f_1} \frac{\partial f_1}{\partial x} + 1 \cdot 0 = 2f_1 \cdot 1 = 2x.$$

This may at first seem overly cumbersome, but for a complex vector-valued function such as the simulator in Section 2.4, efficient computer implementations of AD lets us evaluate the gradient of the function at specific points in the codomain, without specifying an analytic expression for the gradient.

For the purposes of this thesis, we have used the Application Programming Interface (API) of TensorFlow (Abadi et al. 2016) to compute the log-gradient of the full conditional distribution in Eq. (3.12) using AD. The continuous parameters θ have then been sampled using HMC, by calling on the API of TensorFlow Probability (Dillon et al. 2017).

3.4.2. Metropolis-Hastings on Discrete Parameters

As the DAGs $G \in \Omega_G$ are discrete, we are unable to use HMC to sample the parameters from their full conditional distribution in Eq. (3.13). Instead, we assume a mixture of transition kernels that is able to make both local- and global updates to G , and use regular Metropolis-Hastings (Chib and Greenberg 1995) to move the Markov chain. In particular, if we assume that $G^{(\ell)}$ denotes the current state of the chain, we consider proposal kernels of the form $Q_{\text{local}}(G' | G^{(\ell)})$ and $Q_{\text{global}}(G')$, with G' denoting the proposed sample. We then define

$$Q_{\text{global}}(G') = p(G') = \prod_{i=1}^{k-2} p(\mathcal{J}'_i | \kappa'_i) p(\kappa'_i), \quad G' \in \Omega_G,$$

to equal the prior on the DAGs in Eq. (3.7), noting that the global proposal kernel is independent of the current state $G^{(\ell)}$. For the local proposals, we instead fix most of the migration pathways, only updating the pathways corresponding to a single reservoir. If we sample this reservoir r uniformly at random amongst the lower $k - 2$ reservoirs who exhibit stochastic migration pathways, i.e. $r \sim \text{Uniform}(1, k - 2)$, we get that

$$Q_{\text{local}}(G' | G^{(\ell)}) = \frac{1}{k-2} \sum_{r=1}^{k-2} p(\mathcal{J}'_r | \kappa'_r) p(\kappa'_r) \prod_{\substack{i=1 \\ i \neq r}}^{k-2} \mathbf{I}(\mathcal{J}'_i = \mathcal{J}_i^{(\ell)}) \mathbf{I}(\kappa'_i = \kappa_i^{(\ell)}), \quad G' \in \Omega_G,$$

by the law of total probability, where $\mathbf{I}(\cdot)$ denotes the indicator function, which is equal to 1 whenever the argument is true and 0 otherwise.

t [years]	$\mathbf{d}(t)$ [m]	DAG(s)
2.5	(15.0 0 0 0 0) [⊤]	G_1 and G_2
25.0	(22.0 20.0 18.0 10.0 0) [⊤]	G_1
25.0	(25.0 18.0 20.0 0 20.0) [⊤]	G_2

Table 3.2.: Synthetic measurements $\mathbf{d}(t)$ used for simulating from the full conditional distribution in Eq. (3.12). Note in particular that the 2nd measurement at $t = 25.0$ years differ between the 2 DAGs under consideration. Figure 3.6 shows a diagram of both of these DAGs.

By defining the proposal kernels in this particular way, the acceptance probability α_{accept} of the Metropolis-Hastings update simplifies in both cases to include only the ratio of the simulator likelihood, i.e.

$$\alpha_{\text{accept}} = \min \left\{ 1, \prod_{\mathbf{d}(t) \in \mathcal{D}} \frac{p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G')}{p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G^{(\ell)})} \right\},$$

a proof of which is included in Appendix A. Note that by occasionally proposing a global update $G' \sim Q_{\text{global}}(G')$, we ensure that the Markov chain on G does not get stuck in a region of low posterior probability, whilst also giving the chain ample time to explore smaller neighbourhoods in Ω_G where the posterior probability is high.

3.5. Inference Using Observed CO₂ Plume Heights

Using the sampling techniques of the previous section, we are now able to make inference on the posterior distribution $p(\boldsymbol{\theta}, G \mid \mathcal{D})$ in Eq. (3.11). Due to the time constraint under which the thesis was written, we are however only going to present results pertaining to the sampling of the full conditional distribution in Eq. (3.12). Effectively, this means that we keep the DAG G fixed and only consider how the observations $\mathcal{D}$ affect the posterior distribution of the continuous parameters $\boldsymbol{\theta}$. To illustrate how the DAGs affect the full conditional distribution, we are however going to repeat the sampling for 2 distinct DAGs. For that purpose, we recall from Section 3.2 the definition of the Markovian and non-Markovian DAG G_1 and G_2 , whose diagrams are shown in Figure 3.6.

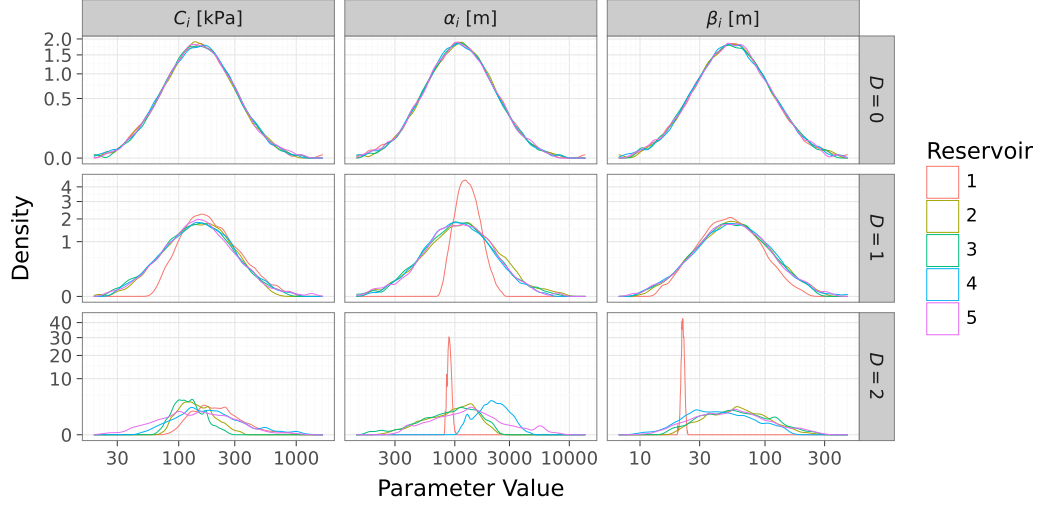
Throughout this section, we assume that 2 measurements have been made for each of the 2 stratigraphic systems, the 1st of which is identical for both G_1 and G_2 , as seen in Table 3.2. Ideally, we would want the 2nd measurement to be identical as well, so that the full conditional distributions only differ in the DAGs we condition on. In that case, it would be easier to discuss the effect provided by the DAGs. This has proven difficult in practice however, as it is hard to find synthetic measurements $\mathbf{d}(t)$ at a late time that have a high likelihood under both G_1 and G_2 . Consequently, the Markov chain on $\boldsymbol{\theta}$ regularly got stuck in the starting state, even when using HMC. Although we have not examined the performance of the joint sampler, it stands to reason that if we could update G as well, the chain on $\boldsymbol{\theta}$ would likely get unstuck once the transition kernel in Section 3.4.2 have had time to move the Markov chain on G into a region of high posterior probability. That being said, we still want to determine how the posterior changes when accounting for additional observations, and so we condition on a 2nd measurement for both DAGs, even though the measurement in question is not identical.

We also note that the particular values of the synthetic measurements used, as seen in Table 3.2, were chosen roughly based on the median plume heights $\mathbf{h}(t)$ in Figure 3.7. The synthetic measurements therefore represent a set of likely plume heights $\mathbf{h}(t)$ under each DAG. We considered other synthetic measurements as well, in addition to varying the parameter settings of both the prior-, transition kernel and acquisition model. These experiments resulted in various degrees of success however. Some settings resulted in fast mixing and robust posterior assessment, while in other cases, the Markov chains showed signs consistent with poor mixing. To illustrate the potential of the method, we have therefore chosen to illustrate the results of one of the experiments in which the Markov chain on θ exhibited a high degree of mixing. The parameter settings of this particular experiment is included in Appendix B.

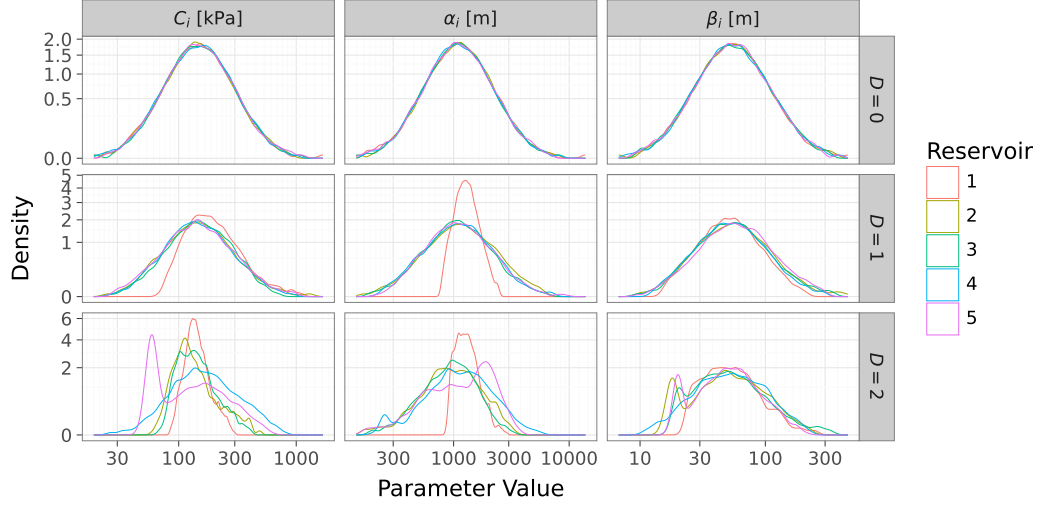
Figure 3.11 shows the estimated kernel density of $M = 10,000$ prior- and posterior samples of the CTPs $\mathbf{C}$ and reservoir extents α and β , corresponding to this experiment. For each DAG G_1 and G_2 , the figure shows how the full conditional distribution $p(\theta \mid \mathcal{D}, G)$ changes based on the number D of observations in the observation set $\mathcal{D}$. In the case where $D = 0$, i.e. $\mathcal{D} = \emptyset$, samples are instead taken from the prior distribution on the continuous parameters $p(\theta)$. This enables us to determine how the observations affect the prior uncertainty of $\mathbf{C}$, α and β , by comparing the prior- to the posterior uncertainty.

Note that when we condition on the 1st observation in Figure 3.11, only the full conditional distribution of the 1st reservoir seems to have been affected by the conditioning. This is expected, since we are not really learning anything new about the other reservoirs by making a censored observation of the plume heights contained within them. The censored observation is essentially saying that after 2.5 years, the injected CO_2 has only accumulated within the 1st reservoir. This has the effect of slightly shifting the posterior mode of the CTP of the 1st reservoir C_1 to the right, as it would be unlikely for the generalised IP condition of Section 2.3.3 to be met for the 1st reservoir at this time, given the observation. Had this been the case, migration from the 1st reservoir would have been possible, and so CO_2 would have accumulated in the other reservoirs. As the measurement threshold $\mathbf{h}^{\text{thres}}$ is similarly small for every reservoirs in our case, see Appendix B, we would likely have detected this accumulation, unless it happened right before the measurement was made at $t = 2.5$ years. The likelihood of the 1st reservoir being compromised after e.g. 1 year, is therefore extremely low given the censored measurement of the other reservoirs and the low detection threshold $\mathbf{h}^{\text{thres}}$.

We also observe that the posterior mode of the lateral extent of the 1st reservoir α_1 shifts slightly to the right, and that the spread of the distribution decreases, consistent with a smaller uncertainty in the value of α_1 . If we recall from Section 2.2.1, and Eq. (2.6) in particular, the ellipsoidal cap model used to represent the reservoirs in our model, this should come as no surprise. In particular, we have that the volume $V_1(t)$ of accumulated CO_2 within the 1st reservoir at time t is proportional to the square of the reservoir's lateral extent α_1 . As the injection rate due to external processes $\mathbf{q}^{\text{ext}}$ is fixed, we have that $V_1(2.5 \text{ years}) = q_1^{\text{ext}} \cdot 2.5 \text{ years}$, assuming that no inter-reservoir migration have taken place prior to $t = 2.5$ years. In reality, CO_2 could have accumulated in the other reservoirs as well prior to the time of the observation (as previously mentioned), and so $V_1(2.5 \text{ years}) \lesssim q_1^{\text{ext}} \cdot 2.5 \text{ years}$. The value of α_1 therefore directly affects the plume height within the 1st reservoir at the time of the observation $h_1(2.5 \text{ years})$, which was observed to equal $d_1(2.5 \text{ years}) = 15.0 \text{ m}$. It is therefore natural that the uncertainty in α_1 is reduced when we condition on the censored measurement. As the observations in Table 3.2 were all chosen based on the plume heights $\mathbf{h}(t)$ that were likely under the prior distribution $p(\theta)$, the full conditional distribution does not need to move far to account for the likelihood of



(a) Densities for the stratigraphic system corresponding to G_1 , the diagram of which is shown in Figure 3.6a.



(b) Densities for the stratigraphic system corresponding to G_2 , the diagram of which is shown in Figure 3.6b.

Figure 3.11.: Estimated prior- and posterior density of the CTPs $\mathbf{C}$ and the reservoir extents $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, for 2 stratigraphic systems comprising $k = 5$ reservoirs each. The kernel estimates are based on $M = 10,000$ samples of the prior $p(\boldsymbol{\theta})$ and full conditional distribution $p(\boldsymbol{\theta} \mid \mathcal{D}, G)$. The figure shows the densities that results from conditioning on D synthetic measurements, as described in Table 3.2, where we take $D = 0$ to mean that the estimates are based on the prior samples. The parameter settings used to obtain the samples are listed in Appendix B.

either the 1st or the 2nd observations.

When conditioning on a 2nd observation, we see that in the case of G_1 (Display a) the spread of the posterior distribution on the reservoir extents α_1 and β_1 decreases significantly, even when measured against the posterior where we condition on a single observation. This is typical for Bayesian posterior models, where we usually see a decrease in the uncertainty of the parameters for each subsequent observation included in the likelihood. As this is the 1st time we have gathered any "useful" information pertaining to the other

reservoirs, it is natural that we do not see a similar decrease in the uncertainty of the parameters corresponding to these reservoirs. We do however see that the mode of some parameters, like the lateral extent of the 4th reservoir α_4 , have shifted somewhat relative to their prior counterparts.

We do not see the same decrease in uncertainty for the case with G_2 however, as seen in Figure 3.11b, where the posterior distribution on α_1 and β_1 are seemingly unaffected by conditioning on a 2nd observation. We do however observe that the distribution of some parameters, such as that vertical extent of the 2nd, 3rd and 5th reservoirs β_2 , β_3 and β_5 and the CTP of the 5th reservoir C_5 , apparently have 2 modes. This could be a result of us not running the Markov chain long enough, in which case the samples could become biased towards specific regions of the sample space Ω_{θ} . As we have used HMC to sample the full conditional distribution, that might not be the most likely explanation however. Instead, a more probable explanation might be that the parameters of our model are unidentifiable, or at most weakly identifiable, in the sense that different continuous parameters θ have the same, or at least similar, likelihoods $p(\mathcal{D} \mid \theta, G)$ (Lehmann and Casella 1998, def. 5.2). This is evident when we consider for instance how the CTP C_i and vertical extent β_i of a particular reservoir i enters into the simulator of Section 2.4. Specifically, it appears as a ratio C_i/β_i in the expression for the scaled critical volume y_{ij}^{crit} in Eq. (2.19), determining when the migration pathway between the i th and j th reservoirs activate. The pairs $(s \cdot \beta_i, s \cdot C_i)$, where $s \in \mathbb{R}$ such that $s > 0$, therefore produces the exact same activation thresholds. Of course, such identification problems, where parameter distributions focus on ridges or in multiple modes, yield posterior sampling challenges.

This is precisely the reason why we chose to consider a common lateral extent for the ellipsoidal caps in Section 2.2.1. Had we instead chosen to use the exact model proposed by Dutta et al. (2026), the resulting collection of continuous parameters θ would certainly have been unidentifiable, given the likelihood model in Eq. (3.10). By scaling 1 of the lateral axes by some constant $s \in \mathbb{R}$, such that $s > 0$, and simultaneously scaling the other lateral axis by its inverse, i.e. $1/s$, the likelihood of θ would stay constant. This would in turn affect our ability to make inference on the parameters and would possibly also result in poorer mixing of the Markov chain (Wang et al. 2026).

Parameters $\mathbf{C}$, α and β are common for both DAGs G_1 and G_2 . Although the saturated weights $\mathbf{q}^{\text{sat}}$ and $\mathbf{e}^{\text{sat}}$ are common as well, the (non-saturated) weight matrices $\mathbf{Q}$ and $\mathbf{E}$ used by the simulator $\mathcal{S}$ differ between the DAGs, as they are given by a Hadamard product of the saturated weight matrices $\mathbf{Q}^{\text{sat}}$ and $\mathbf{E}^{\text{sat}}$ and the adjacency matrix of each DAG $\mathbf{A}(G_1)$ and $\mathbf{A}(G_2)$, as seen in Sections 2.3.1 and 2.3.3. In the case of G_1 , $q_{13} = 0$ for instance, as there is no migration pathway between the 1st and 3rd reservoirs in the stratigraphic system represented by this DAG, see Figure 3.6. Under G_2 , there is such a pathway however, and so $q_{13} = 1 - q_{12}$ is stochastic. Note also that the 1st reservoir in the system represented by G_2 is the only reservoir in both systems having multiple migration pathways. Per its construction in Section 2.3.1, the relative migration rates of the other pathways in the systems are therefore deterministically equal to 1. In the case where we condition on the systems represented by G_1 and G_2 , the posterior samples of $\mathbf{C}$, α and β are therefore much more interesting from an inference perspective.

Note that the challenges posed by poorly identifiable models is well known within geostatistics, see for instance Diggle et al. (2003). This need not be a huge problem however, if the particular values of the parameters are of limited value. As for our purposes, we are really only interested in the physical properties of the CO₂ plume, in particular its height distribution over time. By conditioning on the posterior samples of θ and propagating

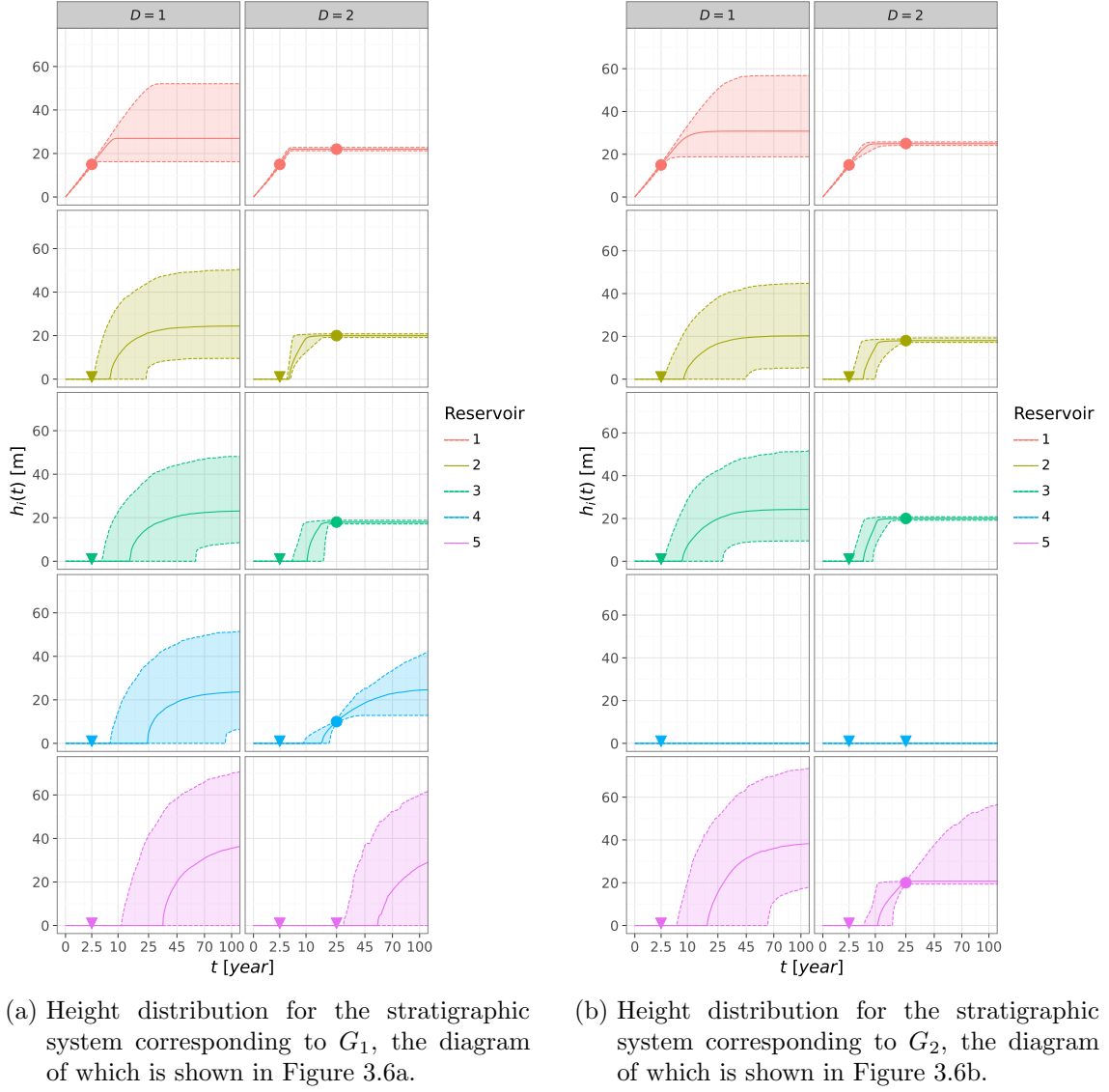


Figure 3.12.: Empirical percentiles of the posterior induced distribution on plume heights $\mathbf{h}(t)$ as a function of time t , for 2 stratigraphic systems comprising $k = 5$ reservoirs each. The figure shows the distribution that results from conditioning on D synthetic measurements, as described in Table 3.2. Censored measurements are indicated in the figure as downwards facing arrows, whilst regular measurements are indicated using dots. Solid lines correspond to the median (50th percentile) of $M = 10,000$ samples, with dotted lines indicating the 5th and 95th percentiles. The shaded regions therefore cover 90 % of the simulated plume heights $\mathbf{h}(t)$ at each time t . The parameter settings used to obtain the samples are listed in Appendix B.

the posterior uncertainty through the differentiable simulator $\mathcal{S}$, similar to what we did in Section 3.2, we can quantify how the prior induced uncertainty changes once we condition on a single- or multiple observations. This is illustrated in Figure 3.12, where we for both G_1 (Display a) and G_2 (Display b) illustrate how the posterior changes once we condition on the measurements in Table 3.2. In particular, the figure shows the empirical median of the plume heights resulting from propagating the posterior uncertainty of $M = 10,000$ samples through the differentiable simulator $\mathcal{S}$, along with the 5th and 95th empirical

percentile.

Similar to what was observed in Figure 3.11 when we conditioned on a single early measurement only, we observe from Figure 3.12 that the uncertainty of all but the lowermost reservoir is seemingly identical to the prior induced uncertainty shown in Figure 3.7. This is expected, as the early censored observation provides very little information pertaining to the reservoirs higher up in the strata. It is still worth mentioning, that the lower 5th percentile of the plume heights within these reservoirs only becomes strictly positive, thus indicating the accumulation of CO₂ within the reservoirs, once $t > 2.5$ years, i.e. the time of the 1st measurement. The posterior induced uncertainty have therefore decreased somewhat at the lower end for these reservoirs, likely a result of the posterior mode of the CTP of the 1st reservoir C_1 being shifted a tiny bit to the right relative to the prior, as seen in Figure 3.11.

For the 1st reservoir, the uncertainty seems to have vanished almost all together in the time prior to the 1st measurement however. This is likely a result of the detection threshold being particularly small, coupled with the 1st observation being censored for all but the lowermost reservoir. As previously mentioned, this lets us determine the volume of accumulated CO₂ within the 1st reservoir leading up to the 1st observation with a high degree of certainty. Hence, when we invert the volume using Eq. (2.6), the resulting estimate of the plume height $h_1(t)$ for $t \leq 2.5$ years can be made with a high degree of certainty.

Note that when we condition on the 2nd measurement as well, the posterior induced uncertainty collapses around the median for $t \geq 25.0$ years, i.e. after the 2nd measurement, for all 3 lowermost reservoirs. This is a result of the particular way in which we have defined the relative flow rates q_{ij} for systems where the pathway between the i th and j th reservoirs is the only pathway emanating from the i th reservoir. In particular, conservation of mass required us to set $q_{ij} = 1$ in this case, and so the net injection rate $\mathbf{q}^{\text{net}}(t)$ in the lower reservoirs, once a higher reservoir starts accumulating CO₂, is 0. Hence, when making an observation indicating that CO₂ has started accumulating in the 4th- or 5th reservoirs, as seen in Figure 3.12, Display a and Display b respectively, the height of the CO₂ plumes within deeper reservoirs are no longer able to increase. This is illustrated particularly well for the 1st reservoir, which has a single migration pathway under G_1 and an additional pathway to the 3rd reservoir under G_2 . Note that in Display a, representing the system described by G_1 , the plume height $h_1(t)$ immediately flattens out the moment $h_2(t) > 0$. As for Display b, representing the system described by G_2 , we notice that the uncertainty does not immediately collapse once the median $h_1(t)$ flattens out. This is precisely due to the fact that $q_1^{\text{net}}(t) > 0$ until both $h_2(t), h_3(t) > 0$. The residual uncertainty still visible for the 3 lowermost reservoirs after the 2nd measurement is made, is then directly tied to measurement noise ζ of the accusation model, described in Section 3.3. As the measurements are not exact, there is still some uncertainty as to the exact height of the CO₂ plumes following the observation of CO₂ accumulating in the 4th- and 5th reservoirs at time $t = 25.0$ years, which subsequently persist as time goes on.

We also note that for the 4th reservoir in Figure 3.12 Display a and the 5th reservoir in Display b, the uncertainty in the plume heights seems to increase as time goes on, following the 2nd measurement at $t = 25.0$ years. This is typically observed in Bayesian posterior models just like this one, as the observation constrains the underlying quantities locally in time, not globally. Once time progresses and we move farther away from the time of the observation, uncertainty accumulates and we obtain a characteristic outward fanning.

4. Closing Remarks

We have in this thesis expanded upon existing probabilistic surrogate models for modelling inter-layer CO₂ migration in multi-layered stratigraphic systems. In particular, by drawing inspiration from graph theory and representing the migration pathways of multi-layered stratigraphic systems in terms of DAGs, we have developed a simulator $\mathcal{S} : t \mapsto \mathbf{h}(t)$, that to each time $t < T_k^{\text{mig}}$ leading up to the migration time of the topmost reservoir, associates a particular CO₂ plume height $\mathbf{h}(t)$ to each of the system's k reservoirs. This lets us characterise the development of the CO₂ plume, noting that other aspects of the plume, such as its volume $\mathbf{V}(t)$ and pressure head $\mathbf{P}(t)$, are functions of $\mathbf{h}(t)$. Moreover, by introducing smoothness parameters τ^{act} and τ^{min} to the simulator, we have demonstrated that it can be made differentiable everywhere. This is not trivial, as the underlying IP theory, describing how CO₂ migrates in multi-layered stratigraphic systems, relies on time events and is inherently non-differentiable at certain points in time, corresponding to the event changes in the migration pattern. As the simulator $\mathcal{S}$ is differentiable everywhere, additional information can be inferred from the process, such as its gradient with respect to the model parameters. Inference techniques that require the evaluation of this gradient, including HMC, can therefore be used to increase the efficiency of the proposed framework.

We originally set out to answer whether it is possible to reduce the uncertainty associated with inter-layer CO₂ migration in a general multi-layered stratigraphic system. Current frameworks that use a Markovian assumption on the migration of CO₂, including for instance Callioli Santi et al. (2025), Dutta et al. (2026) and Eidsvik et al. (to appear), show that this is indeed possible in situations where the inter-layer migration is restricted to happen between subsequent layers only. In this thesis, we have expanded upon the results of these authors, showing that it is possible to reduce the uncertainty in the continuous parameters of our model $\boldsymbol{\theta}$, also in cases where the CO₂ is allowed to migrate between distant reservoirs. We note however that the likelihood model for seismic data appears to be at most weakly identifiable in $\boldsymbol{\theta}$, and so direct inference on $\boldsymbol{\theta}$ using Bayesian inversion via posterior sampling can be difficult. Inference on the migration process should therefore ideally be performed indirectly (Diggle et al. 2003), by propagating the posterior uncertainty through the differentiable simulator $\mathcal{S}$. We have shown that by doing so, observations $\mathbf{d}(t)$ of the plume heights $\mathbf{h}(t)$ can be used to restrict the prior induced uncertainty on $\mathbf{h}(t)$, hence also the migration process as a whole. Moreover, by making additional observations pertaining to the same system, we have shown that the posterior induced uncertainty on $\mathbf{h}(t)$ is further reduced.

Although we have shown how $\boldsymbol{\theta}$ affect both the prior- and posterior induced distribution on $\mathbf{h}(t)$ when we fix the underlying DAG G , we have, due to the limited time under which the thesis was written, not been able to carry out the same analysis on G for a fixed $\boldsymbol{\theta}$, or to sample both $\boldsymbol{\theta}$ and G from their joint posterior distribution. It is important to be aware of this limitation when considering the results of the simulations. When simulating posterior values of $\boldsymbol{\theta}$, we were unable to find a common late measurement $\mathbf{d}(t)$ with high likelihood under both the Markovian- and non-Markovian DAG under consideration. This resulted in the Markov chain on $\boldsymbol{\theta}$ being stuck in its starting state, which seemingly indicates that the likelihood model is highly sensitive in G . This is to be expected when

using an accurate acquisition procedure, as non-censored observations of some reservoir i are highly unlikely to occur if for instance the i th reservoir has no migration pathways entering it, in which case the joint-likelihood of θ and G would be close to 0. Since the sample space of G , which we denoted by Ω_G , grows extremely fast with the number of reservoirs k , the transition kernel of the McMC sampler might struggle to propose a DAG whose likelihood is not vanishingly small. As a result, we might need unrealistically many samples from the joint-posterior of θ and G for the inference based on these samples to be approximately unbiased, especially if the posterior distribution is multimodal. One might therefore decide to restrict the amount of DAGs considered, and carry out McMC only on a subset of Ω_G whose samples are expected to have a moderate to high likelihood for a given set of measurements $\mathcal{D}$. A future study on the inter-layer migration of CO₂, using graph-based models, should preferentially try and answer some of these questions.

We also note that the proposed framework has only been applied to a set of synthetic measurements, where the parameter settings were chosen to allow for efficient sampling from the prior- and posterior distributions, whilst also reflecting some of the general uncertainty found in multi-layered stratigraphic systems. This enables us to illustrate the potential of such a framework, but it does not answer whether the proposed framework can be used to accurately model specific stratigraphic systems of interest. Both Callioli Santi et al. (2025) and Dutta et al. (2026) applied their models to the stratigraphic system at the Sleipner field, to show how a probabilistic surrogate model can be used in practice. Since our proposed framework does not depend on a Markovian assumption to model the migration of CO₂, it stands to reason that our framework might be better equipped to model the migration process at Sleipner. Future studies on graph-based models would, as a result, likely benefit from conducting a similar case-based study. Not just to demonstrate the feasibility of the framework in a real world scenario, but also to determine the strengths and weaknesses associated with the framework, compared to the Markovian based approach of previous studies.

The proposed framework successfully demonstrates that the migration structure is of great importance if accurate inference is to be made. Specifically, we observe a clear difference in the behaviour of Markovian and non-Markovian systems, which mirrors what we know from previous studies on CO₂ sequestration in saline aquifers. In particular, when microfractures are present in the cap formations of the reservoirs, we observe a plume that rapidly rises towards the topmost reservoir (Cavanagh and Haszeldine 2014; Wang et al. 2021). As for our framework, this rapid ascent is really only possible in the non-Markovian case, where the migrating CO₂ is allowed to bypass some of the intermediate reservoirs. The proposed framework therefore provides a natural and elegant way of describing this phenomena mathematically, which is currently not possible using existing probabilistic surrogate models for inter-layer CO₂ migration in multi-layered stratigraphic systems.

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Appendix

A. Mathematical Derivations

Derivation of Eq. (2.6)

Eq. (2.5) states that all $(x, y, z) \in \mathbb{R}^3$ making up ψ_i must satisfy,

$$\left(\frac{x}{\alpha_i}\right)^2 + \left(\frac{y}{\alpha_i}\right)^2 = 1 - \left(\frac{z}{\beta_i}\right)^2, \quad i = 1, 2, \dots, k,$$

in addition to $z \geq 0$ being non-negative.

If we define the function $\lambda_i(z) = \alpha_i \sqrt{1 - (z/\beta_i)^2}$ and assume that $z \in [0, \beta_i] \subset \mathbb{R}$, this becomes equivalent to $x^2 + y^2 = \lambda_i^2(z)$, which we recognise as the formula for a circle of radius $\lambda_i(z)$, centred at the origin.

Hence, if we fix $z \in [0, \beta_i] \subset \mathbb{R}$, the intersection between ψ_i and the horizontal plane at z becomes a circle, encompassing an area of size

$$A_i = \pi \lambda_i^2(z) = \pi \alpha_i^2 \left[1 - \left(\frac{z}{\beta_i}\right)^2 \right], \quad i = 1, 2, \dots, k.$$

Integrating A_i over z from $\beta_i - \gamma_i$ to β_i , where $\gamma_i \in [0, \beta_i] \subset \mathbb{R}$, we find that

$$V_i = \int_{\beta_i - \gamma_i}^{\beta_i} A_i dz = \pi \alpha_i^2 \int_{\beta_i - \gamma_i}^{\beta_i} 1 - \left(\frac{z}{\beta_i}\right)^2 dz = \pi \alpha_i^2 \left[\gamma_i - \frac{\beta_i^3 - (\beta_i - \gamma_i)^3}{3\beta_i^2} \right], \quad i = 1, 2, \dots, k,$$

noting that $z \geq 0$ remains non-negative over the whole integration interval.

Using the identity $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$, some of the terms in the above expression cancels, resulting in

$$V_i = \pi \alpha_i^2 \left[\frac{\gamma_i^2}{\beta_i} - \frac{\gamma_i^3}{3\beta_i^2} \right] = \frac{\pi \alpha_i^2}{3\beta_i^2} \gamma_i^2 [3\beta_i - \gamma_i], \quad i = 1, 2, \dots, k.$$

As the above relation holds for all fixed $\gamma_i \in [0, \beta_i] \subset \mathbb{R}$, it must also hold for any function having $[0, \beta_i] \subset \mathbb{R}$ as its image, including $h_i(t)$. It follows that,

$$V_i(t) = \frac{\pi \alpha_i^2}{3\beta_i^2} h_i^2(t) [3\beta_i - h_i(t)], \quad i = 1, 2, \dots, k,$$

thus completing the derivation.

Derivation of the Structure of $\mathbf{A}(G)$

In Section 2.2.2, $\mathbf{A}(G) = (a_{ij})$ was defined element-wise, by stating that a_{ij} is the number of arcs in G having $v_i \in V(G)$ as its tail and $v_j \in V(G)$ as its head. As G has $\nu(G) = k$ vertices in total, it follows that both indices i, j run from 1 to k , resulting in $\mathbf{A}(G)$ being a $k \times k$ matrix.

Moreover, as G is strict, Eq. (2.7) implies that no 2 arcs in G join the same vertices with the same orientation. The number of arcs in G having $v_i \in V(G)$ as its tail and $v_j \in V(G)$ as its head is therefore at most 1, implying that $\mathbf{A}(G)$ is binary.

We showed in Section 2.2.2, that by assuming positive buoyancy of CO_2 in the underlying stratigraphic system, G would be topologically sorted. This implies that $i < j$ for all $a_{i,j} \in \mathbf{A}(G)$, or equivalently, $a_{i,j} \notin \mathbf{A}(G)$ if $i \geq j$. From the definition of $\mathbf{A}(G) = (a_{ij})$, it follows that $a_{ij} = 0$ for all $i \geq j$, which in turn implies that $\mathbf{A}(G)$ is strictly upper-triangular (Wheeler 1980).

As G is connected, we know per section Section 2.2.2, that we are unable to partition $V(G)$ into non-empty subsets $V_1(G), V_2(G), \dots, V_n(G)$, such that the vertices in any one partition are not connected to the vertices in any of the other partitions. As G is acyclic, this is only made possible if each but the topmost vertex have at least a single arc to a vertex of higher topological order. It follows that the first $k - 1$ rows of $\mathbf{A}(G)$ must have at least a single 1 element each.

This demonstrates that $\mathbf{A}(G)$ is $k \times k$ upper-triangular with binary entries, and that the first $k - 1$ rows of $\mathbf{A}_G$ contain at least a single 1 element each, thus completing the derivation.

Derivation of the Inverse of $\mathbf{I}_k - \mathbf{F}^\top(G, t)$

Recall from Section 2.4.1 that the determinant of $\mathbf{I}_k - \mathbf{F}^\top(G, t)$ is equal to 1 and that the matrix admits an inverse. For a matrix $\mathbf{A} \in \mathbb{R}^{k \times k}$ to have an inverse $\mathbf{A}^{-1}$, the following condition must hold,

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_k = \mathbf{A}^{-1}\mathbf{A}.$$

We note, for the sake of completeness, that any such inverse must be unique (Axler 2023, ch. 3D).

Suppose for the moment, that

$$(\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1} = \sum_{\ell=0}^{k-1} (\mathbf{F}^\top(G, t))^\ell = \mathbf{I}_k + \mathbf{F}^\top(G, t) + (\mathbf{F}^\top(G, t))^2 + \dots + (\mathbf{F}^\top(G, t))^{k-1}$$

is the unique inverse of $\mathbf{I}_k - \mathbf{F}^\top(G, t)$.

By substitution, we then have that

$$\begin{aligned} (\mathbf{I}_k - \mathbf{F}^\top(G, t))(\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1} &= (\mathbf{I}_k - \mathbf{F}^\top(G, t)) \sum_{\ell=0}^{k-1} (\mathbf{F}^\top(G, t))^\ell \\ &= \sum_{\ell=0}^{k-1} (\mathbf{F}^\top(G, t))^\ell - \sum_{\ell=1}^k (\mathbf{F}^\top(G, t))^\ell \\ &= \mathbf{I}_k - (\mathbf{F}^\top(G, t))^k. \end{aligned}$$

Similarly, we find that

$$\begin{aligned}
(\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1}(\mathbf{I}_k - \mathbf{F}^\top(G, t)) &= \left(\sum_{\ell=0}^{k-1} (\mathbf{F}^\top(G, t))^\ell \right) (\mathbf{I}_k - \mathbf{F}^\top(G, t)) \\
&= \mathbf{I}_k - \mathbf{F}^\top(G, t) + \mathbf{F}^\top(G, t) - \cdots - (\mathbf{F}^\top(G, t))^k \\
&= \mathbf{I}_k - (\mathbf{F}^\top(G, t))^k.
\end{aligned}$$

For $(\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1}$ to be the inverse of $(\mathbf{I}_k - \mathbf{F}^\top(G, t))$, we must therefore prove that all entries of $(\mathbf{F}^\top(G, t))^k$ equates to 0. In terms of linear algebra, that would make $\mathbf{F}^\top(G, t)$ nilpotent, with index of nilpotence smaller than or equal to k (Herstein 1973, ch. 6.5).

Assume to the contrary, that there is at least a single element in $(\mathbf{F}^\top(G, t))^k$ that is non-zero. It follows that

$$\left[(\mathbf{F}^\top(G, t))^k \right]^\top = (\mathbf{F}^\top(G, t))^\top \cdots (\mathbf{F}^\top(G, t))^\top = \mathbf{F}(G, t) \cdots \mathbf{F}(G, t) = \mathbf{F}^k(G, t)$$

has at least a single non-zero element as well.

Using the definition of matrix multiplication, we have that

$$\begin{aligned}
(\mathbf{F}^k(G, t))_{ij} &= \sum_{\ell_1=1}^k f_{i\ell_1} (\mathbf{F}^{k-1}(G, t))_{\ell_1 j} \\
&= \sum_{\ell_1, \ell_2=1}^k f_{i\ell_1} f_{\ell_1 \ell_2} (\mathbf{F}^{k-2}(G, t))_{\ell_2 j} \\
&\vdots \\
&= \sum_{\ell_1, \ell_2, \dots, \ell_{k-1}=1}^k f_{i\ell_1} f_{\ell_1 \ell_2} \cdots f_{\ell_{k-1} j}.
\end{aligned}$$

As we assumed that $(\mathbf{F}^\top(G, t))^k$ has at least a single non-zero element, we know that there exists at least a single combination of indices $i, j, \ell_1, \ell_2, \dots, \ell_{k-1} \in [1, k] \subset \mathbb{N}$, such that $f_{i\ell_1} f_{\ell_1 \ell_2} \cdots f_{\ell_{k-1} j} \neq 0$.

Recalling from Section 2.4.1, that $\mathbf{F}(G, t)$ is strictly upper-triangular, we also know that $f_{ij} = 0$ for all $i \geq j$. Hence, for $f_{i\ell_1} f_{\ell_1 \ell_2} \cdots f_{\ell_{k-1} j} \neq 0$, we must have that $i < \ell_1, \ell_1 < \ell_2, \dots, \ell_{k-1} < j$, or equivalently, $i < \ell_1 < \ell_2 < \cdots < \ell_{k-1} < j$.

Notice however, that this requires a strictly increasing sequence of $k + 1$ indices in the interval $[1, k] \subset \mathbb{N}$. As the interval in question only contains k elements in total, this is a contradiction.

For $\mathbf{F}(G, t)$ to be strictly upper-triangular, our assumption on the existence of at least a single non-zero element in $(\mathbf{F}^\top(G, t))^k$ must therefore be wrong. It follows that $\mathbf{F}^\top(G, t)$ is nilpotent, with index of nilpotence smaller than or equal to k . Hence,

$$(\mathbf{I}_k - \mathbf{F}^\top(G, t))^{-1} = \sum_{\ell=0}^{k-1} (\mathbf{F}^\top(G, t))^\ell = \mathbf{I}_k + \mathbf{F}^\top(G, t) + (\mathbf{F}^\top(G, t))^2 + \cdots + (\mathbf{F}^\top(G, t))^{k-1}$$

is the inverse of $\mathbf{I}_k - \mathbf{F}^\top(G, t)$.

Derivation of the Limit of $\mathbf{S}^{\text{act}}(G, t)$ as $\tau^{\text{act}} \rightarrow 0$ From Above

Using Eq. (2.13), we have that for finite $\tau^{\text{act}} > 0$,

$$P_i(t) = e_{i,j}C_i \implies s_{ij}^{\text{act}} = a_{ij} [1 + e^0]^{-1} = \frac{a_{ij}}{2}, \quad i, j = 1, 2, \dots, k.$$

Hence, for all $i, j = 1, 2, \dots, k$, if we let $\tau^{\text{act}} \rightarrow 0$ from above (as $\tau^{\text{act}} \in (0, \infty) \subset \mathbb{R}$), we get that

$$s_{ij}^{\text{act}} \rightarrow \begin{cases} a_{ij}, & \text{if } P_i(t) > e_{i,j}C_i. \\ a_{ij}/2, & \text{if } P_i(t) = e_{i,j}C_i. \\ 0, & \text{if } P_i(t) < e_{i,j}C_i. \end{cases}$$

A result of the exponential function in Eq. (2.13) tending to 0 or (positive) infinity, for all $i, j = 1, 2, \dots, k$, depending on the sign of $P_i(t) - e_{i,j}C_i \neq 0$ as $\tau^{\text{act}} \rightarrow 0$ from above.

Using the definition of $\mathbf{A}^{\text{act}}(G, t)$ in Section 2.4.1, we furthermore note that if $P_i(t) - e_{i,j}C_i \neq 0$, the supremum of the absolute difference between s_{ij}^{act} and a_{ij}^{act} , for all $i, j = 1, 2, \dots, k$, tends to 0 in the limit as $\tau^{\text{act}} \rightarrow 0$ from above, i.e.

$$\sup_{P_i(t) - e_{i,j}C_i \neq 0} |s_{ij}^{\text{act}} - a_{ij}^{\text{act}}| \rightarrow 0 \quad \text{as } \tau^{\text{act}} \rightarrow 0, \quad i, j = 1, 2, \dots, k.$$

For any fixed $\varepsilon > 0$, we may therefore define a set $A_\varepsilon \subset \mathbb{R}$ whose measure is strictly less than ε , i.e. $\mu(A_\varepsilon) < \varepsilon$, and where $s_{ij}^{\text{act}} \rightarrow a_{ij}^{\text{act}}$ uniformly on the complement $\mathbb{R} \setminus A_\varepsilon$, for all $i, j = 1, 2, \dots, k$, as $\tau^{\text{act}} \rightarrow 0$ from above.

In particular, if we consider the set $A_\varepsilon = [-\varepsilon/3, \varepsilon/3] \subset \mathbb{R}$ with measure $\mu(A_\varepsilon) = 2\varepsilon/3 < \varepsilon$ and complement $\mathbb{R} \setminus A_\varepsilon = (-\infty, -\varepsilon/3) \cup (\varepsilon/3, \infty) \subset \mathbb{R}$, we have that

$$\sup_{P_i(t) - e_{i,j}C_i \in \mathbb{R} \setminus A_\varepsilon} |s_{ij}^{\text{act}} - a_{ij}^{\text{act}}| \rightarrow 0 \quad \text{as } \tau^{\text{act}} \rightarrow 0, \quad i, j = 1, 2, \dots, k.$$

In terms of measure theory, s_{ij}^{act} is said to converge almost uniformly to a_{ij}^{act} on $\mathbb{R}$, for all $i, j = 1, 2, \dots, k$, as $\tau^{\text{act}} \rightarrow 0$ from above (Roussas 2014). Noting that almost uniform convergence implies convergence almost everywhere (Roussas 2014), we also have that $s_{ij}^{\text{act}} \rightarrow a_{ij}^{\text{act}}$ almost everywhere, for $i, j = 1, 2, \dots, k$, as $\tau^{\text{act}} \rightarrow 0$ from above. In particular, we have that $s_{ij}^{\text{act}} \rightarrow a_{ij}^{\text{act}}$ everywhere, apart from on an arbitrarily small set with 0 measure containing $0 \in \mathbb{R}$.

As $s_{ij}^{\text{act}} \rightarrow a_{ij}^{\text{act}}$ almost uniformly, for all $i, j = 1, 2, \dots, k$, as $\tau^{\text{act}} \rightarrow 0$ from above, it follows that $\mathbf{S}^{\text{act}}(G, t) \rightarrow \mathbf{A}^{\text{act}}(G, t)$ almost uniformly, as $\tau^{\text{act}} \rightarrow 0$ from above.

Derivation of the Bounds and Limit of Eq. (2.17) as $\tau^{\text{min}} \rightarrow 0$ From Above

Let $x_1, x_2, \dots, x_n \in \mathbb{R}$ be real numbers, and consider the quantities

$$y_i = \exp\left(-\frac{x_i}{\tau^{\text{min}}}\right), \quad i = 1, 2, \dots, n,$$

noting that $\tau^{\text{min}} \in (0, \infty) \subset \mathbb{R}$ as defined in Section 2.4.2.

As the exponential function is positive everywhere, we have that

$$\max_{i=1,2,\dots,n} y_i \leq \sum_{i=1}^n y_i \leq n \max_{i=1,2,\dots,n} y_i.$$

Moreover, as the exponential is strictly monotone, we can take the maximum over the exponents as opposed to the exponential itself, implying that

$$\exp\left(\max_{i=1,2,\dots,n}\left\{-\frac{x_i}{\tau^{\min}}\right\}\right) \leq \sum_{i=1}^n y_i \leq n \exp\left(\max_{i=1,2,\dots,n}\left\{-\frac{x_i}{\tau^{\min}}\right\}\right).$$

Using the relation

$$\max_{i=1,2,\dots,n}\left\{-\frac{x_i}{\tau^{\min}}\right\} = -\left(\min_{i=1,2,\dots,n}\left\{\frac{x_i}{\tau^{\min}}\right\}\right) = -\frac{1}{\tau^{\min}} \min_{i=1,2,\dots,n} x_i,$$

we therefore have

$$\exp\left(-\frac{1}{\tau^{\min}} \min_{i=1,2,\dots,n} x_i\right) \leq \sum_{i=1}^n y_i \leq n \exp\left(-\frac{1}{\tau^{\min}} \min_{i=1,2,\dots,n} x_i\right),$$

which upon taking logarithms, implies that

$$-\frac{1}{\tau^{\min}} \min_{i=1,2,\dots,n} x_i \leq \ln \sum_{i=1}^n y_i \leq \ln n - \frac{1}{\tau^{\min}} \min_{i=1,2,\dots,n} x_i.$$

If we now multiply throughout by $-\tau^{\min}$, we get that

$$\left(\min_{i=1,2,\dots,n} x_i\right) - \tau^{\min} \ln n \leq -\tau^{\min} \ln \sum_{i=1}^n y_i \leq \min_{i=1,2,\dots,n} x_i,$$

noting that we must flip the inequalities following $-\tau^{\min} < 0$.

By shifting the inequalities to the right by an amount equal to $\tau^{\min} \ln n$ and substituting our expression for y_i , we obtain the lower bound of Eq. (2.17) reported in Section 2.4.2, i.e.

$$\min_{i=1,2,\dots,n} x_i \leq \text{LSE}(x_1, x_2, \dots, x_n) \leq \tau^{\min} \ln n + \min_{i=1,2,\dots,n} x_i,$$

where

$$\text{LSE}(x_1, x_2, \dots, x_n) = \tau^{\min} \left[\ln n - \ln \sum_{i=1}^n \exp\left(-\frac{x_i}{\tau^{\min}}\right) \right].$$

Note in particular that for all $x_1, x_2, \dots, x_n \in \mathbb{R}$,

$$\sup_{x_1, x_2, \dots, x_n \in \mathbb{R}} \left| \text{LSE}(x_1, x_2, \dots, x_n) - \min_{i=1,2,\dots,n} x_i \right| \leq \tau^{\min} \ln n.$$

Thus, for any fixed $n \in \mathbb{N}$ that is also finite, as $\tau^{\min} \rightarrow 0$ from above, Eq. (2.17) converges uniformly to $\min_{i=1,2,\dots,n} x_i$. 0

Derivation of the pdfs of the Quantities Depicted in Figures 3.2 to 3.4

Recall that if $\mathbf{Y} = h(\mathbf{X})$ is a bijective and differentiable transformation of a random vector $\mathbf{X}$, then its pdf is given by the equation

$$p_{\mathbf{Y}}(\mathbf{y}) = p_{\mathbf{X}}(h^{-1}(\mathbf{y})) \cdot |\mathcal{J}|,$$

where $p_{\mathbf{X}}(\cdot)$ is the pdf of $\mathbf{X}$ and

$$|\mathcal{J}| = \left| \frac{\partial \mathbf{x}}{\partial \mathbf{y}} \right|,$$

the Jacobian determinant corresponding to the transformation $\mathbf{X} \mapsto \mathbf{Y}$.

Moreover, in the special case where $\mathbf{Y} = h(\mathbf{X}) = \mathbf{A}\mathbf{X} + \mathbf{b}$ is an affine transformation of a normal vector $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, the transformation is also normal with mean vector $\mathbf{A}\boldsymbol{\mu} + \mathbf{b}$ and covariance matrix $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. In other words,

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Deriving the Marginal pdf of C_i

Consider for the moment the prior distribution on the unconstrained CTPs $\tilde{\mathbf{C}}$. From Section 3.1.1 we know that $\tilde{\mathbf{C}} \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{C}}}, \boldsymbol{\Sigma}_{\tilde{\mathbf{C}}})$. Hence if we let $\mathbf{A}$ be the $1 \times k$ matrix whose i th component is 1 with all other components being 0, we have that

$$\tilde{C}_i \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_{\tilde{\mathbf{C}}}, \mathbf{A}\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}\mathbf{A}^\top) = \mathcal{N}(\mu_i, \sigma_i^2),$$

where $\mu_i = (\boldsymbol{\mu}_{\tilde{\mathbf{C}}})_i$ and $\sigma_i^2 = (\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}})_{ii}$.

As

$$C_i = \exp \tilde{C}_i \iff \tilde{C}_i = \ln C_i$$

is a differentiable bijection, with Jacobian determinant

$$|\mathcal{J}| = \left| \frac{\partial \tilde{C}_i}{\partial C_i} \right| = \left| \frac{\partial \ln C_i}{\partial C_i} \right| = \frac{1}{C_i}, \quad C_i \in (0, \infty),$$

we get that

$$p(C_i) = p_{\tilde{C}_i}(\ln C_i) \cdot \frac{1}{C_i}, \quad C_i \in (0, \infty).$$

Deriving the Joint pdf of C_i and C_j

If we instead consider $\mathbf{A}$ to be the $2 \times k$ matrix whose components are all equal to 0 except for the components $a_{1i} = a_{2j} = 1$, we have that

$$\begin{pmatrix} \tilde{C}_i \\ \tilde{C}_j \end{pmatrix} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_{\tilde{\mathbf{C}}}, \mathbf{A}\boldsymbol{\Sigma}_{\tilde{\mathbf{C}}}\mathbf{A}^\top) = \mathcal{N}\left(\begin{pmatrix} \mu_i \\ \mu_j \end{pmatrix}, \begin{pmatrix} \sigma_i^2 & \rho\sigma_i\sigma_j \\ \rho\sigma_i\sigma_j & \sigma_j^2 \end{pmatrix}\right),$$

where ρ is the correlation between $\tilde{C}_i$ and $\tilde{C}_j$.

As

$$\begin{pmatrix} C_i \\ C_j \end{pmatrix} = \exp \begin{pmatrix} \tilde{C}_i \\ \tilde{C}_j \end{pmatrix} \iff \begin{pmatrix} \tilde{C}_i \\ \tilde{C}_j \end{pmatrix} = \ln \begin{pmatrix} C_i \\ C_j \end{pmatrix}$$

is a differentiable bijection, with Jacobian determinant

$$|\mathcal{J}| = \left| \frac{\partial(\tilde{C}_i, \tilde{C}_j)}{\partial(C_i, C_j)} \right| = \left| \begin{pmatrix} \frac{1}{C_i} & 0 \\ 0 & \frac{1}{C_j} \end{pmatrix} \right| = \frac{1}{C_i C_j}, \quad C_i, C_j \in (0, \infty),$$

we get that

$$p\left(\begin{pmatrix} C_i \\ C_j \end{pmatrix}\right) = p_{(\tilde{C}_i, \tilde{C}_j)^\top}\left(\ln \begin{pmatrix} C_i \\ C_j \end{pmatrix}\right) \cdot \frac{1}{C_i C_j}, \quad C_i, C_j \in (0, \infty).$$

Deriving the Marginal pdf of e_{ij}^{sat}

Consider for the moment the prior distribution on the unconstrained relative eCTPs of the saturated system $\tilde{\mathbf{e}}^{\text{sat}}$. From Section 3.1.1 we know that $\tilde{\mathbf{e}}^{\text{sat}} \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{e}}^{\text{sat}}}, \boldsymbol{\Sigma}_{\tilde{\mathbf{e}}^{\text{sat}}})$. Hence if we let $\mathbf{A}$ be the $1 \times [(k-1) + (k-2) + \dots + 1]$ matrix that extracts $\tilde{e}_{ij}^{\text{sat}}$ from $\tilde{\mathbf{e}}^{\text{sat}}$, we have that

$$\tilde{e}_{ij}^{\text{sat}} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_{\tilde{\mathbf{e}}^{\text{sat}}}, \mathbf{A}\boldsymbol{\Sigma}_{\tilde{\mathbf{e}}^{\text{sat}}}\mathbf{A}^\top) = \mathcal{N}(\mu_{ij}, \sigma_{ij}^2),$$

where we define μ_{ij} and σ_{ij}^2 in terms of the above products.

Using Eq. (3.2), we have that

$$\exp(-\tilde{e}_{ij}^{\text{sat}}) = \frac{1}{e_{ij}^{\text{sat}}} - 1 \iff \tilde{e}_{ij}^{\text{sat}} = -\ln\left(\frac{1}{e_{ij}^{\text{sat}}} - 1\right) = \ln\left(\frac{e_{ij}^{\text{sat}}}{1 - e_{ij}^{\text{sat}}}\right),$$

and so

$$e_{ij}^{\text{sat}} = \frac{1}{1 + \exp(-\tilde{e}_{ij}^{\text{sat}})} \iff \tilde{e}_{ij}^{\text{sat}} = \ln\left(\frac{e_{ij}^{\text{sat}}}{1 - e_{ij}^{\text{sat}}}\right)$$

is a differentiable bijection, with Jacobian determinant

$$|\mathcal{J}| = \left| \frac{\partial \tilde{e}_{ij}^{\text{sat}}}{\partial e_{ij}^{\text{sat}}} \right| = \left| \frac{1}{e_{ij}^{\text{sat}}} - \frac{1}{1 - e_{ij}^{\text{sat}}}(-1) \right| = \frac{1}{e_{ij}^{\text{sat}}(1 - e_{ij}^{\text{sat}})}, \quad e_{ij}^{\text{sat}} \in (0, 1).$$

We therefore get that,

$$p(e_{ij}^{\text{sat}}) = p_{\tilde{e}_{ij}^{\text{sat}}}\left(\ln\left(\frac{e_{ij}^{\text{sat}}}{1 - e_{ij}^{\text{sat}}}\right)\right) \cdot \frac{1}{e_{ij}^{\text{sat}}(1 - e_{ij}^{\text{sat}})}, \quad e_{ij}^{\text{sat}} \in (0, 1).$$

Deriving the Joint pdf of e_{ij}^{sat} and e_{il}^{sat}

If we instead consider $\mathbf{A}$ to be the $2 \times [(k-1) + (k-2) + \dots + 1]$ matrix that extracts both $\tilde{e}_{ij}^{\text{sat}}$ and $\tilde{e}_{il}^{\text{sat}}$ from $\tilde{\mathbf{e}}^{\text{sat}}$, we have that

$$\begin{pmatrix} \tilde{e}_{ij}^{\text{sat}} \\ \tilde{e}_{il}^{\text{sat}} \end{pmatrix} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_{\tilde{\mathbf{e}}^{\text{sat}}}, \mathbf{A}\boldsymbol{\Sigma}_{\tilde{\mathbf{e}}^{\text{sat}}}\mathbf{A}^\top) = \mathcal{N}\left(\begin{pmatrix} \mu_{ij} \\ \mu_{il} \end{pmatrix}, \begin{pmatrix} \sigma_{ij}^2 & \rho\sigma_{ij}\sigma_{il} \\ \rho\sigma_{ij}\sigma_{il} & \sigma_{il}^2 \end{pmatrix}\right),$$

where ρ is the correlation between $\tilde{e}_{ij}^{\text{sat}}$ and $\tilde{e}_{il}^{\text{sat}}$.

As the transformation in Eq. (3.2) can be applied to each component separately, we have that

$$\begin{pmatrix} e_{ij}^{\text{sat}} \\ e_{il}^{\text{sat}} \end{pmatrix} = \begin{pmatrix} \frac{1}{1 + \exp(-\tilde{e}_{ij}^{\text{sat}})} \\ \frac{1}{1 + \exp(-\tilde{e}_{il}^{\text{sat}})} \end{pmatrix} \iff \begin{pmatrix} \tilde{e}_{ij}^{\text{sat}} \\ \tilde{e}_{il}^{\text{sat}} \end{pmatrix} = \ln\left(\frac{e_{ij}^{\text{sat}}}{1 - e_{ij}^{\text{sat}}}, \frac{e_{il}^{\text{sat}}}{1 - e_{il}^{\text{sat}}}\right)$$

is a differentiable bijection, with Jacobian determinant

$$|\mathcal{J}| = \left| \frac{\partial(\tilde{e}_{ij}^{\text{sat}}, \tilde{e}_{il}^{\text{sat}})}{\partial(e_{ij}^{\text{sat}}, e_{il}^{\text{sat}})} \right| = \begin{vmatrix} \frac{1}{e_{ij}^{\text{sat}}(1 - e_{ij}^{\text{sat}})} & 0 \\ 0 & \frac{1}{e_{il}^{\text{sat}}(1 - e_{il}^{\text{sat}})} \end{vmatrix} = \frac{1}{e_{ij}^{\text{sat}}(1 - e_{ij}^{\text{sat}})e_{il}^{\text{sat}}(1 - e_{il}^{\text{sat}})},$$

for $e_{ij}^{\text{sat}}, e_{il}^{\text{sat}} \in (0, 1)$.

It follows that

$$p\left(\begin{pmatrix} e_{ij}^{\text{sat}} \\ e_{il}^{\text{sat}} \end{pmatrix}\right) = p_{(\tilde{e}_{ij}^{\text{sat}}, \tilde{e}_{il}^{\text{sat}})^\top} \left(\ln \begin{pmatrix} \frac{e_{ij}^{\text{sat}}}{1 - e_{ij}^{\text{sat}}} \\ \frac{e_{il}^{\text{sat}}}{1 - e_{il}^{\text{sat}}} \end{pmatrix} \right) \cdot \frac{1}{e_{ij}^{\text{sat}}(1 - e_{ij}^{\text{sat}})e_{il}^{\text{sat}}(1 - e_{il}^{\text{sat}})},$$

where $e_{ij}^{\text{sat}}, e_{il}^{\text{sat}} \in (0, 1)$.

Deriving the Joint pdf of q_{ij}^{sat} and q_{il}^{sat}

Consider for the moment the prior distribution on the unconstrained relative flow rates of the saturated system $\tilde{\mathbf{q}}^{\text{sat}}$. From Section 3.1.1 we know that $\tilde{\mathbf{q}}^{\text{sat}} \sim \mathcal{N}(\boldsymbol{\mu}_{\tilde{\mathbf{q}}^{\text{sat}}}, \boldsymbol{\Sigma}_{\tilde{\mathbf{q}}^{\text{sat}}})$. Hence if we let $\mathbf{A}$ be the $3 \times [(k-1) + (k-2) + \dots + 1]$ matrix that extracts both $\tilde{q}_{ij}^{\text{sat}}$, $\tilde{q}_{il}^{\text{sat}}$ and $\tilde{q}_{im}^{\text{sat}}$ from $\tilde{\mathbf{q}}^{\text{sat}}$, we have that

$$\begin{pmatrix} \tilde{q}_{ij}^{\text{sat}} \\ \tilde{q}_{il}^{\text{sat}} \\ \tilde{q}_{im}^{\text{sat}} \end{pmatrix} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_{\tilde{\mathbf{q}}^{\text{sat}}}, \mathbf{A}\boldsymbol{\Sigma}_{\tilde{\mathbf{q}}^{\text{sat}}}\mathbf{A}^\top) = \mathcal{N}\left(\begin{pmatrix} \mu_{ij} \\ \mu_{il} \\ \mu_{im} \end{pmatrix}, \begin{pmatrix} \sigma_{ij}^2 & \rho_1\sigma_{ij}\sigma_{il} & \rho_2\sigma_{ij}\sigma_{im} \\ \rho_1\sigma_{ij}\sigma_{il} & \sigma_{il}^2 & \rho_3\sigma_{il}\sigma_{im} \\ \rho_2\sigma_{ij}\sigma_{im} & \rho_3\sigma_{il}\sigma_{im} & \sigma_{im}^2 \end{pmatrix}\right),$$

where we define μ_{ij} , μ_{il} , μ_{im} , σ_{ij} , σ_{il} and σ_{im} in terms of the above products, noting that ρ_1 , ρ_2 and ρ_3 are all correlations between different components in $(\tilde{q}_{ij}^{\text{sat}}, \tilde{q}_{il}^{\text{sat}}, \tilde{q}_{im}^{\text{sat}})^\top$. As for the illustration in Figure 3.3, we assumed a common pairwise correlation of the form $\rho = \rho_1 = \rho_2 = \rho_3$.

In earlier derivations, we obtained an expression for the pdf of the transformed variable(s) using the well known transformation formula. Note that in this case, we cannot naively apply the transformation formula to the transformation in Eq. (3.1), as it is not bijective. In particular, we have that $(\tilde{q}_{ij}^{\text{sat}} + s, \tilde{q}_{il}^{\text{sat}} + s, \tilde{q}_{im}^{\text{sat}} + s)^\top$ gets mapped to the same vector for all $s \in \mathbb{R}$, as mentioned in Section 3.1.1. This becomes apparent when recognising that the transformation in Eq. (3.1) maps $\mathbb{R}^3 \rightarrow \Delta^2$ in this case, and so the transformation has 1 less degree of freedom.

Following Aitchison and Shen (1980), we instead consider the log-ratio transformation of the relative flow rates $(q_{ij}^{\text{sat}}, q_{il}^{\text{sat}})^\top$, defined as

$$\begin{pmatrix} v_{ij} \\ v_{il} \end{pmatrix} = \ln \begin{pmatrix} \frac{q_{ij}^{\text{sat}}}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}} \\ \frac{q_{il}^{\text{sat}}}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}} \end{pmatrix} = \ln \begin{pmatrix} \frac{q_{ij}^{\text{sat}}}{q_{im}^{\text{sat}}} \\ \frac{q_{il}^{\text{sat}}}{q_{im}^{\text{sat}}} \end{pmatrix} = \begin{pmatrix} \ln q_{ij}^{\text{sat}} - \ln q_{im}^{\text{sat}} \\ \ln q_{il}^{\text{sat}} - \ln q_{im}^{\text{sat}} \end{pmatrix},$$

which is bijective with inverse

$$\begin{pmatrix} q_{ij}^{\text{sat}} \\ q_{il}^{\text{sat}} \end{pmatrix} = \frac{1}{1 + \exp(v_{ij}) + \exp(v_{il})} \exp \begin{pmatrix} v_{ij} \\ v_{il} \end{pmatrix}.$$

Using Eq. (3.1), we have that

$$\begin{aligned} \begin{pmatrix} v_{ij} \\ v_{il} \end{pmatrix} &= \begin{pmatrix} \tilde{q}_{ij}^{\text{sat}} - \ln \left(\sum_{s \in \{j, \ell, m\}} \exp \tilde{q}_{is}^{\text{sat}} \right) \\ \tilde{q}_{il}^{\text{sat}} - \ln \left(\sum_{s \in \{j, \ell, m\}} \exp \tilde{q}_{is}^{\text{sat}} \right) \end{pmatrix} - \begin{bmatrix} \tilde{q}_{im}^{\text{sat}} - \ln \left(\sum_{s \in \{j, \ell, m\}} \exp \tilde{q}_{is}^{\text{sat}} \right) \\ \tilde{q}_{im}^{\text{sat}} - \ln \left(\sum_{s \in \{j, \ell, m\}} \exp \tilde{q}_{is}^{\text{sat}} \right) \end{bmatrix} \\ &= \begin{pmatrix} \tilde{q}_{ij}^{\text{sat}} - \tilde{q}_{im}^{\text{sat}} \\ \tilde{q}_{il}^{\text{sat}} - \tilde{q}_{im}^{\text{sat}} \end{pmatrix}, \end{aligned}$$

and so the log-ratio transformation of the relative flow rates $(q_{ij}^{\text{sat}}, q_{il}^{\text{sat}})^{\top}$ is a linear transformation of the unconstrained relative flow rates $(\tilde{q}_{ij}^{\text{sat}}, \tilde{q}_{il}^{\text{sat}}, \tilde{q}_{im}^{\text{sat}})^{\top}$.

In particular, by defining

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix},$$

and using the result on affine transformations of normal random vectors, we get that

$$\begin{pmatrix} v_{ij} \\ v_{il} \end{pmatrix} \sim \mathcal{N} \left(\mathbf{B} \begin{pmatrix} \mu_{ij} \\ \mu_{il} \\ \mu_{im} \end{pmatrix}, \mathbf{B} \begin{pmatrix} \sigma_{ij}^2 & \rho_{1ij}\sigma_{il} & \rho_{2ij}\sigma_{im} \\ \rho_{1ij}\sigma_{il} & \sigma_{il}^2 & \rho_{3il}\sigma_{im} \\ \rho_{2ij}\sigma_{im} & \rho_{3il}\sigma_{im} & \sigma_{im}^2 \end{pmatrix} \mathbf{B}^{\top} \right) = \mathcal{N}(\boldsymbol{\mu}_v, \boldsymbol{\Sigma}_v),$$

where

$$\boldsymbol{\mu}_v = \begin{pmatrix} \mu_{ij} - \mu_{im} \\ \mu_{il} - \mu_{im} \end{pmatrix}$$

and

$$\boldsymbol{\Sigma}_v = \begin{pmatrix} \sigma_{ij}^2 - 2\rho_{2ij}\sigma_{im} + \sigma_{im}^2 & \rho_{1ij}\sigma_{il} - \rho_{2ij}\sigma_{im} - \rho_{3il}\sigma_{im} + \sigma_{im}^2 \\ \rho_{1ij}\sigma_{il} - \rho_{2ij}\sigma_{im} - \rho_{3il}\sigma_{im} + \sigma_{im}^2 & \sigma_{il}^2 - 2\rho_{3il}\sigma_{im} + \sigma_{im}^2 \end{pmatrix}.$$

Moreover, the log-ratio transformation is also differentiable, having Jacobian determinant

$$\begin{aligned} |\mathcal{J}| &= \left| \frac{\partial(v_{ij}, v_{il})}{\partial(q_{ij}^{\text{sat}}, q_{il}^{\text{sat}})} \right| \\ &= \begin{vmatrix} \frac{1}{q_{ij}^{\text{sat}}} - \frac{1}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}}(-1) & -\frac{1}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}}(-1) \\ -\frac{1}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}}(-1) & \frac{1}{q_{il}^{\text{sat}}} - \frac{1}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}}(-1) \end{vmatrix} \\ &= \begin{vmatrix} \frac{1}{q_{ij}^{\text{sat}}} + \frac{1}{q_{im}^{\text{sat}}} & \frac{1}{q_{im}^{\text{sat}}} \\ \frac{1}{q_{im}^{\text{sat}}} & \frac{1}{q_{il}^{\text{sat}}} + \frac{1}{q_{im}^{\text{sat}}} \end{vmatrix} \\ &= \left(\frac{1}{q_{ij}^{\text{sat}}} + \frac{1}{q_{im}^{\text{sat}}} \right) \left(\frac{1}{q_{il}^{\text{sat}}} + \frac{1}{q_{im}^{\text{sat}}} \right) - \left(\frac{1}{q_{im}^{\text{sat}}} \right)^2, \quad q_{ij}^{\text{sat}}, q_{il}^{\text{sat}}, q_{im}^{\text{sat}} \in (0, 1) \\ &= \frac{q_{ij}^{\text{sat}} + q_{il}^{\text{sat}} + q_{im}^{\text{sat}}}{q_{ij}^{\text{sat}} q_{il}^{\text{sat}} q_{im}^{\text{sat}}} \\ &= \frac{1}{q_{ij}^{\text{sat}} q_{il}^{\text{sat}} (1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}})}, \end{aligned}$$

where we note that Eq. (3.1) implies $q_{ij}^{\text{sat}} + q_{il}^{\text{sat}} + q_{im}^{\text{sat}} = 1$.

We can therefore use the transformation formula on the log-ratio transformation to obtain the wanted pdf, i.e.

$$p\left(\begin{pmatrix} q_{ij}^{\text{sat}} \\ q_{il}^{\text{sat}} \end{pmatrix}\right) = p_{(v_{ij}, v_{il})}^\top \left(\ln \begin{pmatrix} \frac{q_{ij}^{\text{sat}}}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}} \\ \frac{q_{il}^{\text{sat}}}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}} \end{pmatrix} \right) \cdot \frac{1}{q_{ij}^{\text{sat}} q_{il}^{\text{sat}} (1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}})},$$

where $q_{ij}^{\text{sat}}, q_{il}^{\text{sat}} \in (0, 1)$, $q_{ij}^{\text{sat}} + q_{il}^{\text{sat}} < 1$ and $q_{im}^{\text{sat}} = 1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}$.

Deriving the Marginal pdf of q_{ij}^{sat}

To obtain the marginal pdf of q_{ij}^{sat} , we then marginalise over the joint density of q_{ij}^{sat} and q_{il}^{sat} . In particular, we compute

$$\begin{aligned} p(q_{ij}^{\text{sat}}) &= \int_0^{1-q_{ij}^{\text{sat}}} p\left(\begin{pmatrix} q_{ij}^{\text{sat}} \\ q_{il}^{\text{sat}} \end{pmatrix}\right) dq_{il}^{\text{sat}} \\ &= \frac{1}{q_{ij}^{\text{sat}}} \int_0^{1-q_{ij}^{\text{sat}}} p_{(v_{ij}, v_{il})}^\top \left(\ln \begin{pmatrix} \frac{q_{ij}^{\text{sat}}}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}} \\ \frac{q_{il}^{\text{sat}}}{1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}}} \end{pmatrix} \right) \cdot \frac{dq_{il}^{\text{sat}}}{q_{il}^{\text{sat}} (1 - q_{ij}^{\text{sat}} - q_{il}^{\text{sat}})} \end{aligned}$$

on the computer using some quadrature, e.g. the trapezoidal rule.

We note for the sake of being complete that the normal density appearing in the above integral is parametrised by

$$\boldsymbol{\mu}_v = \begin{pmatrix} \mu_{ij} - \mu_{im} \\ \mu_{il} - \mu_{im} \end{pmatrix}$$

and

$$\boldsymbol{\Sigma}_v = \begin{pmatrix} \sigma_{ij}^2 - 2\rho_2\sigma_{ij}\sigma_{im} + \sigma_{im}^2 & \rho_1\sigma_{ij}\sigma_{il} - \rho_2\sigma_{ij}\sigma_{im} - \rho_3\sigma_{il}\sigma_{im} + \sigma_{im}^2 \\ \rho_1\sigma_{ij}\sigma_{il} - \rho_2\sigma_{ij}\sigma_{im} - \rho_3\sigma_{il}\sigma_{im} + \sigma_{im}^2 & \sigma_{il}^2 - 2\rho_3\sigma_{il}\sigma_{im} + \sigma_{im}^2 \end{pmatrix},$$

and so the marginal density of q_{ij}^{sat} generally depends on the hyperparameters corresponding to q_{il}^{sat} and q_{im}^{sat} as well, not just those corresponding to q_{ij}^{sat} .

Derivation of $\mathbb{E}[\kappa_i]$ and its Limits as $\lambda_i \rightarrow 0$ From Above and $\lambda_i \rightarrow \infty$

From its definition in Section 3.1.2, we have that

$$\kappa_i \sim \text{Poisson}(\lambda_i) \mid 1 \leq \kappa_i \leq k - i, \quad i = 1, 2, \dots, k - 2.$$

This can equivalently be written as $\kappa_i = \nu_i \mid 1 \leq \nu_i \leq k - 1$, where we define the helper variable $\nu_i \sim \text{Poisson}(\lambda_i)$, for $i = 1, 2, \dots, k - 2$.

For $x \in \{1, 2, \dots, k - i\}$, it follows that

$$\begin{aligned} \Pr(\kappa_i = x) &= \Pr(\nu_i = x \mid 1 \leq \nu_i \leq k - 1) \\ &= \frac{\Pr(\nu_i = x \cap 1 \leq \nu_i \leq k - 1)}{\Pr(1 \leq \nu_i \leq k - 1)} \\ &= \frac{\Pr(\nu_i = x)}{\Pr(1 \leq \nu_i \leq k - 1)}, \end{aligned}$$

using the definition of conditional probability and the fact that if $\nu_i = x$, then we have that $1 \leq \nu_i \leq k - i$.

As ν_i is Poisson distributed with rate λ_i , we have that

$$\begin{aligned} \Pr(\kappa_i = x) &= \frac{\lambda_i^x e^{-\lambda_i}}{x!} \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell e^{-\lambda_i}}{\ell!} \right), \quad x \in \{1, 2, \dots, k-i\} \\ &= \frac{\lambda_i^x}{x!} \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right). \end{aligned}$$

Per the definition of the expectation of a discrete random variable, we therefore have that

$$\begin{aligned} \mathbb{E}[\kappa_i] &= \sum_{x=1}^{k-i} x \Pr(\kappa_i = x) \\ &= \left(\sum_{x=1}^{k-i} x \frac{\lambda_i^x}{x!} \right) \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right) \\ &= \lambda_i \left(\sum_{x=1}^{k-i} \frac{\lambda_i^{x-1}}{(x-1)!} \right) \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right) \\ &= \lambda_i \left(\sum_{\ell=0}^{k-i-1} \frac{\lambda_i^\ell}{\ell!} \right) \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right), \end{aligned}$$

by making the change of variables $\ell = x - 1$.

Note that if $\lambda_i \rightarrow 0$ from above, then all terms in $\mathbb{E}[\kappa_i]$ containing a strictly positive power of λ_i vanishes. As

$$\mathbb{E}[\kappa_i] = \lambda_i \left(\sum_{\ell=0}^{k-i-1} \frac{\lambda_i^\ell}{\ell!} \right) \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^\ell}{\ell!} \right) = \left(\sum_{\ell=0}^{k-i-1} \frac{\lambda_i^\ell}{\ell!} \right) \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^{\ell-1}}{(\ell-1)!} \right)$$

by factorising the denominator, we have that

$$\mathbb{E}[\kappa_i] \rightarrow \frac{1}{0!} \bigg/ \frac{1}{1!} = 1.$$

We can construct a similar expression for $\mathbb{E}[\kappa_i]$ where most terms cancel in the other limit as well. In particular, using one of the earlier expressions for $\mathbb{E}[\kappa_i]$, we have that

$$\mathbb{E}[\kappa_i] = \frac{\lambda_i^{-(k-i)}}{\lambda_i^{-(k-i)}} \mathbb{E}[\kappa_i] = \left(\sum_{x=1}^{k-i} x \frac{\lambda_i^{x-(k-i)}}{x!} \right) \bigg/ \left(\sum_{\ell=1}^{k-i} \frac{\lambda_i^{\ell-(k-i)}}{\ell!} \right).$$

If we then let $\lambda_i \rightarrow \infty$, we see that all but the last term in both the numerator and denominator vanishes. As a result, we have that

$$\mathbb{E}[\kappa_i] \rightarrow \frac{k-i}{(k-i)!} \bigg/ \frac{1}{(k-i)!} = k-i.$$

Derivation of the Asymptotic Behaviour of $\binom{k-i}{\kappa_i}$

Note that if $\kappa_i = 1, 2, \dots, k-i$ is fixed, we have by the definition of the binomial coefficient,

$$\begin{aligned} \binom{k-i}{\kappa_i} &= \frac{(k-i)!}{\kappa_i!(k-i-\kappa_i)!} \\ &= \frac{(k-i)(k-i-1)\cdots(k-i-\kappa_i+1)(k-i-\kappa_i)!}{\kappa_i!(k-i-\kappa_i)!} \\ &= \frac{(k-i)(k-i-1)\cdots(k-i-\kappa_i+1)}{\kappa_i!}. \end{aligned}$$

Hence, by factorisation of $k-i$,

$$\binom{k-i}{\kappa_i} = (k-i)^{\kappa_i} \left[\frac{1}{\kappa_i!} \left(1 - \frac{1}{k-i} \right) \left(1 - \frac{2}{k-i} \right) \cdots \left(1 - \frac{\kappa_i-1}{k-i} \right) \right].$$

Note that for $k-i \gg 0$, the term in brackets is approximately equal to $1/\kappa_i!$. Indeed, if we let $k-i \rightarrow \infty$, the term in brackets approaches $1/\kappa_i!$, as the other fractions tend to 0.

Asymptotically, we therefore have

$$\mathcal{O}\left(\binom{k-i}{\kappa_i}\right) = \mathcal{O}\left(\frac{(k-i)^{\kappa_i}}{\kappa_i!}\right) = \mathcal{O}((k-i)^{\kappa_i}),$$

as κ_i is a fixed constant.

If we instead treat κ_i as a function of $k-i$, and consider the case where $\binom{k-i}{\kappa_i}$ attains its maximum at $\kappa_i = \lfloor (k-i)/2 \rfloor$, the asymptotic behaviour changes.

In particular, if $k-i = 2n$ is even, we have by the definition of the binomial coefficient,

$$\binom{k-i}{\lfloor (k-i)/2 \rfloor} = \binom{2n}{\lfloor 2n/2 \rfloor} = \frac{(2n)!}{(n!)^2},$$

as $\lfloor 2n/2 \rfloor = n$.

If instead $k-i = 2n+1$ is odd, we have that

$$\binom{k-i}{\lfloor (k-i)/2 \rfloor} = \binom{2n+1}{\lfloor (2n+1)/2 \rfloor} = \frac{(2n+1)!}{n!(n+1)!} = \left[\frac{2n+1}{n+1} \right] \cdot \frac{(2n)!}{(n!)^2},$$

as $\lfloor (2n+1)/2 \rfloor = n$.

In the limit as $n \rightarrow \infty$, the factor in brackets becomes constant, i.e.

$$\frac{2n+1}{n+1} = \frac{2+n^{-1}}{1+n^{-1}} \rightarrow 2,$$

and so the asymptotic behaviour of $\binom{k-i}{\lfloor (k-i)/2 \rfloor}$ is in both cases determined by the behaviour of $(2n)!/(n!)^2$, where $2n = k-i$.

Using Stirling's formula (see for instance Cao et al. (2025)), we have that

$$\begin{aligned}\frac{(2n)!}{(n!)^2} &\sim \left[\sqrt{2\pi(2n)} \left(\frac{2n}{e} \right)^{2n} \right] / \left[\sqrt{2\pi n} \left(\frac{n}{e} \right)^n \right]^2 \\ &= \frac{\sqrt{4\pi n}}{2\pi n} \left(\frac{2n}{e} \right)^{2n} \left(\frac{n}{e} \right)^{-2n} \\ &= \sqrt{\frac{2}{\pi(2n)}} 2^{2n},\end{aligned}$$

as $n \rightarrow \infty$.

It follows by substitution of $2n = k - i$, that

$$\mathcal{O}\left(\binom{k-i}{\lfloor (k-i)/2 \rfloor}\right) = \mathcal{O}\left(\sqrt{\frac{2}{\pi(k-i)}} 2^{k-i}\right) = \mathcal{O}\left(\frac{2^{k-i}}{\sqrt{k-i}}\right),$$

as $\sqrt{2/\pi}$ is a constant not depending on the value of $k - i$.

Derivation of the Size of a Sphere of $4.532 \cdot 10^{56}$ Water Molecules

Using Avogadro's number $N_A \approx 6.0221415 \cdot 10^{23} \text{ mol}^{-1}$ (Fox et al. 2007), $4.532 \cdot 10^{56}$ water molecules is approximately equal to

$$M = \frac{4.532 \cdot 10^{56}}{N_A} = 7.525 \dots \cdot 10^{32} \text{ mol}.$$

Assuming then that the molecular weight of water is given by $M_{\text{H}_2\text{O}} = 18.015 \text{ g/mol}$ (National Institute of Standards and Technology 2025), the total mass of all molecules is approximately equal to

$$m = M \cdot M_{\text{H}_2\text{O}} = 1.355 \dots \cdot 10^{31} \text{ kg}.$$

If we then assume standard density $\rho_{\text{H}_2\text{O}} = 1,000 \text{ kg/m}^3$, it follows that the water molecules occupy a volume approximately equal to

$$V = \frac{m}{\rho_{\text{H}_2\text{O}}} = 1.355 \dots \cdot 10^{28} \text{ m}^3.$$

Noting that the volume of a sphere V is related to its radius r , through the relation $V = 4\pi/3 \cdot r^3$, the above volume corresponds to a sphere with radius

$$r = \sqrt[3]{\frac{3}{4\pi} V} = 1.479 \dots \cdot 10^6 \text{ km}.$$

As the nominal solar radius is $R_{\odot}^N = 695,700 \text{ km}$ (Fox et al. 2007), the radius of the sphere of water molecules is

$$\frac{r}{R_{\odot}^N} = 2.126 \dots$$

times larger than the radius of the sun. It follows that the diameter of a sphere of $4.532 \cdot 10^{56}$ water molecules, is approximately 4.252 times the size of the diameter of the sun.

Derivation of α_{accept} in Section 3.4.2

From Chib and Greenberg (1995), it follows that the acceptance probability α_{accept} associated with a proposal kernel $Q(G' | G)$ is given by

$$\alpha_{\text{accept}} = \min \left\{ 1, \frac{Q(G^{(\ell)} | G') p(G' | \mathcal{D}, \boldsymbol{\theta})}{Q(G' | G^{(\ell)}) p(G^{(\ell)} | \mathcal{D}, \boldsymbol{\theta})} \right\},$$

where we use $G^{(\ell)}$ to denote the current state of the chain.

If we substitute the full conditional distribution $p(G | \mathcal{D}, \boldsymbol{\theta})$ by Eq. (3.13), we therefore get that

$$\alpha_{\text{accept}} = \min \left\{ 1, \frac{Q(G^{(\ell)} | G') p(G') \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \boldsymbol{\theta}, G')}{Q(G' | G^{(\ell)}) p(G^{(\ell)}) \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \boldsymbol{\theta}, G^{(\ell)})} \right\}.$$

Consider for the moment the global proposal kernel, defined by

$$Q_{\text{global}}(G') = p(G') = \prod_{i=1}^{k-2} p(\mathcal{J}'_i | \kappa'_i) p(\kappa'_i),$$

A simple substitution in the expression for α_{accept} then yields

$$\begin{aligned} \alpha_{\text{accept}} &= \min \left\{ 1, \frac{p(G^{(\ell)}) p(G') \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \boldsymbol{\theta}, G')}{p(G') p(G^{(\ell)}) \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) | \boldsymbol{\theta}, G^{(\ell)})} \right\} \\ &= \min \left\{ 1, \prod_{\mathbf{d}(t) \in \mathcal{D}} \frac{p(\mathbf{d}(t) | \boldsymbol{\theta}, G')}{p(\mathbf{d}(t) | \boldsymbol{\theta}, G^{(\ell)})} \right\}. \end{aligned}$$

If we instead consider the local proposal kernel, defined by

$$Q_{\text{local}}(G' | G^{(\ell)}) = \frac{1}{k-2} \sum_{r=1}^{k-2} p(\mathcal{J}'_r | \kappa'_r) p(\kappa'_r) \prod_{\substack{i=1 \\ i \neq r}}^{k-2} \mathbf{I}(\mathcal{J}'_i = \mathcal{J}_i^{(\ell)}) \mathbf{I}(\kappa'_i = \kappa_i^{(\ell)}),$$

then we know, per its definition in Section 3.4.2, that G' and $G^{(\ell)}$ differ by the migration pathways that are emanating from at most a single reservoir, which we denote by s .

Using Eq. (3.7), we therefore have that

$$\frac{p(G')}{p(G^{(\ell)})} = \prod_{i=1}^{k-2} \frac{p(\mathcal{J}'_i | \kappa'_i) p(\kappa'_i)}{p(\mathcal{J}_i^{(\ell)} | \kappa_i^{(\ell)}) p(\kappa_i^{(\ell)})} = \frac{p(\mathcal{J}'_s | \kappa'_s) p(\kappa'_s)}{p(\mathcal{J}_s^{(\ell)} | \kappa_s^{(\ell)}) p(\kappa_s^{(\ell)})}.$$

Similarly, in the case of $Q_{\text{local}}(G' | G^{(\ell)})$, we have that the product of indicator functions equals 1 only in the case where $r = s$. Hence, we get that

$$\frac{Q_{\text{local}}(G^{(\ell)} | G')}{Q_{\text{local}}(G' | G^{(\ell)})} = \frac{\frac{1}{k-2} p(\mathcal{J}_s^{(\ell)} | \kappa_s^{(\ell)}) p(\kappa_s^{(\ell)})}{\frac{1}{k-2} p(\mathcal{J}'_s | \kappa'_s) p(\kappa'_s)} = \frac{p(\mathcal{J}_s^{(\ell)} | \kappa_s^{(\ell)}) p(\kappa_s^{(\ell)})}{p(\mathcal{J}'_s | \kappa'_s) p(\kappa'_s)}.$$

A simple substitution of these fractions in the expression for α_{accept} then yields

$$\begin{aligned}\alpha_{\text{accept}} &= \min \left\{ 1, \frac{p(\mathcal{J}_s^{(\ell)} \mid \kappa_s^{(\ell)})p(\kappa_s^{(\ell)})p(\mathcal{J}'_s \mid \kappa'_s)p(\kappa'_s) \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G')}{p(\mathcal{J}'_s \mid \kappa'_s)p(\kappa'_s)p(\mathcal{J}_s^{(\ell)} \mid \kappa_s^{(\ell)})p(\kappa_s^{(\ell)}) \prod_{\mathbf{d}(t) \in \mathcal{D}} p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G^{(\ell)})} \right\} \\ &= \min \left\{ 1, \prod_{\mathbf{d}(t) \in \mathcal{D}} \frac{p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G')}{p(\mathbf{d}(t) \mid \boldsymbol{\theta}, G^{(\ell)})} \right\}.\end{aligned}$$

B. Parameter Settings

Parameter	Value	Unit
$\mu_{\tilde{C}}$	$(5.0 \ 5.0 \ 5.0 \ 5.0 \ 5.0)^\top$	kPa
$\Sigma_{\tilde{C}}$	$\begin{pmatrix} 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0.25 \end{pmatrix}$	$(\text{kPa})^2$
$\mu_{\tilde{\alpha}}$	$(7.0 \ 7.0 \ 7.0 \ 7.0 \ 7.0)^\top$	m
$\Sigma_{\tilde{\alpha}}$	$\begin{pmatrix} 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0.25 \end{pmatrix}$	m^2
$\mu_{\tilde{\beta}}$	$(4.0 \ 4.0 \ 4.0 \ 4.0 \ 4.0)^\top$	m
$\Sigma_{\tilde{\beta}}$	$\begin{pmatrix} 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0.25 \end{pmatrix}$	m^2
$\mu_{\tilde{q}^{\text{sat}}}$	$(0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0)^\top$	-
$\Sigma_{\tilde{q}^{\text{sat}}}$	$\begin{pmatrix} 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 \end{pmatrix}$	-
$\mu_{\tilde{e}^{\text{sat}}}$	$(0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0)^\top$	-
$\Sigma_{\tilde{e}^{\text{sat}}}$	$\begin{pmatrix} 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.25 \end{pmatrix}$	-

Table B.1.: Parameter settings/hyperparameters for the prior on continuous parameters $p(\theta)$, used for simulation in Sections 3.2 and 3.5.

Parameter	Value	Unit
Σ_{ζ}	$\begin{pmatrix} 0.25^2 & 0 & 0 & 0 & 0 \\ 0 & 0.25^2 & 0 & 0 & 0 \\ 0 & 0 & 0.25^2 & 0 & 0 \\ 0 & 0 & 0 & 0.25^2 & 0 \\ 0 & 0 & 0 & 0 & 0.25^2 \end{pmatrix}$	m^2
$\mathbf{h}^{\text{thres}}$	$(1.0 \ 1.0 \ 1.0 \ 1.0 \ 1.0)^{\top}$	m

Table B.2.: Parameter settings for the acquisition/likelihood model $p(\mathbf{d}(t) \mid \mathbf{h}(t))$, used for simulation in Section 3.5.

Parameter	Value	Notes
Initial step size	$(0.01 \ 0.01 \ 0.01 \ 0.01 \ 0.01)^{\top}$	Initial step size used by the HMC transition kernel. The kernel automatically updates the step size during the first 80 % of the burn-in period, targeting an acceptance probability of 0.75. For details, see TensorFlow developers (2023b).
Number of leapfrog steps	5	Number of steps taken by the leapfrog integrator. For details, see TensorFlow developers (2023a).
Number of burn-in samples	2,000	Number of burn-in samples discarded by the HMC kernel. If M samples are drawn from the posterior distribution using HMC, then a total of $M + 2,000$ samples are realised, with the 1st 2,000 samples being discarded in this case. For details, see TensorFlow developers (2023a).

Table B.3.: Parameter settings for the HMC transition kernel, used for simulation in Section 3.5.

Parameter	Value	Unit
$\Delta\rho$	300.0	kq/m^3
$\mathbf{q}^{\text{ext}}$	$(0.04 \ 0 \ 0 \ 0 \ 0)^{\top}$	Mtpa
τ^{min}	$1.0 \cdot 10^{-3}$	-
τ^{act}	$1.0 \cdot 10^{-6}$	-
N	75,000	-
M	10,000	-

Table B.4.: Additional parameter settings used for simulation in Sections 3.2 and 3.5.

C. Source Code

The source code of both the differentiable simulator $\mathcal{S}$, MCMC kernel and various functions needed to reproduce the figures of this thesis have been consolidated and uploaded to GitHub, to provide transparency and ensure reproducibility. To access the repository, the reader may use the following link, <https://github.com/GroedelandR/TMA4900-Industrial-Mathematics-Master-s-Thesis>, noting that the repository also provides a general description of the files contained within it.

